



- Proprietary -

File No.:	RT79	87630	RFP	R005	0
	Doc. Collection ID	SI	Document Type	Serial No.	Revision No.

2025 Feb 21

Attention: Sales

Subject: Candu Request for Proposal

The material described herein is required by Candu Energy Inc. for its Cernavoda Unit 1 Refurbishment project.

You are invited to provide to Candu Energy Inc. (hereafter Candu) your written proposal containing your best price(s) and delivery, including all discounts and incentives, for the following in accordance with the requirements described in this RFP:

This RFP consists of the following documents:

1. This Request for Proposal, including the attached Acknowledgement of Receipt/ Confirmation of Intention to Submit Proposal;
2. C.161 Terms and Conditions, and
3. Form of Proposal, including all attachments thereto, to be completed and submitted by the Supplier.

Failure to comply with the RFP may cause your Proposal to be rejected, disqualified or not be considered.

This RFP, including all of the documents listed above, remains the property of Candu, and is disclosed to the recipient in accordance with the provisions of the non-disclosure agreement entered into by the recipient and Candu.

The supplier shall bear its own costs of preparing the proposal and participating in any proposal reviews and negotiations with Candu and otherwise in respect of this procurement process, and Candu shall not be responsible to pay, reimburse, or compensate the supplier in any manner whatsoever for any expenses, costs, losses or any direct or indirect damages incurred or suffered by any supplier or any third party, regardless of the reasons or circumstances, as set out in the Request for Proposal.





Proposal Submission Instructions:

Proposals shall be addressed to:

Candu Energy Inc.
Procurement Department
2251 Speakman Drive
Mississauga, Ontario
L5K 1B2
Attention: Ashna Irani-kavasji

Your proposal shall be submitted electronically to Candu's Procurement Proposal Submission email at pps@atkinsrealis.ca by no later than 5:00 pm . Proposals submitted after this date may be deemed non-responsive or rejected by Candu.

****Note: If your email submission is greater than 10 MB in size, please contact pps@atkinsrealis.ca for access to Candu's SFTP site.**

Proposal Preparation:

Proposals shall be provided in the **Form of Proposal Document** along with related Appendices and shall have straightforward concise description of your abilities to satisfy the requirements in the RFP.

Exceptions to any requirements contained within this RFP must be clearly identified per the attached Appendix D Summary of Exceptions.

Acceptance of Proposal:

Candu reserves the right to accept or reject any or all proposals received as a result of this RFP, and may at its sole discretion, negotiate any portion of this RFP with any Bidder(s). Candu is not obligated to accept, or recommend, the lowest value submitted. Proposals will be judged on which one best meets Candu's technical and commercial requirements.

Candu reserves the right to place Purchase Order(s) for all or part of the Scope of Work specified in this RFP.

Candu may cancel the Request for Proposal at any time, for any reason.





ATKINSRÉALIS SUPPLIER CODE OF CONDUCT AND ANTI-CORRUPTION LAWS

The Supplier acknowledges having obtained and read the AtkinsRéalis Supplier Code of Conduct (hereinafter the “Code”), which is available at <https://www.snclavalin.com/~media/Files/S/SNC-Lavalin/download-centre/en/policy/supplier-code-conduct-en.pdf>, which is deemed incorporated into this Contract. Furthermore, the Consultant undertakes to adhere and comply with the principles set forth in the Code.

The Supplier will conduct itself, and will cause its employees, directors, officers, sub-suppliers, representatives and agents to conduct themselves, in all matters concerning (or that may reasonably be perceived as concerning) the Project or the Contractor, in compliance with the Code. Any breach of this article shall, in addition to any other rights or remedies, be considered a material breach and entitle the Contractor to terminate this Agreement or any other relationship it may have with the Consultant. In addition, the Consultant undertakes to include these provisions in its agreements with any third party rendering services related to the Project or the Contractor.

When used herein, the term “Corrupt Practices” means: (i) “bribery of a foreign public official”, as such term is defined in the OECD Convention on Combating Bribery of Foreign Public Officials in International Business Transactions, dated 21 November 1997 (the “Convention”); and/or (ii) bribery of public official under applicable local legislation; and/or (iii) an offence under any applicable anti-bribery legislation.

The Consultant represents and warrants that:

- i) It has not engaged, and shall not be engaging, directly or indirectly through its officers, directors, employees, agents, consultants and subcontractors (the last three hereinafter “Agents”), in any Corrupt Practices in connection with the Project; and that,
- ii) To the best of its knowledge, none of its or its affiliates’ officers, directors, employees or Agents involved in the Project or in the execution of the purposes of this Agreement is a foreign or local public official..

The Supplier undertakes to promptly notify if:

- i) It learns or has reason to believe that Corrupt Practices or a breach of the Code may have occurred in connection with the Project; or if,
- ii) It learns that any of its or its affiliates’ directors, employees or Agents involved in the Project or in the execution of the purposes of this Agreement, becomes a foreign or local public official; or that a foreign or local public official becomes an officer, director, employee or Agent of such Party or of an affiliate thereof.
- iii) It has any question or concerns with respect to the Code and contact its AtkinsRéalis contract manager representative for any clarification.

The Consultant shall fully cooperate in good faith with the Contractor in investigating into an alleged Corrupt Practice or any other breach of compliance legislation; promptly furnish to the Contractor such information as it reasonably requests; and permit the Contractor or its representative to have access to the books and account and records as may be relevant for such investigation.

The Consultant recognizes that any breach of this section and violation of the Code will be treated in accordance with the provisions of the terms and conditions agreed between the Consultant and the Contractor.





Intellectual Property (IP) Management

Suppliers are expected to have appropriate business processes in place to ensure that IP and confidential information is appropriately managed. We encourage our suppliers to contact us if there is uncertainty with respect to any particular IP or business practices. Awareness of IP issues, adherence to contractual terms, implementation of proper safeguards and consultation with Candu Energy will help you mitigate your risk of IP infringement and the supply of counterfeit, fraudulent or suspect items.

The industry as a whole also benefits from transparency in the proper and legal use of IP, which in turn, mitigates such risks as non-conformance of items or services to quality requirements (e.g., codes, standards, technical specifications) leading to removal from the approved bidders list, or in an extreme case, effect station shutdown and damage to individual and vendor professional viability. Improper management or unauthorized use of IP is a serious concern and Candu Energy intends to protect its commercial interests and enforce its rights with respect to IP.

Suppliers may not use or share the information provided or technology or work product generated during prior AECL or Candu Energy contracts for any purpose beyond the scope of the respective contracts, unless you have obtained the express prior written consent of Candu Energy (that is since Oct 2, 2011), regardless of whether such use is for work relating to the same or a different utility. Note that the foregoing applies both with respect to information provided to the vendor, as well as information or technology developed by the vendor in the course of its work for AECL, Candu Energy. For further certainty, AtkinsRéalis is the owner to the rights of use and title of information where the purchaser was SNC, SLN, NPM, Canatom NPM and Canatom.

Use of AtkinsRéalis Formal Document Template

To fulfill the Purchase Order requirements, including documents, drawings, software and other work products, per Appendix F– PDCL, the supplier shall provide information within the AtkinsRéalis template, per Appendix H - Formal Document Template. In the event that the information to be provided is background intellectual property owned by the supplier or its sub-suppliers, the supplier shall declare the background IP in the bid process via Appendix D - Summary Exceptions.

The Buyer shall be notified by formal letter of any new additional back ground intellectual property discovered during the course of the work.

(Background IP: information that is generated under a commercial agreement where the right and title belongs solely with the supplier).

If the supplier or sub-suppliers will require the use of contract information from a previous AECL or Candu Energy PO, they must clearly document this information in their bid submission, by identifying the previous Purchase Order Number.

Human Performance Program:

Please conduct a daily Pre-job Briefing with workers performing work on our products:

A Pre-Job briefing is a meeting of workers and supervisors conducted before performing a job to discuss the tasks involved, hazards, and related safety precautions. This meeting helps individuals to better understand





what to accomplish and what to avoid. Pre-job briefings help participants avoid surprises in the field and reinforce the idea that there are no routine activities.

Please encourage a Questioning Attitude from the workers executing tasks on our products:

This tool alerts people to imminent hazards, warning signs, and uncertainties in the work environment or with the work plan. It encourages the user to stop and resolve those hazards, warnings, or uncertainties before proceeding with the job. Doubt must be followed up with the discovery of facts, not assumptions, to reveal more about the situation, which eliminates doubt.

Please encourage workers to STOP When Unsure, while working on our products:

Stop When Unsure is a communication technique. It prompts performers to gain more accurate information about the work situation from other knowledgeable persons before proceeding with the activity. It involves a brief stoppage of work to allow workers, the Supervisor, or other knowledgeable persons to discuss and resolve the issue before resuming the task. This removes an individual from knowledge based mode and returns the individual to rule based mode, thereby significantly reducing the likelihood of error.

Short List:

Candu may, after reviewing all the proposals submitted in response to this RFP, create a short list of those proposals that are judged most qualified to meet the requirements of the work. Candu may enter into negotiations with these short-listed suppliers to further seek mutual opportunities.

Scope of Work:

Design, prepare and provide drawings and required documentation, Supply all material, manufacture, assemble, inspect, test, identify, prepare, pack and mark for shipment, deliver and guarantee the following in accordance with Appendix B – RFP Price Schedule and Appendix C – Technical Requirements and Specifications.

Country of End Use: Romania

Quality Assurance Requirements:

The supplier must have a quality assurance program that meets, or exceeds, the requirements of N299.4-16. Quality Assurance Program Requirements for the Supply of Items and Services for Nuclear Power Plants, Category 4. The Supplier confirms the Goods will be manufactured in compliance with such quality program(s).

Item Identification Requirements:

The supplier shall ensure Identification requirements in accordance with C.161 Section 16.

Reporting of Non-Conformances:





Supplier shall submit any non-conformances with the corresponding proposed dispositions to Candu for acceptance/approval of disposition, as required.

Non-Delivery of Counterfeit, Fraudulent, and Suspect Items (CFSI):

The Supplier shall take all measures including quality systems and controls to ensure that no counterfeit materials or parts are incorporated into any of the Goods or the Work in accordance with C.161 Terms and Conditions, Section 14. The Supplier shall supply the Purchaser with a certificate to this effect confirming same as part of its Certificate of Compliance/Conformance.

Certificate of Compliance:

A Certificate of Compliance for the above materials is required to accompany the goods at time of delivery. The certificate must be on the Supplier's letterhead, describe the item(s) supplied, reference the supplier's packing slip number, reference the Candu Purchase Order number, be signed by an authorized Supplier representative and contain the statement "The material supplied herein fully satisfies the requirements of Candu's Purchase Order". All the above information must be shown on the certificate.

(Use when CRN is specified)

A Certificate of Compliance for the above material is required to accompany the goods at time of delivery. The Certificate must have the Canadian Registration Number (CRN) clearly identified, together with a written certification to the validity of the CRN as of the current date.

Alternately, the CRN may be provided together with the proof of TSSA acceptance. The Certificate of Compliance should also be traceable to the Supplier's packing slip number and to the Candu PO number. All information must be shown on the certificate.

Quality Records – History Docket/History File:

A history docket or history file shall be prepared in accordance with the Purchase Order requirements.

A History Docket shall be supplied for all Goods which are specified in accordance with CSA-N285.0 and CSA B51.

The history docket or History File should be compiled as the work progresses so that the complete history docket/file will be available for acceptance on completion of the work. Drawings, specifications, procedures and reports included in the history docket/file shall bear evidence of their acceptance by the Purchaser and acceptance/approval by the Authorized Inspector where such acceptance/approval is a requirement. Only permanent records shall be included in the history docket/file.

For CSA B51 pressure retaining components where applicable, the Certificate of Compliance/Conformance must have the Canadian Registration Number (CRN) clearly identified, together with a written certification of the validity of the CRN as of the current date. Alternately, the CRN may be provided together with the proof of Jurisdictional acceptance.

All documents shall be adequately identified to the Purchase Order and to the particular items in the Purchase Order to which they are related.

The Supplier shall be responsible for the maintenance of these records while they are in the Supplier's possession. This responsibility includes protection from damage while in storage and during handling. The





original copy of the Quality Records shall be controlled by the Supplier, in accordance with the applicable codes and standards.

The history docket/file shall include the History Docket/File Acceptance form, to be provided at a later date, as a cover sheet and a Table of Contents listing the enclosed document.

Contents of the History Dockets/Files shall be in accordance with Appendix C – Technical Requirements and Specifications, Specification 00-00660-TS-001.

All Quality Records provided under the Purchase Order shall be legible and reproducible.

The history docket/file shall be completed, presented to and accepted by the Purchaser's Quality Assurance representative before the Goods are released for shipment. A Quality Assurance Release is required for the history docket/file.

After acceptance by the Purchaser's representative, the history docket/file shall be delivered as directed in the Purchase Order.

Copies of History Docket/History Files required:

One (1) PDF Copy with OSR to The Purchaser; and

One (1) PDF Copy with OSR to each Free Issue Recipient, If Applicable.

CSA/ULC Approval: *(Use with electrical devices)*

All devices shall be CSA certified and must identify CSA or ULC certification clearly shown on each device.

Quality Assurance, Surveillance and Audits

Quality Assurance, Surveillance and Audits shall be in accordance with C.161 Section 13.

Quality Assurance Release:

The work will be subject to expediting, quality surveillance acceptance and release by Candu and/or its authorized representative or the purchaser's client.

Delivery Requirements:

Please identify your best achievable delivery in number of days/weeks from receipt of PO.

Destination/Delivery Point:

The Delivery Terms for the Goods and Documentation shall be DAP in accordance with the latest Incoterms. Refer to the Price Schedule's End Use/Destination column for specific delivery destination.

All costs associated in delivering under these terms as well as costs for any Freight Forwarders' Services shall be borne by the Supplier.





Goods destined to: Candu Energy Inc.
2251 Speakman Drive
Mississauga, Ontario L5K 1B2
Attention: Shipping & Receiving, Building SP3 (2233 Speakman Drive)

Pricing Requirements:

Prices are to be submitted in Canadian funds, fixed, not subject to adjustment, taxes (HST) excluded. All other costs and taxes, including, but not limited to duty, tariffs and other taxes payable to a taxation authority shall be included in the offered price. Suppliers shall identify, in their proposal, any known additional potential costs that would be applicable to fully deliver the scope of work. Failure by the Supplier to disclose any such additional costs may cause rejection of the Supplier's proposal.

Liquidated Damages:

Liquidated damages shall apply and are in accordance with C.161 Section 2

Warranty

The supplier warrants that the goods/work furnished hereunder will be new and free from defects in design, workmanship and materials and shall conform to the technical specifications set out herein. This warranty will expire 18 months from the date of delivery or 12 months from the date of installation, whichever is earlier.

The supplier shall promptly correct all defects in design, workmanship and materials of the goods appearing within the warranty period with replacement, repairs or modifications at its own expense. When an item is replaced, repaired or modified, that specific item will be further warranted until the later date of (i) the expiration of the above applicable warranty period or (ii) the expiration of a period of one year from the date of such replacement, repair or modification.

Information to be provided with the Proposal:

The Supplier shall provide in the Form of Proposal the following information:

Schedule by when documentation identified in the Scope of Work, Technical Specification or Procedures/Document Checklist will be delivered to Candu. Please specify in number of weeks from a pre-determined milestone ie. receipt of PO, acceptance of test plan, etc.

Identify the QA Program, or details of any QA system currently implemented at your facility. Submit a copy of your QA Manual, if not already in Candu's possession, along with your proposal.

Copies of correspondence or certification from any Authority or nationally recognized body accepting the QA Program that your Company may have implemented.





The name of representative within your organization with whom we may discuss the current state of your QA Program. i.e. QA Manager.

The name(s), phone numbers and contacts of other firms to whom you have performed work of a similar nature to that contained in this RFP.

A list of sub-contractors to be used in the delivery of the scope of work (material(s) supply, processes, etc.), their role and quality certificates/qualification they possess to perform the required work.

Any descriptive literature, specifications or data sheets of the equipment/devices offered. Documentation must be suitable for Candu to make a technical assessment of the product(s).

Identify any deviations and/or exceptions to our technical or commercial terms & conditions.

Documents Supplied to Suppliers/Confidentiality:

All documents, whether technical or otherwise, provided to suppliers are confidential and remain the property of Candu. No exploitation or transfer of any information contained therein is permitted in the absence of an agreement with Candu, and neither the information nor any such information may be released without written consent of Candu. Likewise, all information submitted by Suppliers and identified as confidential, will be treated similarly.

Those suppliers not submitting a proposal in response to this RFP shall return all documents supplied to them to Ashna Irani-Kavasji immediately, stating the reason for decline. Alternatively, the supplier may issue a document to Candu declaring they will not be responding to the RFP and that all documents have been destroyed.

Clarification of Request for Proposal:

No officer, agent or employee of Candu is authorized to explain, or interpret orally any part of the Request for Proposal.

Any Supplier requiring clarification of any part of the Request for Proposal, whether from a technical or commercial prospective, shall do so in writing directed to Candu, Attention: Ashna Irani-Kavasji, Procurement Department. Inquiries shall be submitted only by Suppliers and not by prospective subcontractors or suppliers.

Where required, additions, deletions or changes to the Request for Proposal required as a result of such inquiries will be issued to all Suppliers.

Alternate Proposals:

Proposals shall be submitted in accordance with the requirements of the Request for Proposal, however, Candu wishes to make clear that notwithstanding, the accompanying Technical Specification, Candu may give consideration to alternatives or deviations which the supplier considers advisable by reason of his own





experience and which is in Candu's interests, provided attention is clearly drawn to such alternatives or deviations and descriptive matter submitted, pointing out wherein the recommended arrangement or device is equal to or superior to that required by the Technical Specification.

All alternatives or deviations initiated by the supplier shall include all particulars, including price, delivery, details of alternate, manufacturer and component performance, and shall be clearly identified separately in the proposal.

Discrepancies, Omissions, Clarification to RFP Documents:

Suppliers who find discrepancies in, omissions from, or requiring clarification to, the RFP documents, or having any doubt to the meaning or intent of any part thereof, should be brought to the attention of Ashna Irani-Kavasji, by email. The buyer will send written instruction, clarification or explanations in reply. All such requests and their responses will be provided to all the other Suppliers without divulging the name of the firm requesting the information.

Candu will not be responsible for any consequence instructions given to a supplier by any Candu employee, either orally or in writing, that change the requirements, other than Ashna Irani-Kavasji, identified in this RFP.

Disqualification:

Any attempt by the Supplier, or its employees, to approach Candu employees, other than Ashna Irani-Kavasji, during the RFP process with a view to extract information on a preferential basis or to influence decisions, may result in a withdrawal of Candu's invitation or disqualification of the proposal.

Repeated breach in this protocol may give Candu cause to remove the supplier's name from its supplier listing and consequently eliminate future potential business opportunities.

Validity of Proposals:

Proposals shall remain valid for acceptance by Candu for a period of 90 days from the date of the proposal.

Proposal Expenses:

Suppliers will not be reimbursed for any costs of any kind whatsoever incurred in connection with the preparation and presentation of their proposal.

Terms & Conditions:

Any Purchase Order issued as the result of this invitation will be subject to Candu's Terms & Conditions C.161 Rev.2, identified and attached hereto.

Acknowledgement:





Please acknowledge receipt of this RFP and indicate to Ashna Irani-Kavasji, identified below, your intention to submit a proposal, or to decline our invitation.

Yours truly,

Ashna Irani-Kavasji

Ashna Irani-Kavasji

Buyer

Phone: 905-369-2616

E-Mail: Ashna.Irani-Kavasji@atkinsrealis.ca

Authorized by:

Tim
Zabel

Tim Zabel -Procurement Lead

Digitally signed by Tim Zabel
DN: cn=Tim Zabel, o=Candu Energy Inc., ou=Procurement, email=tim.zabel@atkinsrealis.ca
Reason: I am the author of this document
Date: 2025.02.24 09:40:12 -0500
Using PKCS#7 syntax version 1.2.0

Appendices:

Appendix A – Transmittal

Appendix B – RFP Price Schedule

Appendix C – Technical Requirements & Specifications

Appendix D – Summary of Exceptions

Appendix E – Candu Terms and Conditions C.161 Rev2

Appendix F – Procedures Documentation Checklist (PDCL)

Appendix G – Procurement Specifications

Appendix H – Formal Document Template





APPENDIX A

Transmittal

Please complete and email back to confirm receipt of this Request for Proposal

From:

Tel.:

Date: _____

To - Procurement

Candu Energy Inc.

2251 Mississauga, Ontario,

L5K 1B2

Attention: Ashna Irani-kavasji/Procurement Proposal Submission

Email: ashna.irani-Kavasji@atkinsrealis.ca/pps@atkinsrealis.ca

ACKNOWLEDGEMENT

Subject: Request for Proposal	
Request for Proposal Ref#	RT79-87630-RFP-R005
Title:	VRS MANUAL RETRIEVAL TOOL FOR C1R

We hereby acknowledge receipt of your inquiry and confirm our intention to:

☐ Respond to your RFP as requested by

☐ Not to respond to your RFP. (Please specify reason below)

Request for Proposal and all related Document(s) have been disposed of by:

☐ Returning to Candu Energy Inc., Attn: Ashna Irani-Kavasji

or

☐ Destroyed at Supplier's facility

Reason for Decline: _____

Received: _____

Name: _____

(Date)

(Print Name)

Signature: _____



AtkinsRéalis



Candu Energy Inc.
An AtkinsRéalis company
2251 Speakman Drive
Mississauga, Ontario,
L5K 1B2, Canada
 +1 905.823.9040

APPENDIX B
RFP Price Schedule
ATTACHED

This document has been prepared for an AtkinsRéalis company. The information contained herein is subject to the terms and conditions of an agreement, which may be referenced on this document. No use, disclosure or reproduction of this document or the information contained herein is permitted, except in accordance with the applicable agreement.

Nuclear





CANDU Material Management System (CMMS)

RFP Price Schedule

RFP: RT79-87630-RFP-R005

Cernavoda Unit 1 Refurbishment

Item	Qty	Material Code	Description	End Use / Destination	Related Documents	Unit Price	Total Price
------	-----	---------------	-------------	-----------------------	-------------------	------------	-------------

Unit: 1 Price Schedule

01	1 EA	304A2800	TOOL VRS MANUAL RETRIEVAL TOOL DRAWING #: RT-87630-71254-1000-1-GA-D Nuclear Class: NA QA Program: CSA N299.4-16 Seismic Qual: Not Required	- Retube - SP3 Warehouse			
10	1 LO	909A9900	DOCUMENT - DOCUMENTATION PACKAGE <i>Remarks: DOCUMENTS AS REQUESTED IN RT79-87000-PDCL-003</i>	- Retube - SP3 Warehouse			
11	1 LO	909A0061	SERVICE - EXPORT PACKAGING, MARKING & SHIPPING DOCS <i>Remarks: AS PER PROCEDURE CE-904100-PRO-008</i>	- Retube - SP3 Warehouse			

Cernavoda Unit 1 Refurbishment Unit 1 Items - Total Fixed Price: \$ _____

Total Fixed Price: \$ _____



Candu Energy Inc.
An AtkinsRéalis company
2251 Speakman Drive
Mississauga, Ontario,
L5K 1B2, Canada
 +1 905.823.9040

APPENDIX C –
Technical Requirements and Specifications

Document Number	Revision	Document Name
		Documents will be sent via FTP site

This document has been prepared for an AtkinsRéalis company. The information contained herein is subject to the terms and conditions of an agreement, which may be referenced on this document. No use, disclosure or reproduction of this document or the information contained herein is permitted, except in accordance with the applicable agreement.





Candu Energy Inc.
An AtkinsRéalis company
2251 Speakman Drive
Mississauga, Ontario,
L5K 1B2, Canada
 +1 905.823.9040

APPENDIX D
Summary of Exceptions
ATTACHED



Summary of Exceptions – Commercial			
Document Number:	Project: C1 Refurb	Buyer:	Date:

Package Description:	Bidder:
----------------------	---------

Item No.	Description	Bidder’s Comment	Candu’s Reply Comment	Bidder’s Reply Comment	Candu’s Reply Comment	Bidder’s further Comment
1						
2						
3						
4						
5						
6						

Summary of Exceptions - Technical			
Document Number:	Project:	Buyer:	Date:

Package Description:	Bidder:
----------------------	---------

Item No.	Description	Bidder’s Comment	Candu’s Reply Comment	Bidder’s Reply Comment	Candu’s Reply Comment	Bidder’s further Comment
1						
2						
3						
4						
5						
6						



APPENDIX E
Terms and Conditions

ATTACHED

Document Number	Revision	Document Name
C.161	3	Terms and Conditions
Supplementary Terms and Conditions to C.161	2	





CANDU PURCHASE ORDER
TERMS AND CONDITIONS
Equipment, Materials and Services
(Nuclear & Non-Nuclear)

REVISION 3

CANDU ENERGY INC.



SECTION 1 DEFINITIONS

The following terms wherever used in the Purchase Order, unless the context otherwise requires, shall have the following meaning:

1) “Applicable Laws” means

- a) Federal, provincial or municipal laws, statutes, orders-in-council, by-laws, regulations, codes, orders, or permits; and/or
- b) Rulings and conditions of any licence (including without limitation, Candu’s licence(s), certificate, registration, authorization, consent and approval issued by a government authority;

Relating to the performance or completion of the Work or the obligations of either Party under the Purchase Order;

2) “Authorized Inspector” means a person designated as such by an appropriate legal authority to perform inspections, more fully described in CSA Standard N285.0;

3) “Certificate of Compliance” has the meaning set out in Section 13.10;

4) “Change Notice” has the meaning set out in Section 20.1;

5) “Change Order” has the meaning set out in Section 20.3;

6) “Commodity Taxes” means all taxes which are levied on or measured by, referred to as sales, use, consumption, GST, HST, QST, RST, value-added, excise, stamp, all customs duties, countervail, anti-dumping and special import measures, import and export taxes, including any interest, penalties, administrative monetary penalties or their equivalent as imposed by a governmental authority, fines or other amounts imposed in respect hereof;

7) “Contract” or “Purchase Order” means this legally binding agreement entered by the Purchaser and the Supplier on the Contract Effective Date;

8) “Contract Delivery Point” means the specified locations for delivery of the Goods as referred to in Purchase Order, Appendix A, *Price Schedule*;

9) “Contract Effective Date” means the date the Purchase Order takes effect;

10) “Supplier” means the supplier who enters into the Contract with the Purchaser;

11) “Supplier Parties” has the meaning set out in Section 23.1;

12) “Correspondence Procedure” means the correspondence procedure established pursuant to Section 6.2 and set out in Appendix E, *Correspondence Procedure*;

13) “CSA” means the Canadian Standards Association;

14) “Daily Delay LD Amount” means an amount equal to the product of:

a) 0.15%; and

b) The Contract Price;

15) “Delivery Date” means the date listed in the Purchase Order, “Deliv Date” column of Appendix A, *Price Schedule* for the applicable Goods, as modified from time to time by a Change Order;

16) “Deviation Disposition Request” or “DDR” has the meaning set out in Section 19.1;

17) “Documentation” means all Manuals, documents, reports, analyses, data, drawings and specifications required under the Purchase Order to be provided by the Supplier to the Purchaser;

18) “First Party” has the meaning set out in Section 27.8;

19) “Force Majeure” has the meaning set out in Section 30.1;

20) “Free Issue Material” means material supplied by and owned by the Purchaser without charge to the Supplier to be incorporated into the Work;

21) “Free Issue Recipient” means the Supplier, where it receives the Free Issue Material, subject to the Supplier acknowledging that title remains in the name of the Purchaser;



- 22) “Goods” means all materials, commodities, articles, services, equipment, goods, things, spares, consumables, and Documentation to be supplied by the Supplier in fulfilment of the Work;
- 23) “Government Authority” means any government, regulatory authority, government department, agency, commission, bureau, official, minister, crown corporation, court, board, tribunal, dispute settlement panel or body or other law, rule or regulation making entity (a) having or purporting to have jurisdiction on behalf of any nation, province, state or other geographic or political subdivision, or (b) exercising, or entitled or purporting to exercise any administrative, executive, judicial, legislative, policy, regulatory or taxing authority or power;
- 24) “GST” means the goods and services tax levied under Part IX of the *Excise Tax Act* (Canada);
- 25) “HST” means the goods and services tax levied under the [*Harmonized Sales Tax Act*](#) or similar act in any Province in Canada, as applicable;
- 26) “In-Service Date” means the date on which the Reactor Unit is capable of providing reliable electrical output to the grid, whether or not it is actually in-service and regardless of the capacity level that can be provided, as advised by Candu;
- 27) “Liquidated Damages” has the meaning set out in Section 2;
- 28) “Loss or Losses” means any and all claims, demands, losses, liabilities, damages, obligations, payments, fines and penalties, costs and expenses, including without limitation, accrued interest thereon, and the costs and expenses of any and all actions, suits, proceedings, assessments, judgments, settlements and compromises relating thereto and lawyers' fees and disbursements in connection therewith;
- 29) “Manuals” means operation and maintenance manuals provided with the Purchase order when required;
- 30) “Manufacturing Schedule” has the meaning set out in Section 6.3.2;
- 31) “Other Party” has the meaning set out in Section 27.8;
- 32) “Parties” means the Purchaser and the Supplier;
- 33) “Party” means either the Purchaser or the Supplier;
- 34) “Plant” means the pressurized heavy water nuclear power generation station;
- 35) “Purchaser” means Candu Energy Inc. or Candu;
- 36) “Purchaser Indemnified Parties” has the meaning set out in Section 23.1;
- 37) “Purchaser’s Proprietary Information” has the meaning set out in Section 27.1;
- 38) “Purpose” has the meaning set out in Section 27.3;
- 39) “Reactor Unit” means a single CANDU® pressurised heavy water reactor or a particular unit at the Plant;
- 40) “RST” means the Ontario retail sales tax levied under the *Retail Sales Tax* (Ontario);
- 41) “Site” means the Client’s site where the Plant is situated;
- 42) “Software Schedule” means the detailed list of Documentation to be submitted by the Supplier to the Purchaser, as per Purchase Order, Appendix F, *Software Schedule* as modified from time to time by the parties through formal change request to the Order Administrator;
- 43) “Subcontractor” means a person, firm or corporation, sometimes referred to as supplier, distributor, or vendor, having a contract of any tier with the Supplier for part of the Work, including the furnishing of labour, materials, equipment, apparatus or services therefor;
- 44) “Technical Exceptions” are contained in Appendix I *Completed and Approved Summary of Exceptions*;
- 45) “Technical Requirement Package” or “TRP” or “Technical Specifications” or any “Technical Documentation” means documents as per Appendix G, *Technical Requirements and Specifications*;
- 46) “Technical Information” means information relating to the Work, or designed, written, produced, developed or first actually reduced to practice by the Supplier in the course of the



Work including without limitation all inventions conceived, Documentation, technical documents, software (in both source code and compiled code formats), analyses, blueprints and as built plans and specifications, installation, operating, safety and maintenance manuals, documentation of procedures and processes used or intended to be used in connection with the Work, warranty records relating to the Work, reports relating to the Work, copies of design, user and maintenance documentation and bug lists on any medium and other related documentation, whether in paper, electronic or other form;

- 47) “Technical Requirements” means all of the technical requirements hereunder that the Supplier is required to meet in the course of performing the Work, more specifically being all of the requirements set out in Appendix G *Technical Requirements and Specifications*;
- 48) “Unit Rates” means the unit rates, if any, for individual Goods as set out in Appendix A *Price Schedule*, “Unit Price”.
- 49) “Value Added Taxes” means such sum as shall be levied upon the Contract Price by the Federal or any Provincial Government and is computed as a percentage of the Contract Price and includes the GST, QST, HST and any similar tax, the payment or collection of which is by the legislation imposing such tax an obligation of the Supplier;
- 50) “Warranty Period” means the period of time for which the Supplier shall warrant the Goods and the Work, being the period of time set out in Section 26.1; and
- 51) “Work” means the whole of the work, labour, acts, services, deliverables, and things required to be done or supplied by the Supplier as set out in the Purchase Order and the appendices, including, but not limited to, the supply of the Goods.

SECTION 2 LIQUIDATED DAMAGES

If specified in the Purchase Order, Liquidated Damages apply per the following:

- 2.1 The Supplier shall deliver the Goods in accordance with the Delivery Dates and the Software Schedule. Where the delivery of all or any portion of the Goods to the Contract Delivery Point is late other than due to Force Majeure, or all or any portion of the Goods delivered to the Purchaser becomes unavailable or unusable after the Delivery Date but prior to the end of the Warranty Period due to defective quality or due to damage attributable to the Supplier, the Supplier shall pay to the Purchaser as liquidated damages (hereinafter “Liquidated Damages”), and not as a penalty, the amounts as applicable as set out and calculated in accordance with Section 2.2. Such Liquidated Damages are agreed to reflect a reasonable and genuine pre-estimate of such damages and are without limitation and in addition to any other remedies the Purchaser may have under the Purchase Order, including where the cap on Liquidated Damages may have been reached pursuant to Section 2.4.
- 2.2 For the purposes of Section 2.1, Liquidated Damages shall accrue in favour of the Purchaser in an amount equal to the Daily Delay LD Amount for each complete day until the Purchaser has received complete delivery of such Goods or until the unavailability or un-usability has been rectified to the satisfaction of the Purchaser, as applicable. Liquidated Damages accrued from time to time shall be due and payable by the Supplier to the Purchaser within thirty (30) days after notice to the Supplier.
- 2.3 The Contract Price shall be reduced by the amount of the Liquidated Damages as such amounts accrue. The Purchaser may set-off such reduction against any or all obligations remaining to be paid by the Purchaser to the Supplier under the Purchase Order. The reduction of the Contract Price by the amount of Liquidated Damages in accordance with this Section 2.3 shall not relieve the Supplier from its obligations to provide the Goods and perform and complete the Work in accordance with the Purchase Order.



- 2.4 Under no circumstances shall the Supplier be liable for Liquidated Damages in excess of ten percent (10%) of the total Contract Price. Once such Liquidated Damages cap of ten percent (10%) has been reached, and the Good or Goods remain undelivered, unavailable or unusable, within the time frames required in Section 2.2, or any additional Goods are undelivered, unavailable or unusable, the Supplier shall be in breach of the Purchase Order, and the Purchaser may terminate the Purchase Order in accordance with Section 34.
- 2.5 Notwithstanding a reduction in Contract Price pursuant to Section 2.3 above, progress payments, if applicable, shall be calculated disregarding such reduction.

SECTION 3 TAXES

- 3.1 The Purchaser shall pay Value Added Taxes eligible on all amounts invoiced by the Supplier to the Purchaser in accordance with this Agreement. The Contract Price is inclusive of all Commodity Taxes.
- 3.2 The Supplier and the Subcontractors shall be responsible for and shall pay all Commodity Taxes incurred by them in performing the Work.
- 3.3 The Supplier shall not include in the calculation of, nor be entitled to receive from the Purchaser in respect of, the value of any Work not performed pursuant to this Agreement:
- 3.4 Any amount on account of Commodity Taxes that are or were recoverable by the Supplier or any Subcontractors by way of exemption or input tax credit, input tax refund or any other refund, rebate, remission or other tax recovery mechanism under Applicable Laws;
- 3.5 Any amount on account of the payment of any interest, penalties, administrative monetary penalties, or their equivalent as imposed by a governmental authority, fines or similar amounts incurred by Supplier or Subcontractors; or
- 3.6 Any RST on tangible personal property to be incorporated into the Plant.
- 3.7 The Supplier shall do all things necessary to assist the Purchaser to recover any Commodity Taxes the Purchaser may be eligible to recover, where applicable, including without limitation providing the Purchaser with completed *Revenue Canada Drawback Certificate of Importation, Sale or Transfer K32* forms.
- 3.8 The Contract Price is inclusive of all income/capital taxes or taxes on profits of the Supplier and the Subcontractors and any non-resident withholding taxes.

SECTION 4 CONTRACT BONDING OR LETTER OF CREDIT

- 4.1 If requested by the Purchaser, the Supplier shall, within fifteen (15) days after the Contract Effective Date, provide a supply bond or Letter of Credit to the Purchaser. The supply bond shall be in the sum of one hundred percent (100%) of the Contract Price plus the value of Free Issue Material, if any, specified by the Purchaser.
- 4.2 Should the Contract Price be increased through a Change Order the Supplier shall on request of the Purchaser, unless otherwise provided in the Change Order, increase the value and alter the term of the supply bond to the amended Contract Price and Delivery Date.

SECTION 5 TERMS OF PAYMENT

- 5.1 The Purchaser shall pay the Supplier in accordance with the payment schedule set out in Appendix B, *Payment Schedule*. The Purchaser shall pay all invoices within thirty (30) days after:
- 1) The Supplier has delivered the Goods (including but not limited to applicable Documentation) in accordance with the Purchase Order and the Purchaser has accepted same, such acceptance demonstrated by the Purchaser's quality assurance release form executed by both Parties, to the specified Contract Delivery Point;
 - 2) Receipt by the Purchaser of the Supplier's invoice supported by all required documentary evidence as set out by the Purchase Order.



- 5.2 The Purchaser shall have the right to pay directly and offset against payment to the Supplier, as required to discharge any liens, claims or demands related to the Work.

SECTION 6 MANAGEMENT, CORRESPONDENCE, SCHEDULE AND REPORTING CONTRACT REVIEW MEETINGS

6.1 CONTRACT REVIEW MEETINGS

- 6.1.1 The Purchaser, when required, shall arrange an Initial “Kick-off” meeting and subsequent contract review meetings in order to review the progress of the Supplier’s work and discuss any commercial or technical issues arising under this Contract. The Supplier shall attend such meetings and if applicable, cause its Subcontractors to attend same.

6.2 CORRESPONDENCE PROCEDURE

- 6.2.1 The Parties’ respective order administrators, technical representatives, and quality representatives are listed in Appendix E, *Correspondence Procedure*. This form must be updated by both Parties as soon as staff changes are made. Each Party shall ensure that all correspondence and Documentation is issued pursuant to Appendix E, *Correspondence Procedure*.
- 6.2.2 The Supplier shall promptly notify the Purchaser of any changes to its named representatives listed in the Correspondence Procedure.

6.3 MONTHLY PROGRESS REPORTS/SCHEDULES

- 6.3.1 If specified by the Purchase Order, the Supplier shall submit monthly progress reports as specified in Appendix H, *Procurement Specifications*. Such monthly progress reports shall be submitted within the first ten days of each calendar month and shall cover the previous month’s activities plus forecasted activities for the following month, or as otherwise requested by the Purchaser.
- 6.3.2 If requested by the Purchase Order, the Supplier shall submit a “Manufacturing Schedule”, of the detailed schedule for the Work which shall include but is not limited to the preparation, issue and acceptance of drawings, procedures and quality assurance manuals, developing and testing programs, supply and testing of tooling, calibration of fixtures and gauges, Subcontractor commitments and activities, material procurement and supply, production including assembly, testing and inspection, packaging and delivery.
- 6.3.3 The provision of monthly progress reports or manufacturing schedules does not relieve the Supplier from giving any notice required under the Purchase Order.

6.4 NON-CONFORMANCE REPORTS

- 6.4.1 The Supplier shall provide the Purchaser any non-conformances on components. Any non-conformances shall also include all Deviation Disposition Requests.
- 6.4.2 The Supplier shall notify the Purchaser immediately of any critical non-conformances, that is, those that may have a severe impact on quality or schedule, as well as provide within five (5) working days of such notice of the critical non-conformance, a report documenting the critical non-conformance along with planned corrective action.
- 6.4.3 In the event of occurrence of incidents affecting quality, such as fire and flooding, the Supplier shall report the situation to the Purchaser on the date of occurrence and report its effect on quality and planned corrective action within five (5) days.

SECTION 7 SUBCONTRACTORS

- 7.1 Subcontracting any portion of the Work shall not relieve the Supplier from its responsibilities under the Purchase Order, and the Supplier shall be responsible for the performance of the Subcontractors as if the Work had been done by the Supplier.



- 7.2 The Supplier may use the Subcontractor listed in Appendix J, *List of Subcontractors*. The Supplier shall request and obtain the Purchaser's prior written approval to any change of Subcontractors from those identified in Appendix J, *List of Subcontractors*. The Supplier shall provide all information the Purchaser, acting reasonably, may request with regards to such new Subcontractors.
- 7.3 The Supplier shall ensure that all of its subcontracts comply with all terms and conditions of the Purchase Order, including without limitation, the quality program standard and those terms ensuring the Purchaser and the Client have access as set out in Appendix C, *Quality Assurance Requirements*.
- 7.4 Nothing contained in this Contract shall create any contractual relation between any Subcontractor and the Purchaser.

SECTION 8 MEASURES

- 8.1 The Parties shall use SI units or British units with SI units in brackets in all documents generated during the implementation of the Purchase Order, except for size (diameter) of pipes, tubes, valves and specialized equipment in British units in drawings, flow sheets and related documents.
- 8.2 The Parties shall use SI units for instruments, instrument calibration, display screens, printed records generated during commissioning, operations, testing and maintenance, and set points of the Plant.

SECTION 9 DRAWINGS, SPECIFICATIONS AND PROCEDURES / DOCUMENTATION TO BE SUPPLIED BY PURCHASER

- 9.1 If required by the Purchase Order, the Purchaser shall furnish to the Supplier copies of all later revisions to the Technical Requirements as included in Appendix G, *Technical Requirements and Specifications*, except for specifications of nationally or internationally recognized standards associations, which the Supplier shall be required to obtain. The Purchaser shall issue such documents in electronic format, as part of the Purchase Order.
- 9.2 If any such revised Technical Requirements result in an increase or decrease in the Work or changes to the Manufacturing Schedule or Delivery Dates, the Supplier shall immediately advise the Purchaser.
- 9.3 Drawings and related specifications and procedures are intended to complement each other so that if anything is shown on the drawing but not mentioned in the specifications or procedures, or vice versa, it shall be of like effect as if shown or mentioned in all documents. If the Supplier notes any errors, omissions, contradictions or if the Supplier discovers any discrepancies in the figures, drawings, specifications or procedures or if the Supplier considers any feature of the drawings or specifications to be indefinite or unclear, the Supplier shall refer same to the Purchaser whose written confirmation or corrections shall be obtained before proceeding with the applicable portion of the Work. Scaled dimensions shall not be used.
- 9.4 All references in the Purchase Order to national or international standards and the Supplier's specifications shall imply the edition or revision thereof in effect at time the Contract Effective Date unless otherwise specified. All references in the Purchase Order to Purchaser-supplied documents shall be to the revision as identified herein unless otherwise specified.

SECTION 10 DOCUMENTATION SUBMITTED BY THE SUPPLIER AND ACCEPTANCE BY PURCHASER

If requested by the Purchase Order:

10.1 DOCUMENTATION TO BE SUBMITTED

The Supplier shall submit electronically to the Purchaser for acceptance all of the Documentation set out in Appendix F, *Software Schedule* in accordance with the dates specified.



10.2

ACCEPTANCE OF DOCUMENTS BY THE PURCHASER

The Supplier shall only use Documentation accepted by the Purchaser in carrying out the Work, and except for *Accepted as Noted* Documentation, which the Supplier may use provided the Supplier agrees to incorporate all of the Purchaser's comments and produces revised Documentation forthwith.

The Purchaser shall review all of the Documentation submitted by the Supplier to the Purchaser for the Purchaser's acceptance. Upon the Purchaser's review, the Purchaser will return to the Supplier an electronic copy of the Documentation under the following headings, which indicate the document status:

1) Accepted by Purchaser

If the Purchaser marks the Documentation with the following notation, "*Accepted at Revision X by [Candu Employee's Name] on [date]*" the Purchaser has accepted the applicable Documentation.

2) Accepted as Noted by Purchaser

If the Purchaser marks the Documentation with the following notation, "*Accepted as Noted at Revision X by [Candu Employee's Name] on [date]*" the Purchaser has conditionally accepted the applicable Documentation. Changes necessary for Acceptance will be noted on the applicable Documentation or on an identified attached sheet if there is insufficient space on same.

"Accepted as Noted" status signifies a working document, and Work by the Supplier may proceed.

The Supplier shall revise and resubmit the applicable Documentation for acceptance by the Purchaser.

3) Not Accepted by Purchaser

If the Purchaser marks the Documentation with the following notation, "*Not Accepted at Revision X by [Candu Employee's Name] on [date]*" the Purchaser has not accepted the applicable Documentation. The reasons for non-acceptance will be noted on the applicable Documentation or on an identified attached sheet if there is insufficient space on same.

The Supplier shall revise and resubmit the applicable Documentation for acceptance by the Purchaser.

The Supplier shall prepare a comprehensive summary of any revisions to any Documentation previously submitted to the Purchaser, which shall be entered in the revision block or sheet for all applicable Documentation. The Supplier shall not use such terms as "general revision" or "up-dated to suit" as these are not acceptable. Descriptions of dimensional changes must include the new and old dimensions.

10.3

The Supplier shall allow a reasonable time for the Purchaser's acceptance of Documentation in its consideration of schedule of Work and in the time allowed for completion of the Purchase Order. The Supplier shall allow a minimum of two (2) weeks in the Manufacturing Schedule for the Purchaser's initial in-depth review of any Documentation. The Purchaser shall provide its in-depth comments to the Supplier, and the Supplier shall incorporate all of the Purchaser's in-depth comments in its subsequent revision of such Documentation. The Supplier shall then allow a minimum of two (2) weeks in the Manufacturing Schedule for subsequent reviews of such Documentation by the Purchaser.

10.4

The Supplier shall be responsible for extra time and cost expended by the Purchaser for review and acceptance of Documentation due to the Supplier's deficiencies or errors. The Supplier shall recover any time lost so as to maintain the Manufacturing Schedule and the Delivery Date. In addition, at Purchaser's option, Purchaser shall charge Supplier and Supplier shall pay for Purchaser's reasonable costs arising from the Supplier's said deficiencies.

10.5

The Supplier shall not change any Documentation accepted by the Purchaser without the prior written consent of the Purchaser. Without derogation to the Section 19, should the Supplier be of the opinion that changes to the Documentation are required, it shall immediately notify the Purchaser in writing. The Supplier shall be responsible for meeting the Manufacturing Schedule and the Delivery Date notwithstanding the duration of time required by the Purchaser to approve or make changes to the Documentation.



- 10.6 Acceptance by the Purchaser of any of the Documentation shall in no way relieve the Supplier from its responsibility for any error or omission therein or from any obligation under the Purchase Order or in law. Acceptance shall not constitute approval of design details, calculations, analyses, tests, or construction methods or materials developed or selected by the Purchaser and shall not relieve the Supplier from full compliance with the requirements of the Purchase Order.
- 10.7 The Supplier is required to submit all Documentation on the Candu Formal Document Template and with a Candu Supplier Document Cover Page in accordance with the requirements set out in Appendix K – *Formal Document Template* and Appendix E – *Correspondence Procedure*.

SECTION 11 LICENSES, PERMITS, APPLICABLE LAWS, STANDARDS AND SAFETY

- 11.1 The Supplier shall be responsible for obtaining all licenses and permits it may require for performance and completion of the Work (except for those licences and permits expressly referred in the Purchase Order for which the Supplier is expressly not responsible) and for complying with all Applicable Laws.
- 11.2 Unless otherwise expressly provided, the Supplier shall ensure the Work being done shall conform to the specifications of accredited members of Canada's National Standards System and any applicable provincial safety and occupational health legislation, or to equivalent specifications, as accepted by the Purchaser, of the recognized standards association in the country in which the Work is performed.

SECTION 12 RIGHTS TO ACCESS BY PURCHASER

- 12.1 The Supplier shall ensure the Purchaser and the Client have unrestricted access to any facilities where the Work is being done at all reasonable times during manufacturing, inspection, and testing of the Work, including to the work of the Subcontractors and to the drawings and procedures involved.
- 12.2 The Supplier shall ensure that with respect to access to the drawings and procedures, it makes available to the Purchaser and the Client copies of all Documentation pertaining to the Purchase Order.
- 12.3 The Purchaser shall have the discretion to have its own employee(s) available to assist in the Work at the Supplier's facility.

SECTION 13 QUALITY ASSURANCE, SURVEILLANCE AND AUDITS

- 13.1 The Supplier shall ensure that no materials or parts be used or processed and no finished Work shall be submitted for acceptance or be delivered, unless and until approved pursuant to the Supplier's quality program. The Supplier shall keep proper and adequate inspection records, which shall at all times, be open to examination by the Purchaser and the Client.
- 13.2 The Supplier shall review all contractual and specification requirements and ensure it understands them. The Supplier shall ensure the manufacturing, testing, and inspection facilities for the Supplier and Subcontractors are adequate for the performance, inspection and testing of the Work.
- 13.3 The Work will be subject to expediting, quality surveillance acceptance and release by the Purchaser and its Client.
- 13.4 The Supplier shall:
- 1) Be responsible for all inspection and testing; inspection or testing by the Supplier shall not relieve the Supplier from responsibility for defects or other failure to meet the requirements of the Purchase Order;
 - 2) Furnish to the Purchaser all required information, including drawings, procedures and inspection results, which may be required for the proper surveillance of the Work;
 - 3) Make available the Supplier's personnel and furnish such calibrated gauges, instruments and other measuring and testing equipment as may be required for the expeditious surveillance of the Work;



- 4) Include as a provision of subcontracts an identical quality program standard and appropriate provision for surveillance, audit and expediting by its own, the Purchaser, the Client, and their duly authorized representatives;
 - 5) If the quality program standard in the supplier's subcontract is different from this Purchase Order, the supplier must document this deviation by submitting a Candu Deviation Disposition Request (DDR) for review and acceptance;
 - 6) Print on the face of all orders to Subcontractors the following notation: *"This agreement is subject to surveillance, expediting, and inspection by Candu Energy Inc., its Client, and their representatives"*;
 - 7) Inform the Purchaser in writing of any significant change to its quality assurance organization within thirty (30) days after such change; should the Purchaser advise the Supplier that the change be considered by the Purchaser to have deleterious effect on the Work, the Supplier shall take the necessary action to correct the unsatisfactory condition;
 - 8) Make all necessary arrangements to facilitate quality surveillance by the Purchaser (or Client), including such arrangements for the Purchaser (or the Client) to be present while applicable Work is being done and applicable tests and inspections are being performed;
 - 9) Where quality surveillance by the Purchaser (or Client) is required, ensure that no Work in progress or applicable materials shall be covered or otherwise obstructed from view, until inspected by the Purchaser;
 - 10) Advise the Purchaser in writing, in the form set out in Appendix L, *Witness Notification*, in advance of witness or hold points as specified in the inspection test plan as follows, unless otherwise specified in the Purchase Order:
 - a) If facility where testing is being done is located within a radius of 150 (one-hundred fifty) km from the Purchaser's head office - three (3) working days;
 - b) If facility where testing is being done is located more than 150 (one-hundred fifty) km from the Purchaser's head office - five (5) working days;
 - c) If Purchaser must reasonably take air transit to travel to facility where testing is being done – ten (10) working days;
 - d) The Purchaser shall not be held responsible for any work stoppage or delay if such prior written notice is not provided.
 - 11) Grant the Purchaser and the Client, or their duly authorized representatives, free access to its works and the facilities where the Work is being done at reasonable times for the purpose of quality surveillance, audit and ascertaining progress of the Work;
 - 12) Demonstrate to the satisfaction of the Purchaser that the Work meets the requirements of the Purchase Order;
 - 13) Hold inspection documents which fall into the category of non-permanent records for a period of a minimum of five (5) years after completion of the Purchase Order, or for two (2) years after the Reactor Unit In-Service Date, whichever is later; and
 - 14) Take the necessary corrective actions to resolve any non-conformances identified by the Purchaser or the Client, including without limitation those identified during reviews, surveillance and audits, in a timely and satisfactory manner.
- 13.5 The Purchaser shall have the right to conduct any additional audits, inspections and testing it may deem necessary during any stage of the Work. In the event such additional audits, inspection or testing:
- 1) demonstrate the Supplier's non-compliance with the Technical Requirements;
 - 2) result in re-work, repair or replacement of parts, as may be required;
 - 3) is the result of the Purchaser being required to re-perform any audits, inspections or testing due to the Supplier's unpreparedness for same; or
 - 4) is necessary due to the corrective actions required under the Purchase Order;



in which case the Supplier shall bear all of its costs and be responsible for all delays associated with the performance of such additional audits, inspections or testing. In addition, at Purchaser's option, Purchaser shall charge Supplier and Supplier shall pay for, Purchaser's reasonable costs for such additional audits, inspections and testing.

- 13.6 The Supplier shall not deliver any Goods until the Purchaser has been advised that the Goods are ready for inspection and the Purchaser has issued the final written quality assurance release.
- 13.7 The Purchaser's waiver of surveillance, or acceptance, or release of the Work shall not relieve the Supplier from the responsibility of performing the Work in accordance with the Purchase Order.
- 13.8 In the event that the Supplier must perform Work at the Site or the Purchaser's facilities, the Supplier shall perform all such Work in accordance with all Site rules and regulations, the Purchaser's quality assurance and quality surveillance programs and requirements, all quality and warranty requirements of the Purchase Order.
- 13.9 The Supplier shall submit to the Purchaser unpriced copies of its purchase orders or subcontracts placed with its Subcontractors for major items of the Goods, materials, equipment and services as required by the Purchaser.
- 13.10 The Supplier shall submit a certificate of compliance to the Purchaser, at the Purchaser's request, confirming its compliance with all Technical Requirements (the "Certificate of Compliance") and the requirements under Section 14.
- 13.11 In addition to the requirements of this Section 13 the Supplier shall also perform the Work in accordance with the requirements set out under Appendix C, *Quality Requirements*.

SECTION 14 NON-DELIVERY OF COUNTERFEIT, FRAUDULENT, AND SUSPECT ITEMS

- 14.1 The Supplier shall (and shall cause its Subcontractors to) not use or deliver counterfeit, fraudulent and/or suspect items ("CFSI") while providing work to Candu. If any items under this agreement are described using a manufacturer part number, a product description and/or specified using an industry standard, the Supplier will ensure that the items supplied under this Purchase Order meet all requirements of the latest version of the applicable manufacturer data sheet, description and/or industry standard. If the Supplier is not the manufacturer of the items, the Supplier will ensure that the items supplied under this Purchase Order or used to manufacture the equipment under this Purchase Order are made by the original equipment manufacturer in compliance with the applicable manufacturer data sheet and/or industry standard. If quality requirements identify nuclear industry quality assurance standards, such as N286, Z299, N299, the Supplier will (and, if applicable, will cause each of its Subcontractors to) include in its quality assurance program a documented process to prevent, detect and disposition any CFSI. If the Supplier desires to supply or use an item that may not meet the requirements of section 17.0, the Supplier will notify Candu of any exceptions and receive Candu written approval prior to shipment or use of such item. If CFSI are furnished under this Purchase Order or are found in any of the work provided under this Purchase Order, such items will be dispositioned by Candu and may be returned to the Supplier. The Supplier will promptly replace such CFSI with items acceptable to Candu and the Supplier will be liable for all costs, including Candu's internal and external costs relating to the removal and replacement of such items. The Supplier will include a documented process to prevent, detect and dispose of CFSI into its quality assurance program. The Supplier shall supply the Purchaser with a certificate to this effect confirming same as part of its Certificate of Compliance, if required pursuant to Section 13.10.

SECTION 15 EQUIPMENT TESTING

- 15.1 The Supplier shall not supply any Goods that have undergone destructive testing, including without limitation, seismic testing, fire retardation testing or other similar tests that would render the Goods unfit for service, or reduce the service life of the Goods.



SECTION 16 IDENTIFICATION

16.1 The Supplier shall identify the Goods in accordance with its identification procedure, as accepted by the Purchaser prior to identification of any Goods by the Contactor. The Supplier's identification procedure shall conform to the Purchaser's minimum requirements for identification as set out below in this Section 16 as well as any additional requirements stipulated in the Technical Requirements.

16.2 The Supplier shall identify all Goods, as applicable, as set out below:

16.3 Material Code Number

The Supplier shall identify all Goods by the Purchaser's material code number. The material code number shall be permanently inscribed, stamped, or attached to the body of the Good. In the case of registered equipment, the material code number may be inscribed on the registration label.

Where it is not possible to inscribe the material code number on the Good, the Supplier shall securely fasten a corrosion resistant tag, on which the material code number is stamped, to such Good with corrosion resistant wire.

16.4 Loose Parts

For all Goods shipped loose, including without limitation, installation fasteners, and parts of a Good, the Supplier shall, when possible, attach such parts or loose Goods to the main Good and be suitably identify such loose Goods or parts of a Good. The Supplier shall tag constituent parts of a Good which cannot be attached to the main Good with the appropriate material code number and identified as "part of" the main Good. The Supplier shall identify all Goods manufactured to CSA N285 in accordance with any requirements set out in CSA N285, Clause 12.3.

16.5 The Supplier shall ensure traceability for all Goods, as set out below:

1) Metallic Materials:

- a) For metallic materials, low stress die stamps shall be used. The metal thickness on which the impression is made must be 20.0 mm or greater. In addition:
- b) For stainless steel:
Electro-chemical or laser etching may be used and performed to a written procedure. The fluids used must be certified to contain less than 100 micrograms/ grams of total halogens, lead or other low melting point metals and sulphur. The process must result in marking which can be demonstrated to be legible and durable.
- c) For carbon steel:
For parts too thin to withstand stamping, electro-chemical etching or vibro-engraving may be used. Vibro engraving tools must have a carbide tip with a radius of approximately 0.12 mm and be adjusted to leave an impression approximately 0.25 mm deep maximum. Such markings must be covered by a preservative acceptable to the Purchaser if preservation for storage is required.
- d) For all other metals, as shall be agreed by the Parties or as set out in the Supplier's identification procedure.

2) Fasteners:

For fasteners 25.00 mm (1 inch diameter) size or less, marking on the package is acceptable.

3) Thermo, Thermal or Electric Arc Marking:

Thermo, thermal or electric arc marking are not acceptable.

16.6 NAMEPLATE

The Supplier shall attach a nameplate to each Good, where appropriate. The Supplier shall ensure that its nameplate attachment method does not impair the integrity of the Good. The Supplier shall ensure that the nameplate is manufactured from a corrosion resistant material.

16.7 BULK MATERIALS



In the absence of an agreed method of identification in the Purchase Order for bulk material such as pipe, tube, fittings, the Supplier shall obtain the prior written approval of the Purchaser on the Supplier's method for identification. Should there be any overage or underage of material, the Supplier is obligated to provide the Purchaser with the exact details to assess if a contract change is required to note the variance.

SECTION 17 SHIPMENT

- 17.1 The Manufacturing Schedule shall contain the shipping schedule for the Goods. At any time after Contract Effective Date or after the issuance of a Change Order, and up to ten (10) days prior to dispatch of any Goods, the Purchaser may review the shipping arrangements proposed by the Supplier. Should the Purchaser so decide, it may instruct the Supplier to use an alternative carrier and/ or routing and the Contract Price and Delivery Date will be revised to reflect the change in freight costs and delivery schedule, if any.
- 17.2 The Supplier shall identify, prepare, pack and mark for shipment, and deliver the Goods in accordance with the requirements set out in Appendix H, *Procurement Specifications*.
- 17.3 The Supplier shall obtain all export permits, licences, approvals, and similar documents from the country of export, as required, for shipment to or within Canada or to Site, as applicable.
- 17.4 When the Purchaser approves partial shipments, the Supplier must submit with each shipment all relevant documentation such as, but not limited to, history dockets or history files.
- 17.5 Within two days (2) days prior to shipping Goods, the Supplier shall issue an electronic copy of the quality assurance release, executed by both Parties, and a packing list to the Purchaser.

SECTION 18 FREE ISSUE MATERIAL

- 18.1 If applicable to the Purchase Order, the Purchaser shall provide the Supplier with the Free Issue Materials identified in the Purchase Order.
- 18.2 The Supplier shall, in respect of the Free Issue Materials which it receives:
 - 1) Examine the shipment prior to unloading and accepting material for any readily apparent damage or shortage; wherever such damage or shortage is apparent, the Supplier shall forthwith advise the Purchaser and not sign carrier pro bills on the Purchaser's behalf without specific instructions; where it is impractical to contact the Purchaser prior to unloading, the Supplier's acceptance shall be provisional with notation of any damage or shortage and the Supplier shall advise the Purchaser at the earliest opportunity;
 - 2) Inspect for damage or shortage in transit, off-loading, and be responsible for its storage; the Supplier shall establish and maintain a control system acceptable to the Purchaser designed to prevent damage to, loss of and deterioration of the Free Issue Material as well as unreasonable wastage of Free Issue Material while in its possession;
 - 3) Be responsible for maintenance of the Free Issue Material and, if applicable, receiving and safe-keeping of any history docket or history file for inclusion in the history docket or history file of the Goods;
 - 4) Advise the Purchaser in writing, with a copy to the party supplying the Free Issue Material, after receipt as soon as possible but not later than five (5) working days after receipt of any Free Issue Materials; the notification shall be in the form of a material received report including information such as quantity, concise material description and Purchaser's material code number, date received, recipient's signature and all available attachments, i.e., copies of packing slips, carrier's pro bill, bill of lading and package or crate inspection report;
 - 5) Report requirements for replacement material to the Purchaser, who shall have the option of supplying the material or instructing the Supplier to obtain the material from sources approved by the Purchaser, the whole at no cost to the Supplier, provided the Supplier was not responsible for the loss;
 - 6) Replace, at no charge to the Purchaser, material lost, damaged, allowed to deteriorate, or used for a purpose outside of the terms of the Purchase Order; the Supplier shall be responsible for all costs



associated with the replacement including quality assurance, quality surveillance and expediting by the Purchaser; and

- 7) Maintain and ship to the same Contract Delivery Point any free issue spare parts that are related and had been included with the other Free Issue Material.
- 18.3 The Purchaser shall advise the Supplier in writing of the replacement cost of such Free Issue Material.
- 18.4 The Supplier shall submit to the Purchaser, within thirty (30) days following the completion of the Purchase Order, a list of surplus Free Issue Materials. The list shall give a brief description and the Purchaser's material code number, sizes, quantity, and indicate whether the material is new or rejected or in the process of manufacture. The Purchaser will instruct the Supplier on the method of disposal, return or delivery of any surplus Free Issue Materials. The Supplier shall dispose, return or deliver such surplus Free Issue Material at no cost to the Purchaser.

SECTION 19 DEVIATION DISPOSITION REQUESTS

- 19.1 A Deviation Disposition Request ("DDR") is required where the Supplier wishes to deviate from the Technical Requirements. For greater certainty, where the Supplier deviates from anything set out in the Documentation that is not a Technical Requirement, a DDR is not required.
- 19.2 The Purchaser has full discretion to approve or reject any or all DDRs. If consideration of or acceptance of a DDR results in any cost to the Purchaser or the Client, this cost shall be chargeable to the Supplier at the Purchaser's option. Should the acceptance of a DDR require the Purchaser to carry out analyses or investigations or to initiate other changes to accommodate the deviation, the cost of such additional changes shall also be chargeable to the Supplier at the Purchaser's option.
- 19.3 A DDR may result in a Contract Change, at the Purchaser's option, where the DDR results in significant costs to either Party. Such Contract Change shall be in accordance with Section 20.
- 19.4 For Goods where a jurisdictional authority is involved, it is the Supplier's responsibility to ensure that all deviations and non-conformances are acceptable to that authority, including making any submissions required under the applicable code.

SECTION 20 CONTRACT CHANGES

- 20.1 The Purchaser may at any time up to the end of the Warranty Period, by written instruction to the Supplier ("Change Notice"), advise the Supplier of changes to the Work the Purchaser desires. The Purchaser may set out in the Change Notice, without limitation, changes to the Technical Requirements, Contract Price, the Work, Manufacturing Schedule and Delivery Date. For the purposes of this Section, a design change instruction or an email request for change issued by the Purchaser to the Supplier is deemed a Change Notice.
- 20.2 At the Purchaser's request, the Supplier shall review the Change Notice and provide to the Purchaser its written notice as to whether it is in agreement of the terms set out in the Change Notice. If the Supplier does not agree with the terms set out in the Change Notice and wishes to submit alternative terms to the Purchaser, it shall do so within fourteen (14) calendar days after the Purchaser issues the Change Notice, in the format prescribed by the Purchaser, specifying any cost increases or decreases, and providing detailed price breakdowns for engineering, material, labour, packing and shipping. If the Supplier submits alternate terms, the Supplier and Purchaser shall enter into good faith negotiations to agree on the terms of the Change Notice.
- 20.3 If the Purchaser and Supplier reach agreement on the terms of the Change Notice, the Purchaser shall prepare a Change Order to record such agreement for the execution by both Parties. Once executed, this document shall become a change order to the Purchase Order ("Change Order").
- 20.4 If the Purchaser and the Supplier cannot reach agreement on the terms of the Change Notice the matter shall be submitted to arbitration in accordance with the following:
 - 1) Such arbitration shall be subject to the *Arbitration Act* of the Province of Ontario;



- 2) There shall be one arbitrator;
 - 3) Such arbitration shall be conducted at Toronto, Ontario;
 - 4) The decision of the arbitrator shall be final and binding upon both Parties; and
 - 5) The arbitration expenses shall be borne by the losing Party unless otherwise awarded by the arbitrator.
- 20.5 The Supplier shall not perform a change in the Work without a Change Order.

SECTION 21 FAULTY OR DEFECTIVE WORK

- 21.1 If as a result of inspection, testing or observation at any time during the progress of the Work, the Purchaser advises the Supplier in writing that the Supplier has executed unsound or imperfect Work or has supplied any materials inferior in quantity and quality to those specified or generally, in any way has failed to conform to requirements of the Purchase Order, then the Supplier shall at no charge to the Purchaser, as soon as practicable and no later than five (5) days after receipt of such advice, initiate corrective action to ensure that the Work complies with the requirements of the Purchase Order, including without limitation completion of such corrective action so as not to delay the Delivery Dates.
- 21.2 In the event that the Supplier fails to take corrective action, the Purchaser shall have the right to employ and pay others to do so, or, at its option, re-execute that part of the Work itself and all expenses consequent thereon or incidental thereto shall be recoverable from the Supplier by the Purchaser or may be deducted by the Purchaser from any monies due or which may become due to the Supplier. Failure by the Purchaser to make such deduction shall not release the Supplier from the obligation to pay to the Purchaser the cost of the re-executed Work.
- 21.3 The provisions of this Section shall be in addition to and not in substitution for other provisions and guarantees under the Purchase Order.

SECTION 22 CONTRACTING DIRECTLY WITH CLIENT

- 22.1 The Supplier shall not communicate with the Client or enter into a contract with the Client for the supply of any materials, commodities, articles, services, equipment, goods, things, consumables or parts thereof, related to the scope of the Purchase Order, including spares, until the expiration of the Warranty Period.

SECTION 23 SUPPLIER'S GENERAL LIABILITY

- 23.1 The Supplier shall indemnify, defend and hold harmless the Purchaser and the Purchaser's affiliates, directors, officers, agents, servants and employees (collectively, the "Purchaser Indemnified Parties") from and against all Losses including all Losses arising from Supplier's legal liability to third parties asserted against or suffered by the Purchaser Indemnified Party as a result of, or arising out of:
- 1) A breach of the Purchase Order, including without limitation a breach of any representation or warranty of the Supplier; or
 - 2) The negligence or wilful misconduct of any one or more of the Supplier, its subcontractors, suppliers, and their respective employees, agents, or invitees (the "Supplier Parties") in connection with the Supplier's performance, purported performance, or non-performance of the Purchase Order, or the presence of the Supplier Parties at the site.



- 23.2 Where the Supplier supplies any materials, commodities, articles, services, equipment, goods, things, spares, consumables or parts thereof, directly to the Client other than through the Purchase Order for inclusion in the Plant without requesting and obtaining the prior written approval of the Purchaser, the Supplier shall indemnify and hold the Purchaser and all of its directors, officers, agents, servants and employees harmless from any claim, loss, damage, liability or expense on account of damage to the Reactor Unit, the Plant, the Site, the property of the Purchaser and of third parties, and injury (including death) to any person to the extent that such claim, loss, damage, liability or expense arises out of or results from such supply by the Supplier.
- 23.3 The total cumulative liability of the Supplier to the Purchaser Indemnified Parties for any Losses of any kind, whether based on tort, negligence, contract, warranty, strict liability or otherwise, arising from or relating to the Purchase Order, including without limitation any liquidated damages provided under the Purchase Order, shall not exceed one hundred percent (100%) of the Contract Price. Notwithstanding the foregoing, the limit on the total cumulative liability of the Supplier to the Purchaser Indemnified Parties shall not apply to any Losses that are in relation to a breach by the Supplier of Section 28 and 23.2, in relation to the negligence of the Supplier Parties, or in relation to any claim, action or proceeding asserted against a Purchaser Indemnified Party by any third party.
- 23.4 The liability disclaimers and liability limits set forth in the Purchase Order shall not preclude the Purchaser Indemnified Parties from claiming under any available insurance coverage placed by a Party in accordance with the Purchase Order, and the liability disclaimers and liability limits are in addition to any available insurance proceeds under such insurance coverage.
- 23.5 Notwithstanding any other provision of the Purchase Order, neither Party shall be liable for any claims for special, consequential, indirect or incidental damages whatsoever (including without limitation loss of profits, loss of anticipated profits, loss of revenue) whether based on tort, negligence, contract, strict liability or otherwise arising from or relating to the Purchase Order, including, without limitation, the performance, purported performance, or non-performance of the Purchase Order; provided however that this Section 23.5 shall not apply to any liquidated damages provided for under the Purchase Order nor to any claim, action or proceeding asserted against a Purchaser Indemnified Party by any third party
- 23.6 In the event that either Party has any claims or disputes against the other Party, the Party with the claim or dispute must notify the Party in writing within thirty (30) working days after the event that gives rise to such claim or dispute or the Party with the claim or dispute is hereby deemed to waive its rights to raise such claim or dispute against the other Party.

SECTION 24 INSURANCE

- 24.1 The Supplier shall provide and maintain in full force and effect from the Contract Effective Date to the end of the Warranty Period including extensions thereto, unless otherwise specified, the following insurance coverage in respect of the Work or the Purchase Order:
- 1) All-Risk Insurance covering the Work, including but not limited to, raw materials, stock and supplies, work in process, finished goods, Free Issue Material, if any, and equipment and tools, owned, rented or leased. Coverage shall be based on prevailing all risk replacement cost terms and conditions including extension of coverage for electrical and mechanical breakdown and shall provide indemnification for direct physical loss or damage to the Work. Coverage shall remain in force until custody and risk of loss to the Goods has transferred from the Supplier to the Purchaser in accordance with Section 26.



SECTION 25 TITLE TO WORK

- 25.1 The Supplier hereby irrevocably grants to the Purchaser all right, title, and interest in and to any Goods, materials, equipment, and other deliverables incorporated as part of the Work, other than Technical Information, immediately as such Work is performed. For greater certainty, legal and beneficial title to such Goods, materials, equipment, and other deliverables delivered, but not incorporated as part of the Work, shall pass to the Purchaser as the Work (including work in progress) is performed, procured or fabricated. The Supplier agrees to promptly execute and deliver to the Purchaser, from time to time as the Purchaser may require, any further documentation required by the Purchaser to identify, evidence, perfect or protect the Purchaser's legal, beneficial, and registered interest in such Goods, materials, equipment, and other deliverables including, without limitation, any registrations pursuant to the applicable personal property security legislation.

SECTION 26 WARRANTY

- 26.1 The Supplier warrants the Work to be in compliance with the requirements of the Purchase Order and free from deficiency or defects in design, material and workmanship, and shall be valid for eighteen (18) months from the date of delivery or twelve (12) months from the date of installation at site, whichever is earlier.
- 26.2 If during the Warranty Period the Work or any part thereof is discovered to be broken or deficient, defective or fails due to faulty or improper design, material, workmanship, manufacture, fabrication, shipment or delivery, or fails to meet the requirements of the Purchase Order, then the Supplier, upon notification and instruction in writing from the Purchaser, shall forthwith make good every such breakage, deficiency, defect or failure without any cost to the Purchaser, such costs including, without limitation, all transportation costs, travel costs to and from Site, all labour costs and living costs at Site, import or export duties or costs, all costs associated with working in a radioactive environment, dismantling existing equipment to gain access to the Goods to be repaired or replaced, and any other direct or indirect costs incurred therewith.
- The Purchaser or the Client shall provide the Supplier access to the Site and the Unit in order for the Supplier to perform such replacement or repairs, but the Purchaser shall be required to dismantle the other Goods or other equipment to provide access for the Supplier to perform such replacement or repairs and to reassemble such Goods or other equipment to as-found condition upon completion of such replacement or repair, all at the Supplier's cost.
- 26.3 If after such notification, the Supplier delays or defaults in diligently commencing, continuing and completing the making good of such breakage, deficiency, defect or failure in a manner satisfactory to the Purchaser, or if, in the opinion of the Purchaser reasonably exercised, the Supplier is not qualified to effect the repair or replacement, in whole or in part, then the Purchaser may proceed to do so and to place the Work in good condition in accordance with the Purchase Order. In such event, the Supplier shall be liable for all costs, charges and expenses incurred by the Purchaser in connection therewith, including but not limited to investigation costs and temporary repairs, and shall forthwith pay the Purchaser an amount equal to such costs, charges and expenses upon receipt of the Purchaser's invoices.
- 26.4 In the event that emergency repair of Goods or spare parts or replacement of spare parts are required during the Warranty Period, the Purchaser, at its own discretion, may proceed to conduct the repair or replacement of the Goods or spare parts. The Supplier shall be liable for all costs, charges and expenses incurred in connection therewith and shall forthwith pay the same to the Purchaser upon receipt of invoices therefore.
- 26.5 Any part of the Work made good under this Section shall be subject to all the provisions of this Section for the balance remaining of the Warranty Period or for a further period of twelve (12) months from the date when the same has been made good as aforesaid, whichever expires later.
- 26.6 The Purchaser is hereby permitted to assign any of the above noted warranties to the Client at any time.



26.7

The Supplier shall immediately notify the Purchaser in writing upon obtaining information reasonably indicating that any component, which it supplied, or any activity which it performed pursuant to the Purchase Order, contains a defect. In such notice, the Supplier shall provide:

- 1) All pertinent details about such defect;
- 2) Impact on safety due to such defect; and
- 3) Details of recommended corrective actions.

SECTION 27 RIGHTS TO INFORMATION, DISCLOSURE AND INFRINGEMENT

27.1

Any information provided by the Purchaser to the Supplier pursuant to the Purchase Order (collectively called the “Purchaser’s Proprietary Information”) shall remain the property of the Purchaser or the third party owner of such Purchaser’s Proprietary Information. Purchaser’s Proprietary Information may not be published, communicated or otherwise disclosed by the Supplier to any third party except a third party which requires the disclosure for the performance of Work and agrees to use the Purchaser’s Proprietary Information only in the performance of the Work.

27.2

All intellectual property rights, including the copyright, in any drawings, documents and other materials furnished to the Supplier:

- 1) Which are owned by the Purchaser shall remain vested in the Purchaser and,
- 2) Which are owned by any third party including suppliers of materials, shall remain vested in such third party.

27.3

Any Technical Information provided by the Supplier to the Purchaser pursuant to the Purchase Order shall be deemed to be works for hire and that sole and exclusive ownership of all right, including the copyright, title and interest shall vest in and transfer to Purchaser and become property of the Purchaser. Such Technical Information may not be published, communicated or otherwise disclosed by the Supplier to any third party, except for the purposes of the Work and installing, commissioning, licensing, operating, maintaining, improving, refurbishing and decommissioning the Goods, Reactor Unit, Plant and Site (collectively called the “**Purpose**”). The Supplier shall not assert any right or establish any claim to Technical Information and/or inventions under any patent, copyright or other law.

27.4

All intellectual property rights, including the copyright, in information which is referenced in, forms part of, or is reasonably required to practice or make the use of all Technical Information furnished to the Purchaser, that was not obtained or conceived or developed or first actually reduced to practice in the performance of the Work (“Background Information”):

- 1) Which is owned by the Supplier shall remain vested in the Supplier and
- 2) Which are owned by any third party including suppliers of materials, shall remain vested in such third party.

The Purchaser shall have a non-exclusive, irrevocable, fully paid up license to copy, modify, adapt, and use all pre-existing Background Information, and documents and materials which incorporate such Background Information, for the Purpose and to sublicense such rights in the pre-existing Background Information to the Client for the Purpose.

27.5

Notwithstanding Section 27.3 above, the Purchaser shall have the right to reverse engineer the Background Information, or a part thereof, use directly or disclose to third parties where the Purchaser has requested the Supplier’s assistance and the Supplier cannot or will not provide timely or reasonable assistance or provide such assistance on reasonable commercial terms, in the sole opinion of the Purchaser.



27.6

The Supplier shall indemnify and hold harmless the Purchaser and the Client from and shall defend at its own expense, including defence costs, and shall cooperate with respect to, any suit or action brought against the Purchaser or the Client based on a claim that any Good, Work or Technical Information furnished hereunder, or any part thereof, or the use of same in accordance with the terms of this Agreement constitutes infringement of any intellectual property right of any third party. The Supplier shall also pay all costs and damages awarded therein against the Purchaser or the Client, subject to the Supplier having agreed in writing to any out-of-court settlement. In case such Good, Work or Technical Information, or any part thereof, is in such suit held to constitute infringement and its use is enjoined, the Supplier shall, at its option and own expense either: procure for the Purchaser and the Client the right to continue to use such Good, Work or Technical Information, or part thereof; or replace it with substantially equivalent non-infringing work; or modify it so it becomes non-infringing. The Purchaser shall provide reasonable notice of the assertion of any claim of infringement to the Supplier and shall make available all other reasonably required information, assistance, and authority it possesses for the defence of any suit or proceeding in which such claim is asserted.

27.7

The Purchaser shall indemnify and hold harmless the Supplier from and shall defend at its own expense, including defence costs, and shall cooperate with respect to, any suit or action brought against the Supplier based on a claim that any Purchaser's Proprietary Information furnished hereunder or any part thereof, or the use of same in accordance with the terms of the Purchase Order constitutes infringement of any intellectual property right of any third party. The Purchaser shall also pay all costs and damages awarded therein against the Supplier, subject to the Purchaser having agreed in writing to any out-of-court settlement. In case such Purchaser's Proprietary Information, or any part thereof, is in such suit held to constitute infringement and its use is enjoined, the Purchaser shall, at its option and own expense either: procure for the Supplier the right to continue to use such Purchaser's Proprietary Information or part thereof; or replace it with substantially equivalent non-infringing information; or modify it so it becomes non-infringing. The Supplier shall provide notice as soon as possible of the assertion of any claim of infringement to the Purchaser and shall make available all other reasonably required information, assistance, and authority it possesses for the defence of any suit or proceeding in which such claim is asserted.

27.8

Neither Party shall be liable to the other Party under this Section 27 for disclosure of Technical Information or Purchaser's Proprietary Information, as applicable, received, where,

- 1) Such information is or becomes part of the public domain through no breach of the Purchase Order;
- 2) Such information is subsequently lawfully obtained by the first party ("First Party") from a third party who is established as a lawful source of the information, without breach of the Purchase Order by the First Party;
- 3) The information was known by the First Party prior to its disclosure by the other party ("Other Party") and such prior knowledge can be established;
- 4) The Other Party has provided its prior written consent to such disclosure; or
- 5) Disclosure was made pursuant to legislation, regulatory requirements or a court order.

SECTION 28 CONSENTS

28.1 The Supplier shall obtain all required consents necessary in order to satisfy its obligation to deliver all of the Goods, including without limitation Documentation, hereunder.

SECTION 29 RELEASE OF INFORMATION

29.1 The Supplier shall not hold or cause or allow to be held any public ceremony or publicity release in connection with the Work, or erect or permit the erection of any sign or advertising referring to the Work without the written permission of the Purchaser.

29.2 The Supplier shall ensure that each subcontractor complies with the provisions of Section 29.



SECTION 30 FORCE MAJEURE

- 30.1 Neither Party shall be deemed to be in breach of the Purchase Order where its failure to perform or its delay in performing any obligation, other than an obligation to pay monies due, is due to a cause beyond its reasonable control ("Force Majeure") including, but not limited to, an act of God, an act of any national, civil, or military authority, governmental action not caused by the Supplier or by the Purchaser, civil commotion, war (declared or undeclared), fires, floods, sabotage, earthquake, storm or epidemic.
- 30.2 For greater certainty, strikes, lockouts and other labour disputes of the Supplier or any Subcontractor shall not entitle the Supplier to claim Force Majeure.
- 30.3 Each Party shall notify the other in writing, with minimum delay and in no case later than five (5) days after the beginning of the occurrence of an event of Force Majeure, of any failure to perform or delay in performing due to Force Majeure and shall provide an estimate as soon as possible of the time when the obligation will be performed. The time for performing the obligation shall be extended for a mutually agreed period of time necessary to overcome or minimize the impact of the Force Majeure. The Parties shall notify each other in writing immediately at the termination of such Force Majeure.
- 30.4 Accompanying the written notification of the occurrence referred to in Section 30.3 above, the Party shall submit with the first notification, confirmation from competent authorities certifying to the reality of the circumstances, facts and dates contained therein, including news articles and bulletins relevant to the Force Majeure.
- 30.5 Both Parties shall exercise due diligence in time and effort and take all practical measures in order to minimize the effects of the Force Majeure and at the earliest possible date restore normal conditions and re-establish working schedules as soon as the impediments in Section 30.1 have ceased, and to agree on an equitable change to the Manufacturing Schedule and the Delivery Date.

SECTION 31 NON-WAIVER

- 31.1 The Purchaser's failure to insist on performance of any term, condition and provision hereof, or failure or delay in exercising any rights or remedies provided herein or by law, or to properly notify the Supplier in the event of breach, or the acceptance of or payment for any Work hereunder, or review of design, shall not release the Supplier from any of the warranties or obligations of the Purchase Order and shall not be deemed a waiver of any right of the Purchaser to insist upon strict performance hereof, or a waiver of any of the Purchaser's rights or remedies as to any Work or Goods regardless when shipped, received or accepted, or as to any prior or subsequent default hereunder. Any termination of the Purchase Order by the Purchaser shall not operate as a waiver of any of the terms hereof.

SECTION 32 SUSPENSION OF WORK

- 32.1 The Purchaser may for any reason suspend the Work, in part or in whole, at any time, upon written notice to the Supplier. The suspension notice shall specify the date of suspension, the scope of the Work suspended, and the duration, if known, of the suspension.
- 32.2 The Supplier shall forthwith suspend performance of the Work in accordance with the Purchaser's instructions and shall suspend all activities pertaining to the suspended Work. If the Supplier considers it necessary to perform certain Work after the date of the suspension, the Supplier shall so notify the Purchaser. The Supplier shall not perform any such Work without the prior written consent of the Purchaser.



32.3

The Supplier shall make every reasonable effort to minimize all costs resulting from the suspension. Save to the extent any such suspension arises from any act or omission of the Supplier, the Purchaser shall reimburse the Supplier for all Work performed and for all reasonable and unavoidable costs incurred by the Supplier as a result of such suspension. For any suspension, the resumption and completion of the Work shall be as scheduled by the Purchaser. Any adjustments to the Manufacturing Schedule, the Delivery Dates and Contract Price shall be made through a Change Order.

SECTION 33 TERMINATION FOR CAUSE

33.1

The Purchaser may, upon written notice to the Supplier, terminate the Purchase Order if the Supplier is in default in carrying out any provisions of the Purchase Order, or upon any of the following events:

- 1) The commencement of any application, petition, suit or other proceeding under any bankruptcy, arrangement, reorganization, dissolution, liquidation, insolvency, winding-up or similar law of any jurisdiction now or hereinafter in effect, for the relief from or otherwise affecting creditors of the Supplier, including without limitation, under the *Bankruptcy and Insolvency Act* (including the filing of a notice of intention to make a proposal), *Companies Creditors Arrangement Act* and *Winding-up and Restructuring Act*, by, against or in respect of the Supplier or all or any part of its assets;
- 2) Any receiver, receiver-manager, interim receiving monitor, liquidator, assignee, custodian, trustee, and others similarly appointed in respect of the Supplier or all or any part of the assets;
- 3) The Supplier (i) makes a general assignment for the benefit of the creditors, including without limitation, any assignment made pursuant to the *Bankruptcy and Insolvency Act*, (ii) acknowledges its insolvency or is declared or becomes bankrupt or insolvent, (iii) fails to meet its liabilities as they generally become due, or (iv) commits an act of bankruptcy under the *Bankruptcy and Insolvency Act* or any similar law of any jurisdiction.
- 4) Any holder of any security interest, mortgage, lien, charge, claim or encumbrance of any kind enforces against delivers any notices of intention to enforce security pursuant to Section 244 of the *Bankruptcy and Insolvency Act*, or becomes entitled to enforce against or otherwise takes possession, management or control of all or any part of the Supplier's property, assets or undertaking or all or any part of the Supplier's interest therein,
- 5) A distress, execution, warrant, garnishment, attachment, sequestration, levy, writ or any similar process is used or enforced upon or against all or any parts of the Supplier's property, assets, or undertaking, or any third party demand is issued by the court or any Governmental Authority, administrative body or any taxation authority in respect of the Supplier or all or any part of its property, assets or undertaking or any other seizure is made in respect of all or any part of the Supplier's property, assets, or undertaking.

33.2

The Supplier shall give the Purchaser written notice immediately upon the occurrence of any one of the events set out in Section 33.1.

33.3

Once the Purchaser provides written notice to the Supplier in accordance with Section 33.1, the Supplier shall cease work immediately, and the Supplier shall have no claim for any further payment including payments already invoiced in accordance with Section 5 herein except as otherwise provided in this Section, but shall remain liable to the Purchaser by reason of the default unless such default is caused by Force Majeure as determined in accordance with Section 30.

33.4

The Purchaser may require the Supplier to deliver any Work, work-in-process, Goods, parts or other material which the Supplier has specifically acquired or produced in accordance with the Purchase Order and after the Work has been completed by or on behalf of the Purchaser, the Purchaser shall credit to the Supplier, subject to the deduction of any claim against the Supplier, the value of all such Work, work-in-process, Goods, parts or other material as determined in accordance with the Contract Price.



- 33.5 Any costs incurred by the Purchaser to complete the Work as a result of the Supplier's default of the Purchase Order including legal and professional fees on a full indemnity basis shall be reimbursed to the Purchaser by the Supplier. The Purchaser shall set-off any such costs from any monies owing to the Supplier, and shall invoice the Supplier for any balance, which the Supplier shall pay forthwith on receiving such invoice.
- 33.6 No payment shall be made in respect of termination under the Purchase Order for Work, which is not acceptable to the Purchaser after inspection.
- 33.7 The total payment made to the Supplier pursuant to this Section taken together with any other payment under the Purchase Order, shall not exceed the Contract Price applicable to the Work or any particular part of the Work.
- 33.8 The Supplier shall, as far as is practicable, place subcontracts on terms that will enable the Supplier to terminate in an identical manner as set forth in this Section.

SECTION 34 TERMINATION FOR CONVENIENCE

- 34.1 The Purchaser may upon written notice to the Supplier terminate the Purchase Order at any time and the Supplier shall immediately cease Work as specified in such notice.
- 34.2 The Supplier shall make commercially reasonable efforts to minimize all costs resulting from the termination of Work. In respect of Work partially completed before the giving of such written notice, the Supplier shall be entitled to be reimbursed for all the direct reasonable and unavoidable actual costs of labour and materials to the Supplier of such partially completed Work, and to receive in addition an amount representing profit in respect of such direct reasonable and unavoidable actual costs of labour and materials incurred. If the Parties cannot agree on the appropriate amount of profit, the matter shall be submitted to arbitration in accordance with Section 20.
- 34.3 The total payment to the Supplier with respect to this Section 34 taken together with any other payment made under the Purchase Order shall not exceed the Contract Price.
- 34.4 The Supplier has the responsibility of substantiating its claims for direct reasonable and unavoidable actual costs of labour and materials in a manner acceptable to the Purchaser.
- 34.5 The Supplier shall, as far as is practicable, place subcontracts on terms that will enable the Supplier to terminate in an identical manner as set forth in this Section.
- 34.6 For greater certainty, unless specifically provided for in this Section 34, the Supplier shall not be entitled to prospective profits, overhead, incidental, direct, indirect or consequential damages, due to such cancellation.

SECTION 35 CONFLICT OF INTEREST

- 35.1 The Supplier shall forthwith disclose in writing to the Purchaser, in reasonable detail, the nature and extent of any real or apparent conflict of interest which exists as of the date of the Purchase Order or which may subsequently arise. All such disclosures, and any uncertainties in this matter, shall be directed to the Purchaser for further clarification and further instruction. In the event of any real or apparent conflict of interest, the Purchaser may, in its sole discretion, terminate the Work or direct the Supplier to take such actions as may prevent or eliminate such conflict, including, without limitation, requiring divestiture of assets or interests constituting conflict.



SECTION 36 GOVERNING LAW

- 36.1 The Purchase Order and the rights and duties, responsibilities and obligations of the Parties shall be governed by, interpreted and construed in accordance with the laws of the Province of Ontario and the federal laws of Canada applicable in the Province of Ontario. Each Party irrevocably attorns to and submits to the exclusive jurisdiction of the Courts of Ontario and all other courts competent to hear appeals therefrom with respect to any matter arising or relating to the Purchase Order.

SECTION 37 SETTLEMENT OF DISPUTES

- 37.1 Except as otherwise provided in the Purchase Order, any dispute concerning a question of fact arising under the Purchase Order, which is not disposed of by agreement, shall be decided by the Purchaser and the Purchaser shall forthwith provide written notice to the Supplier of its decision. The decision shall be final and conclusive unless, within thirty (30) days from the date of receipt of the notice or decision, the Supplier makes a written appeal to Purchaser. In the event that a question of fact cannot be disposed of by agreement between Purchaser and the Supplier after such written appeal by the Supplier, then the question of fact shall be finally settled by dispute resolution as described below.
- 37.2 Any other disputes, disagreements, questions or breach beside a question of fact which shall arise as to the obligation of any Party under the Purchase Order or the interpretation of any provision thereof, or otherwise arise out of or in connection with the Purchase Order, if not settled by mutual agreement, shall, at the option of the initiating Party and upon written notice to the other Party, be finally determined in the courts of Ontario.
- 37.3 From the date of any dispute, disagreement or differences arising, to the date of final determination of the matter in question, Purchaser's interpretation shall prevail; and the Supplier shall comply with Purchaser's interpretation and shall continue to fulfil its obligations under the Purchase Order in good faith during the pendency of any dispute and shall not be entitled to traverse the fulfilment of its contractual obligations.
- 37.4 The foregoing is without prejudice to the rights of either Party to seek injunctive or other interim relief in a court of competent jurisdiction in the event of any actual, threatened, or reasonably anticipated breach of any provision of the Purchase Order where a Party may suffer irreparable harm for which monetary damages are not an adequate remedy.

SECTION 38 SUCCESSORS AND ASSIGNS

- 38.1 The Purchase Order shall enure to the benefit of and be binding upon the Parties hereto and their executors, administrators, successors and assigns. The Supplier shall not assign the Purchase Order without the prior written consent of the Purchaser, which consent may not be unreasonably withheld.

All expenses incurred by the Purchaser for such assignment, including preparation of the necessary documents, shall be billed to the Supplier.

- 38.2 The Purchaser shall have the right to assign the Purchase Order to the Client, as may be required by the Project Contract.

SECTION 39 COUNTERPARTS

- 39.1 The Purchase Order may be signed by email and in counterparts and each of such counterparts will constitute an original document and such counterparts, taken together, will constitute one and the same instrument.

**SECTION 40 ENTIRE AGREEMENT**

- 40.1 The entire agreement of the Parties with respect to the subject matter herein consists solely of the Purchase Order. Except as expressly incorporated herein, any and all previous or collateral agreements, negotiations, proposals, promises, representations and conditions relating to the subject matter of the Purchase Order are superseded by the Purchase Order.
- 40.2 The Purchase Order shall take effect as of the Contract Effective Date and shall remain in effect until fulfillment of obligations under the Purchase Order.



SECTIONS 10 DOCUMENTATION SUBMITTED BY THE SUPPLIER AND ACCEPTANCE BY
PURCHASER

Section 10.3 is replaced in its entirety with:

10.3 The Supplier shall allow a reasonable time for the Purchaser's acceptance of Documentation in its consideration of schedule of Work and in the time allowed for completion of the Purchase Order. The Supplier shall allow a maximum of two (2) weeks in the Manufacturing Schedule for the Purchaser's initial in-depth review of any Documentation. The Purchaser shall provide its in-depth comments to the Supplier, and the Supplier shall incorporate all the Purchaser's in-depth comments in its subsequent revision of such Documentation. The Supplier shall allow a maximum of two (2) weeks in the Manufacturing Schedule for the Purchaser and one 1 week for the Client (SNN) in-depth review of Inspection and Test Plans (ITP), and History Files as applicable. The Supplier shall then allow for the same duration in the initial in the Manufacturing Schedule for subsequent reviews of such Documentation by the Client and/or Purchaser.

SECTION 13 QUALITY ASSURANCE, SURVEILLANCE AND AUDITS

Section 13.10 is amended to add the following:

13.10 e) When the ITP has Witness (W) or Hold (H) point for SNN, ISCIR or CNCAN the Contractor shall notify the Purchaser at least thirty (30) days before the approximate start date of the Witness (W) or Hold (H) operations, to allow the Purchaser to coordinate with the Client to make the necessary travel arrangements.

f) When the ITP has Witness (W) or Hold (H) point for SNN, ISCIR or CNCAN the Contractor shall notify the Purchaser at least ten (10) days before the exact date on which the Witness (W) or Hold (H) is reached, to allow the Purchaser to coordinate with the Client to make the necessary travel arrangements to attend the surveillance operations.

SECTIONS 26 – WARRANTY

Section 26.1 is replaced in its entirety with:

The Supplier warrants the Work to be in compliance with the requirements of the Purchase Order and free from deficiency or defects in design, material and workmanship, and shall be valid for twenty-four (24) months from the date of delivery or twelve (12) months from the date of installation at site, whichever is earlier.





Supplemental Packaging Requirements for Tooling

PROCESS

1. Packaging

The intent is to place Packages inside standard Sea Containers for long distance travel; heavy and oversized packages are exempt from this requirement.

All Goods shall be packed, braced and loaded in such a manner as to prevent physical damage and inadvertent damage during transit and storage of the Goods, as well as degradation from marine and adverse climatic conditions.

When deemed Goods shall be shipped as complete systems and ready for installation or erection, as possible. Should the size of the Goods prevent it from being shipped fully assembled, the Supplier shall ensure the Goods are separated for convenient transportation. If disassembly is required, prepare the tool in modules as large as possible to allow for efficient re-assembly at Client's Project site. All items that are separated or disassembled from the main tooling or equipment shall be noted on the packing list as being separated.

In all instances, the Packaging shall provide means of handling by forklift, slings and other conventional means of lifting and moving, including pushing and pulling, as are normally available at transport terminals, dock facilities and the Client's Project site.

All items shall be identified by the proper Tool ID (or other equivalent catalog format) and all crates shall be identified as necessary.

1.1 Types of Packaging

In accordance with the nature of the contents, the following types of Packaging shall be considered. The structural characteristics will be summarized below.

Fibreboard, cardboard or other similar cartons shall not be used as outside Packaging. Inner or outer Packaging shall not be returned, unless otherwise specified in the Purchase Order.

1.1.1 Wooden Case

Wooden cases must allow for handling by forklift truck; therefore, they must be provided with battens allowing forklift tines to pass at right angles beneath all sides and provide a minimum floor clearance of 10 cm (~3.5"). Base boards must not be covered with tarred paper or polythene to allow for ventilation and water drain from the wooden case to avoid moisture from getting in through the bottom of the case. The base structure must be such as to equally distribute the weight of the apparatus.

Cases weighing greater than 50 kg shall be fitted with base skids, allowing a minimum floor clearance of 10 cm. The base skids shall be parallel to and extend the full width of the box. The distance between skids shall not exceed 120 cm.

Cases shall fully comply with ISPM (International Standard for Phytosanitary Measures) #15 and have clearly marked provisions for lifting by crane.

Cases are subdivided into three categories:

**Cases up to 500 kg**

Carrying built crossbars	Where the contents permit, cases can be used without a base slat, these shall be replaced by carrying crossbars to permit movement with a forklift truck and shall also serve as a batten for lifting ropes.
Planking	Planking must be 2 to 2.3 cm thick.
Base	Base planks must be 3 cm thick.
Bands	Each wooden case must be bound with at least two bands.
Cover	The cover must be lined internally with polyethylene sheet (top) and a sheet of tarred paper (bottom).

Cases up to 2,000 kg

Base Blocks	Cases must be provided with battens on the bottom parallel to the longitudinal axis. The section and the space between battens must be proportionate to the structural characteristics, dimensions and weight of the product. The ends of the slats must be bevelled at 45° to allow for the positioning of ropes.
Slats	Slats 8 cm thick should be placed crossways beneath the beams, raising the Package for lifting ropes or to allow movement with the forklift truck. The position and number of slats shall be determined in accordance with the length of the case and the relative centre of gravity (C of G). The slats must be about twice as wide as they are thick and may be fixed by screw, nails or staples.
Base	The base boarding must be 4 cm thick; nailed, bolted or screwed to base crossbars.
Planks	Planks must be laid vertically to achieve improved resistance to superimposed weights.
Stiffening struts	Stiffening struts must be placed immediately beneath the cover. Since their objective is to protect the case against the pressure imposed by the lifting ropes, the dimension of the struts must be proportionate to the weight of the Package and its size.
Cover	Horizontal bracing may be external or internal. The dimensions and spacing shall be proportionate to the dimensions and weight of the case. The cover must be lined internally with polyethylene sheet (top) and a sheet of tarred paper (bottom)



Cases up to 20,000 kg

They are similar to the previous category (cases up to 2,000 kg). Obviously, they must be sturdier.

Planking	Planking must be 3 cm thick.
Bottom Boards	Closed tight and of the following dimensions: • 12 x 12 cm for weights up to 6,000 kg • 15 x 15 cm for weights up to 9,000 kg • 20 x 20 cm for weights up to 20,000 kg
Base	The base boarding must be at least 5 cm thick; bolted or screwed to the base beams.
Cover	Horizontal bracing may be external or internal. Section 20 x 4 cm (80 cm max spacing), the cover of these cases must be supported by a lattice frame (section approximately 15 x 3 cm). The squares or rectangles in the lattice may not exceed 40 to 50 cm at the side, to obtain a sufficiently resistant cover. The cover must be lined internally with polyethylene sheet (top) and a sheet of tarred paper (bottom)

1.1.2 Wooden Crates

All wood and/or lumber used in Packaging, pallets or for securing within containers must be treated and/or fumigated so that they meet with the entry requirements of the country of export and/or destination, pursuant to ISPM #15 guidelines. These materials must be heat treated or fumigated using methyl bromide. The treatment or process of treatment must be officially endorsed by the NPPO (National Plant Protection Organization) of the exporting country.

The treated Packaging materials must be marked with a universally recognized symbol, or another approved by CFIA (Canadian Food Inspection Agency) or other internationally recognized organization, prior to releasing any shipments, to demonstrate they meet the requirements of the receipt country. Recycled, remanufactured or repaired wood Packaging materials should be re-certified and re-marked. All components of such material should be treated.

Plywood made from veneer which has been rotary cut, sliced or sawed, must be well seasoned, commercially dried (10 to 20% moisture content) and free from defects such as loose knots (no larger than 3 cm in size), decay, mould or insect infestation that would materially lessen the strength. Plywood must be at least 3 ply for boxes when cleated and at least 5 ply for boxes when nailed. Boxes exceeding 17 kg must have four vertical corner posts or other equally suitable devices or fasteners.

1.1.3 Metal Industrial Packaging Container

Where applicable, Metal Industrial Packaging containers may be used to pack Goods for delivery.

Type A containers can be used as Metal Industrial Packaging, only when not meeting the defining regulation criteria under the CNSC (Canadian Nuclear Safety Commission) or IAEA (International Atomic Energy Agency) regulations.



1.1.4 Metal and PVC Barrels

Metal or PVC barrels with a capacity of less than 100 L shall be packaged in crates or arranged on pallets protected with a heat-shrunk cover and cross hoops.

Metal or PVC barrels with a capacity of or greater than 100 L shall be sent individually.

Metal strapping shall be applied with stretching tool and secured with crimped steel seals. Size of strapping used should follow latest practice used by manufacturers depending on weight of package being strapped.

1.1.5 Wooden Drums (Wooden Reels)

Drums should be made of robust construction to provide effective protection during transport and handling. Wooden reels must be complete, in good condition, and have been recently painted with waterproof paint.

Drums must be closed by staves nailed to the flanges with strengthening boards, 8 x 8 cm, placed laterally below the stave at 50 cm.

For drums over 1.5 m in diameter, these reinforcing boards must be replaced by stays bolted to the flanges.

A sheet of tarred paper and polystyrene sheet must be placed beneath the staves.

Drums must have at least 2 complete turns of strapping.

1.1.6 Equipment on Saddles or Skids

Openings in equipment shall be closed by metal sheets or wooden covers and painted a different colour. Sheets and covers shall be firmly attached to equipment or material, not taped but preferably bolted.

All saddle and skid locations shall be indicated by the Supplier. Surfaces areas which may contact slings during lifting are to be covered with wooden hurdles.

The Goods shall be fitted with suitable lifting lugs or be properly marked as to where the slings are placed. The C of G shall be indicated by the Supplier.

1.1.7 Loose and Bundled Items

Only pipes, steel structures and plates shall be packed loose and in bundles.

Each length of pipe of 150 mm or larger diameters shall be treated as an individual Package and marked accordingly. Pipe smaller than 150 mm shall be bundled by strapping cleats or cradles of 50 x 100 mm lumber above and below the load. Ends of the pipe must be capped if not contained in crates.

The Package marking requirements shall be detailed in Section 3.

1.1.8 Fragile Items

Fragile items shall be wrapped in crepe cellulose wadding or some equally efficient cushioning material and/or floated in excelsior. Fragile articles must be packed in substantial wooden cases with special precautions to protect against the risk of breakage.



1.2 General Packaging Practices

1.2.1 Skids

Packages having a gross weight exceeding 50 kg shall be fitted with skids allowing a minimum floor clearance for forklift tines of 10 cm. The skids shall be parallel to and extend the full width of the box. The distance between skids shall not exceed 120 cm.

1.2.2 Load Bearing Base

Packages having a gross weight exceeding 200 kg should have an adequate load-bearing base.

1.2.3 Blocking, Bracing and Anchoring

Goods shall be blocked, braced and/or anchored with material that is non-abrasive to prevent movement of the contents in any direction.

1.2.4 Cushioning

Cushioning shall be used on fragile items where protection from shock and vibration is required. Items should be wrapped in crepe-cellulose or some equally efficient material. Alternatively, blister pack material may be used. Peanut packing shells and Styrofoam popcorn are not acceptable.

Care should be taken to ensure that the cushion material selected shall have no corrosive effect on the Goods and the cushion material should not readily support combustion.

1.2.5 Barriers

All Goods shall be packaged with a sealed barrier to eliminate dust, dirt, moisture and other forms of contamination from penetrating the packaging.

1.2.6 Nesting

In all cases, the Goods shall be nested or packed to reduce the volume as much as possible but keeping the parts module to facilitate re-assembly.

Packaging Teams who do not have adequate facilities or experience are strongly encouraged to engage the services of another capable company that specializes in packaging and preservation.

1.2.7 Batteries

Equipment with internal batteries, power-batteries or accumulator batteries shall be segregated and shipped separately. Depending on quantities, battery type and mode of transportation; batteries can be considered hazardous and/or dangerous goods.

Batteries must be packaged and shipped in compliance with the applicable rules and regulations (i.e. IMDG (International Maritime Dangerous Goods) or IATA (International Air Transport Association)). Batteries must be packed in suitable containers, packaged appropriately and display the proper safety marks, labels, product information, and any other information which is required by the regulations governing the mode of transportation.

1.3 Long Term Storage

Supplier is to be made aware if the Goods are to be prepared for long term storage. When in doubt about storage requirements; confirm storage details with the Project.



-Proprietary-

Crates are to be stored in a climate-controlled storage facility where temperature and humidity are monitored and regulated by a heating and cooling system. Wooden crates are to be stored off the ground, approximately 3 to 3.5 inches.

Proper packaging and preventative measures must be taken to eliminate the risk of rodent infiltration and the perforation of the vacuum packaging.

1.4 Heavy and Oversized Items

Goods exceeding one of the following dimensions and/or weight criteria are to be considered Heavy and/or Oversized cargoes:

Length:	11.5 meters
Width:	2.5 meters
Height:	2.5 meters
Weight:	15 Tons

Suitable lifting lugs, spreader bars or other special devices shall be provided, as required, to allow lifting by facilities that are commonly daily available.

Special cradles for support of heavy equipment that is required for support on land and sea carriers shall be provided and shall travel with the equipment to the ultimate destination.

As a minimum, equipment exceeding 4000 kg (four metric tons) shall be adequately equipped with suitable skids, pulling eyes, jacking bosses, etc., so that inland transportation of the equipment by means of rolling-on/rolling-off and pushing/pulling methods may be performed safely.

2. Preservation In cases where preservation instructions are defined in OMs (Operating and Maintenance Manuals) for specific Engineered Tools and conflict with instructions provided in this procedure, the more stringent instructions shall be followed. Supplier is responsible for providing the specific preservation requirements.

All Goods shall be protected and preserved in such a manner to prevent damage from the entry of foreign objects (such as dust, sand, water, rust, etc.) and environmental conditions (such as precipitation, marine atmosphere, extreme temperatures, etc.). Preservation shall not impede packaging or subsequent handling activities that might threaten the integrity of the applied preservatives.

2.1 General Preservation Recommendations

2.1.1 Handling Goods

The following process shall be used for handling Goods before or during packaging:

- All ferrous metal surfaces shall remain free of contaminants such as fingerprints and dirt.
- Latex and/or rubber gloves shall be worn when handling metal components to eliminate the risk of corrosion.
- Metal surfaces from the Engineered Tools and Tool parts shall not be in direct contact with material from a wooden crate or case. Methyl bromine and other corrosive chemicals are used in the treatment of wood and these chemicals can cause corrosion when in contact.

AVOID DIRECT CONTACT BETWEEN METAL AND WOOD **AT ALL COSTS**



2.1.2 Enclosing Goods

This process to enclose the Goods shall be followed prior to packaging:

1. Where required, metallic components and surfaces susceptible to rust damage must be coated with an anti-corrosion coating, corrosion inhibitor or equivalent product.
2. Components shall be tightly wrapped with VAPPRO 826 VCI 6MIL Film, a poly-wrap or equivalent product.
3. Allow a few extra inches of film on each side. Fold over the excess film a few times and staple the extra film together.
4. Place a piece of quality adhesive tape over the stapled areas to provide better sealing properties.
5. Heat seal the edges and excess film together as it provides better sealing properties.

See any special instructions or contractual requirements (i.e., Contract or Purchase Order) for further recommended guidelines for enclosing the Goods to the minimum mentioned above.

2.1.3 Preservation for Transit and Storage

For the duration of transit and storage not exceeding six months, the following practices for preservation are recommended using the following products (or their equivalent):

- VAPPRO V-5 & V-10 VCI Foam Emitters
 - Placed in all enclosed areas of equipment
- VAPPRO 9868 Rust Inhibitor
 - Spray on all bare metal parts and surfaces
- VAPPRO 826 VCI 6MIL Film
 - Fully enclose all equipment, fold over extra film, staple excess film together and heat seal all edges and stapled areas.
- MIL-PRF-131K
 - Wrap and hermetically seal all equipment

For the duration of transit and storage not exceeding two years, the following practices for preservation are recommended using the following products (or their equivalent):

- VAPPRO V-5 & V-10 VCI Foam Emitters
 - Place in all enclosed areas of equipment
- VAPPRO VCI Pouches
 - Place inside VAPPRO 826 VCI 6MIL Film, next to equipment
- VAPPRO 9868 Rust Inhibitor
 - Spray on all bare metal parts and surfaces
- VAPPRO 870CB Electro Spray
 - Spray all electrical and electronic components
- VAPPRO 826 VCI 6MIL Film
 - Fully enclose all equipment, fold over extra film, staple excess film together and heat seal all edges and stapled areas.
- MIL-PRF-131K



- Wrap and hermetically seal all equipment

The moisture vapour-proof barrier shroud (MIL-B-131, Type 1, Class 1) is a silver polypropylene and aluminium foil bag that is used to encapsulate the crated Goods. This material is used in conjunction with humidity absorbing desiccant packs within the bag. The barrier bags are then heat sealed and vacuum packed. The vacuum bag is used with a Humidity Indicator Plug that gauges moisture build up within the bag. These materials must be used where transit and storage conditions are long term and/or severe.

Other effective preservation materials are Cortec products. These preservation materials are used in the prevention of rust, corrosion or oxidation during transit and storage. Recommended Cortec products (or their equivalent) are:

- VCI 105 Foam Emitters (for electronic cabinets)
- VCI 126 Blue or Transparent Film with VpCI (for multi-metal protection)
- VCI 132 VpCI Impregnated Foam Pads (for multi-metal protection, non-toxic, nitrite-free)
- Cor-Pak VpCI Pipe Strips (for protection of ferrous or non-ferrous tubular products)

2.2 Machinery

For the protection of machinery, heat-insulated bags must be used, and dehydrating agents (desiccant packs) must be placed inside the bags to maintain appropriate humidity levels. All auxiliary threaded connections must be hermetically sealed by means of threaded metal plugs, greased and bound with quality adhesive tape. All critical working surfaces such as close tolerance shafts, bearing seals and spindles shall be protected against the entrance of dirt by wrapping with a greaseproof, waterproof paper having a neutral pH factor. This wrapping shall be secured firmly in place with pressure sensitive tape or wax coated twine.

Machine finished surfaces susceptible to corrosion or deterioration shall be coated with a soft drying corrosion inhibitor compound. Extra caution shall be exercised when applying corrosion inhibitor compounds to ensure that items fabricated of leather, mica, rubber and similar materials are not inadvertently coated.

VPI, VCI or other vapour barrier papers must be used where critical moving components of machinery are subject to rusting due to condensation (moisture). These uses are limited by the various metallic content of certain equipment. When transporting or storing in tropical climates, all Goods shall be vacuum packed to ensure components are not damaged by the elements.

2.3 Electrical Goods

All electrical and instrumentation Goods should be packed in wooden cases, wrapped in an interior moisture vapour-proof bag in conjunction with a humidity absorbing desiccant pack placed inside the bag. A vacuum must be created inside the bag to protect against the elements during shipping, handling and storage. To prevent damage or puncture to the protective barrier bag, any sharp edges or corners on the equipment must be padded.

Precision instruments must be wrapped in anti-corrosive waxed paper, padded and enclosed in protective bags in which humidity absorbing desiccant packs have been placed.

Electrical components which are static sensitive must be enclosed in protective packaging (anti-static wrap, static protective bags, anti-static tape or equivalent anti-static material) and placed in appropriate outer packaging with cushioning material to prevent physical or static electricity damage.

All openings of electric motors, generators and other electrical equipment shall be sealed with pressure sensitive, waterproof tape. Brushes and armatures of motors and generators shall also be wrapped with a



-Proprietary-

piece of Grade A greaseproof paper. All openings in Goods such as vessels, valves and pumps shall be properly sealed.

In cases of vulnerable parts of equipment that cannot be treated with protective substances, self-adhesive waxed paper impregnated with passivity substances can be used.

2.4 Long Term Storage

The following practice is recommended for Goods that will require long term storage (i.e., up to two years). Vapour barriers shall be used for all Goods which are designated for long term storage. Metallic components and surfaces susceptible to corrosion/rust damage shall be coated with an anti-corrosion coating, corrosion inhibitor or equivalent.

Desiccant combination packs shall be used. Depending on the Humidity Indicator Plug ("Dew Point Factor"), and storage and transit environments, desiccant packs placed inside crates and containers must be replaced every six to 12 months to avoid potential water leakage from the desiccant packs into the bottom of the Package enclosure.

See any special instructions or contractual requirements (i.e., Contract or Purchase Order) for further recommended guidelines for long storage to the minimum mentioned above.

3. General Marking Requirements

All packages, boxes, cases, crates, bundles and unpacked Goods shall be clearly and indelibly marked to show necessary shipping information as specified hereunder. Markings shall be applied by means of waterproof paint or ink. Markings shall be applied by a method suitable to the type of product and the packaging involved. Markings shall be stencilled on at least two sides other than the side on which it rests or is intended to be stacked (i.e., markings on opposite sides and the top).

All Packages shall bear the following minimum markings:

- ☐ Shipment No. / Shipping Reference No.
- ☐ Customer Name & Destination
- ☐ Project Name & Purchase Order No.
- ☐ Package No. & Total Number of Packages
- ☐ Package Identification per the Applicable-Identifiers
- ☐ Name of the Goods
- ☐ Gross Weight & Net Weight (units: kg)
- ☐ Overall Dimensions, length x width x height (units: cm)
- ☐ General Handling Instructions, i.e., FRAGILE, HANDLE WITH CARE, KEEP DRY, THIS SIDE UP, CENTRE OF GRAVITY, STACKING LIMITATIONS, SLINGING POSITION, INDOOR STORAGE ONLY, etc. as required. These instructions shall be marked on all four sides
- ☐ Special Instructions, i.e., UNPACKING RESTRICTIONS, DANGEROUS GOODS, HAZARDOUS GOODS, etc. Applicable labels shall be in conformity with IMDG & IATA regulations
- ☐ Transport index, referenced in the IAEA Safety Requirement No. TS-R-1

See any special instructions or contractual requirements (i.e., Contract or Purchase Order) for further recommended guidelines for marking of the Goods to the minimum mentioned above.

All Packages weighing a metric ton or more shall have a marking indicating the maximum load that can be placed on top of them. All Packages weighing five tons or more must be marked with slinging points and clearly show the C of G on all sides.



-Proprietary-

Shipping markings must also appear on all packages and associated shipping documents such as packing lists, invoices, etc.

Where the marking cannot be applied directly to the Package itself, such as loose items, bundled pipe or an open crate; the marking shall be applied by at least two metal, indestructible plaques. Both ends of each plaque must be securely wired, nailed or strapped to the bundle, crate or loose item in such a manner that if one strap or one metal plaque is accidentally pulled off or broken the others will remain.

Hand lettering with felt tip markers are not acceptable.

Fabricated pieces of steel or pipe that are banded, bundled, tied together, etc. shall have each piece marked "XXX Project" where XXX is the designated Client's Project site.

Any precautionary statements shall be stencilled on each Package to indicate proper handling and storage.

4. General Packing List Requirements

The Supplier shall not ship Goods and materials from the departed point of origin, regardless of the terms included in the Purchase Order, until the final release for shipment is issued.

Packing List

Prior to the Purchase Order shipping date, Supplier must submit a Packing List to include the following minimum information:

- ☐ Confirmation of Purchase Order Shipping Date
- ☐ Shipment No. / Shipping Reference No.
- ☐ Package No. & Total Number of Packages
- ☐ Description of all Packaged Goods & Corresponding Identifier
- ☐ Reference Tag No.
- ☐ Gross & Net Weight
- ☐ Overall Dimensions (length x width x height)

Details of the contents shall be listed on the Packing List, with item numbers listed accurately to allow proper identification. Whenever Packages are stuffed into containers, the Packing List shall indicate details of all items, piece by piece, accessory by accessory (Purchase Order line-item level). Indication of sets, lots or the like is not acceptable.

Never indicate monetary value on any packing lists.

One copy of the Packing List shall be placed inside the Package along with the Goods. Two copies of the Packing List shall be fixed to the outside of the Package in a weatherproof container (protected by a plastic envelope or aluminum cover), marked 'Packing List'.

Safety Information

Prior to the Purchase Order shipping date, Supplier must provide a list of safety information for all easily combustible Goods, dangerous Goods and hazardous Goods. The safety instructions must include the following minimum information:

- ☐ Name of the Goods
- ☐ Nature of the Goods
- ☐ Measure of Protection
- ☐ Emergency Treatment



- ☐ Material Safety Data Sheets
- ☐ IMCO (Intergovernmental Maritime Consultative Organization) Code
- ☐ UN Product Code

Prior to the Purchase Order shipping date, Supplier must furnish a description of the special precautions to be taken for the Goods with regards to any special requirements for temperature, shock, loading, etc., during the transportation and storage of Goods.

5. Standard Operating Procedure for Transportation

The following tasks are recommended for the packaging and preservation of the Goods prior to transportation:

- The Supplier is to determine which of the following types of containers are recommended to be used:
 - ☐ Cleated plywood boxes
 - ☐ Wooden boxes – lag screws shall be used for securing walls and lids to each other and to the base. Use of nails shall be limited to the base
 - ☐ Wooden crates - shall conform to entry requirements for wood packaging pursuant to ISPM-15 (International Standard for Phytosanitary Measures) guidelines & appropriate IPPC (International Plant Protection Convention) symbols to be marked on the wood package
 - ☐ Metal Industrial Packaging container
 - ☐ Specially designed containers for special equipment
- The following protection shall be used to avoid any damage to the Goods:
 - ☐ Adequate cushioning provided to protect the Goods from shock and vibration
 - ☐ Blocking, bracing and/or anchoring used to prevent movement of the contents in any direction
 - ☐ Metal strapping used to prevent the equipment from shifting
 - ☐ Corrosion inhibitor used, as required
 - ☐ Moisture vapour-proof barrier used, as required
 - ☐ Desiccant and humidity protection used, as required

Note: if desiccant is used, indications are made to the outside of the package of its presence
 - ☐ Anti-static protection used, as required
 - ☐ Put VAPPRO 5 or 10 VCI Foam Emitters (or equivalent approved) to the inside of all electrical and electronic cabinets
 - ☐ Apply VCI plastic barrier, such as VAPPRO 826 VCI FILM, VAPPRO PAKS or VAPPRO VCI POUCH
- Special instructions:
 - ☐ Boxes with weight exceeding 50 kg are fitted with a skid base designed to allow the package to be lifted by forklift and/or crane and pushed or pulled
 - ☐ 4 in. x 4 in. internal bracing is used for crates that are going to be shipped break bulk or contain loads exceeding 2,000 kg. This is to allow for stacking and slinging
 - ☐ Installation of humidity indicators, impact sensors, inclination sensors or other sensors, as reasonably required, to demonstrate proper handling of the package



-Proprietary-

- ☐ Goods are protected against climatic conditions for transporting via sea

See any special instructions or contractual requirements (i.e., Contract or Purchase Order) for further recommended guidelines for the packaging and preservation of the Goods to the minimum mentioned above.



APPENDIX F
Procedures Documentation Checklist (PDCL)

ATTACHED





Document Number		Revision
RT79-87000-PDCL-003		1
Nuclear Project Number	Contract Number	Page
699241	N/A	1
Customer Document Number	Customer Name	
	SOCIETATEA NATIONALA NUCLEARELECTRICA SA	
Title:		
MANUFACTURE OF ALL N299.4 TOOLS FOR CERNAVODA RETUBE		

Project: Cernavoda 1 Retube Tooling Project

699241	TBD		TENDERING	2024/06/07
699241	TBD	N/A	INFORMATION	2024/11/07
699241	TBD		FABRICATION	2025/02/12
FINANCIAL PROJECT NO	CONTRACT NO	CHANGE ORDER	DESCRIPTION	RELEASE DATE



Document Number		Revision
RT79-87000-PDCL-003		1
Nuclear Project Number	Contract Number	Page
699241	N/A	2
Customer Document Number	Customer Name	
	SOCIETATEA NATIONALA NUCLEARELECTRICA SA	
Title:		
MANUFACTURE OF ALL N299.4 TOOLS FOR CERNAVODA RETUBE		

Project: Cernavoda 1 Retube Tooling Project

Prepared by

Handwritten signature of Ahmad, Rusham.

Ahmad, Rusham - Assistant
Analyst I, Tooling

Reviewed by

Handwritten signature of Brian Magda.

Magda, Brian - Specialist Engineer

Handwritten signature of Akash Tank.

Tank, Akash - Quality Assurance
Specialist

Approved by

Handwritten signature of Christopher Janeczko.

Janeczko, Christopher - C1R
Tooling Engineering Manager



Procedures / Document Checklist

- Proprietary -

Page 1 of 14

CE-505210-FM-002 Rev. 7

Ref. Procedure CE-904026-PRO-210

File No.:

RT79-87000-PDCL-003

Revision No. 1

Equipment / Material Services

EQR Ref. #:	N/A	Supplier (to be filled out by buyer only):
iReq. Ref. #	N/A	
Scope of Work:	MANUFACTURE OF ALL N299.4 TOOLS FOR CERNAVODA RETUBE	

In addition to any other documentation considered during manufacturing, the Supplier should be responsible for submitting all required procedures/documents marked X below. Supplier's schedules and progress/status reports are intended for reporting progress and are not to be used to document changes to the Purchase Order.

For each applicable procedure/document, the Supplier shall enter the proposed submission date on this document. If accepted by the Purchaser, these dates shall determine the procedures/documents schedule and will be binding. At the time of submitting the Manufacturing schedule, the Supplier shall provide a detailed list of all procedures/documents planned to be submitted by the agreed dates. Revisions to this PDCL form following the awarding of the Purchase Order shall be formally documented for acceptance.

In case of conflict between this form and the History Docket/History File, the HD/HF requirements will prevail.

No.	Description	Document Required for Review and Acceptance by Purchaser	Document part of the HD/HF (Technical Specification 00-00660-TS-001)	Document part of documentation package as per client requirements	Comments	Submission Date Offered by the Supplier
Supplier's Schedules and Progress/Status Reports						
1.	Manufacturing Schedule (Bar Chart)	☒	☐	☐	For Information Only	
2.	Procedures/Documents Schedule	☐	☐	☐	For Information Only	
3.	Monthly Progress/Status Report	☒	☐	☐	For Information Only	
Suppliers/Sub-Suppliers Design Documents						
4.	Manufacturing Drawings	☐	☐	☐		
5.	Isometric Drawings	☐	☐	☐		
6.	Design Verification Plan	☐	☐	☐		
7.	G/A Drawing (cross section drawings)	☐	☐	☐		
8.	Proof of Design Registration	☐	☐	☐		

Procedures / Document Checklist**- Proprietary -**

Page 2 of 14

CE-505210-FM-002 Rev. 7

Ref. Procedure CE-904026-PRO-210

No.	Description	Document Required for Review and Acceptance by Purchaser	Document part of the HD/HF (Technical Specification 00-00660-TS-001)	Document part of documentation package as per client requirements	Comments	Submission Date Offered by the Supplier
9.	Specifications	☐	☐	☐		
10.	Bills of Materials/Parts Lists	☐	☐	☐		
11.	Mark ups / as build information on Candu supplied drawings and bills of materials	☐	☐	☒	If changes are required during manufacturing of the tool(s)	
12.	Third Party Certification of Below the Hook Lifting Devices to ASME B30.20	☐	☐	☒	If Applicable	
Suppliers/Sub-Suppliers Reports/Analyses/Tests						
13.	Design Report	☐	☐	☐		
14.	Seismic Test Report/Analysis	☐	☐	☐		
15.	Development Report/Analysis	☐	☐	☐		
16.	Performance Test reports	☐	☐	☐		
17.	E/Q Report	☐	☐	☐		
18.	Lubrication List	☐	☐	☐		
19.		☐	☐	☐		
20.		☐	☐	☐		
Manufacturing Procedures/Documents						
21.	Manufacturing Plan	☐	☐	☐		
22.	Forging Procedures(s)	☐	☐	☐		
23.	Heat Treatment Procedure(s)/Records	☒	☐	☒	If Applicable Exception: procedure(s) are not required to be submitted if accepted previously by another C1 order. Submit the procedure(s) for information and include the acceptance page of the previously accepted procedure(s) attached and referenced. Procedure(s) to be included in the documentation package.	

Procedures / Document Checklist

- Proprietary -

Page 3 of 14

CE-505210-FM-002 Rev. 7

Ref. Procedure CE-904026-PRO-210

No.	Description	Document Required for Review and Acceptance by Purchaser	Document part of the HD/HF (Technical Specification 00-00660-TS-001)	Document part of documentation package as per client requirements	Comments	Submission Date Offered by the Supplier
					If there have been revisions to the previously submitted procedures, then they are to be submitted for review and approval. All records per the procedure required to be maintained as per the Supplier QA program and its applicable procedure. These records are added to the documentation package requested with the items. Records to include: Item heat treated, treatment dates, heat treatment procedure identification and revision that has been accepted by Candu, Approval, signatures and date by Company's competent and qualified inspection personnel.	
24.	Stress Relieving Procedure(s)	☒	☐	☒	If Applicable Exception: Stress Relief procedure(s) are not required to be submitted if accepted previously by another C1 order. Submit the procedure(s) for information and include the acceptance page of the previously accepted procedure(s) attached and referenced. Procedure(s) to be included in the documentation package. If there have been revisions to the previously submitted procedures, then they are to be submitted for review and approval. All records per the procedure required to be maintained as per the Supplier's QA program and its applicable procedure. These	

Procedures / Document Checklist

- Proprietary -

Page 4 of 14

CE-505210-FM-002 Rev. 7

Ref. Procedure CE-904026-PRO-210

No.	Description	Document Required for Review and Acceptance by Purchaser	Document part of the HD/HF (Technical Specification 00-00660-TS-001)	Document part of documentation package as per client requirements	Comments	Submission Date Offered by the Supplier
					records are added to the document package requested with the items. Records to include: Item heat treated, treatment dates, heat treatment procedure identification and revision that has been accepted by Candu. Approval, signature and date by Company's competent and qualified inspection personnel.	
25.	Welding Procedure(s)	☒	☐	☒	If Applicable. Exception: Welding procedure(s) are not required to be submitted if accepted previously by another C1 order. Submit the procedure(s) for information and include the acceptance page of the previously accepted procedure(s) attached and referenced. Procedure(s) to be included in the documentation package. If there have been revisions to the previously submitted procedures, then they are to be submitted for review and approval. All records per the procedure required to be maintained as per the Supplier's QA program and its applicable procedure. These records are added to the document package requested with the items. Records to include: Item welded, welding treatment procedure identification and revision that has been accepted by Candu. Approval, signature and date by Company's competent and qualified inspection personnel.	
26.	Weld Repair Procedure(s)	☒	☐	☒	If Applicable.	

Procedures / Document Checklist

- Proprietary -

Page 5 of 14

CE-505210-FM-002 Rev. 7

Ref. Procedure CE-904026-PRO-210

No.	Description	Document Required for Review and Acceptance by Purchaser	Document part of the HD/HF (Technical Specification 00-00660-TS-001)	Document part of documentation package as per client requirements	Comments	Submission Date Offered by the Supplier
					Exception: Welding Repair procedure(s) are not required to be submitted if accepted previously by another C1 order. Submit the procedure(s) for information and include the acceptance page of the previously accepted procedure(s) attached and referenced. Procedure(s) to be included in the documentation package. If there have been revisions to the previously submitted procedures, then they are to be submitted for review and approval. All records per the procedure required to be maintained as per the Supplier's QA program and its applicable procedure. These records are added to the document package requested with the items. Records to include: Item welded, welding treatment procedure identification and revision that has been accepted by Candu. Approval, signature and date by Company's competent and qualified inspection personnel.	
27.	Brazing Procedure(s)	☒	☐	☒	If Applicable Exception: Brazing procedure(s) are not required to be submitted if accepted previously by another C1 order. Submit the procedure(s) for information and include the acceptance page of the previously accepted procedure(s) attached and referenced. Procedure(s) to be included in the documentation package.	

Procedures / Document Checklist

- Proprietary -

Page 6 of 14

CE-505210-FM-002 Rev. 7

Ref. Procedure CE-904026-PRO-210

No.	Description	Document Required for Review and Acceptance by Purchaser	Document part of the HD/HF (Technical Specification 00-00660-TS-001)	Document part of documentation package as per client requirements	Comments	Submission Date Offered by the Supplier
					If there have been revisions to the previously submitted procedures, then they are to be submitted for review and approval. Reports to be reviewed and accepted	
28.	Plating Procedure(s)	☒	☐	☒	If Applicable Exception: Plating procedure(s) are not required to be submitted if accepted previously by another C1 order. Submit the procedure(s) for information and include the acceptance page of the previously accepted procedure(s) attached and referenced. Procedure(s) to be included in the documentation package. If there have been revisions to the previously submitted procedures, then they are to be submitted for review and approval. Reports to be reviewed and accepted	
29.	Painting/Protection Coating Procedure(s)	☒	☐	☒	Exception: Paint/Protection Coating procedure(s) are not required to be submitted if accepted previously by another C1 order. Submit the procedure(s) for information and include the acceptance page of the previously accepted procedure(s) attached and referenced. Procedure(s) to be included in the documentation package. If there have been revisions to the previously submitted procedures, then they are to be submitted for review and approval. Reports to be reviewed and accepted	

Procedures / Document Checklist

- Proprietary -

Page 7 of 14

CE-505210-FM-002 Rev. 7

Ref. Procedure CE-904026-PRO-210

No.	Description	Document Required for Review and Acceptance by Purchaser	Document part of the HD/HF (Technical Specification 00-00660-TS-001)	Document part of documentation package as per client requirements	Comments	Submission Date Offered by the Supplier
30.	Flushing Procedure(s)	☐	☐	☐		
31.	Cleaning Procedure(s)	☒	☐	☒	Exception: Cleaning procedure(s) are not required to be submitted if accepted previously by another C1 order. Submit the procedure(s) for information and include the acceptance page of the previously accepted procedure(s) attached and referenced. Procedure(s) to be included in the documentation package. If there have been revisions to the previously submitted procedures, then they are to be submitted for review and approval. Reports to be reviewed and accepted	
32.	Passivation Procedure(s)	☒	☐	☒	If Applicable Exception: Passivation procedure(s) are not required to be submitted if accepted previously by another C1 order. Submit the procedure(s) for information and include the acceptance page of the previously accepted procedure(s) attached and referenced. Procedure(s) to be included in the documentation package. If there have been revisions to the previously submitted procedures, then they are to be submitted for review and approval. Reports to be reviewed and accepted	

Procedures / Document Checklist**- Proprietary -**

Page 8 of 14

CE-505210-FM-002 Rev. 7

Ref. Procedure CE-904026-PRO-210

No.	Description	Document Required for Review and Acceptance by Purchaser	Document part of the HD/HF (Technical Specification 00-00660-TS-001)	Document part of documentation package as per client requirements	Comments	Submission Date Offered by the Supplier
33.	Preservation Procedure(s)	☒	☐	☒	Exception: Preservation procedure(s) are not required to be submitted if accepted previously by another C1 order. Submit the procedure(s) for information and include the acceptance page of the previously accepted procedure(s) attached and referenced. Procedure(s) to be included in the documentation package. If there have been revisions to the previously submitted procedures, then they are to be submitted for review and approval. Reports to be reviewed and accepted	
34.	Assembly Procedure(s)	☒	☐	☒	Assembly Procedure(s) and Assembly Report(s) will be separate documents	
35.	Workmanship Standards (Supplier's sub-contractors)	☐	☐	☐		
36.	Identification Procedure(s)	☒	☐	☒		
37.	Packing and Shipping Procedure(s)	☒	☐	☒		
38.	Storage Procedure(s)	☐	☐	☐		
39.	Performance Test Procedures/Reports	☐	☐	☐		
40.	Functional Test Procedures/Reports	☒	☐	☒	Functional Procedure(s) and Functional Report(s) will be separate documents If applicable, include Shielding validation	
41.	Installation Instructions/Procedures	☒	☐	☒	Instructions, Procedure(s), Report(s) to be separate documents	
42.	Pressure Test Procedure(s)/Report(s)	☐	☐	☐		
43.	Lead Pouring and Repair Procedure(s)/Report(s)	☒	☐	☒	If Applicable	

Procedures / Document Checklist**- Proprietary -**

Page 9 of 14

CE-505210-FM-002 Rev. 7

Ref. Procedure CE-904026-PRO-210

No.	Description	Document Required for Review and Acceptance by Purchaser	Document part of the HD/HF (Technical Specification 00-00660-TS-001)	Document part of documentation package as per client requirements	Comments	Submission Date Offered by the Supplier
					Exception: Lead Pouring and Repair procedure(s) are not required to be submitted if accepted previously by another C1 order. Submit the procedure(s) for information and include the acceptance page of the previously accepted procedure(s) attached and referenced. Procedure(s) to be included in the documentation package. If there have been revisions to the previously submitted procedures, then they are to be submitted for review and approval. Reports to be reviewed and accepted	
44.	Load Test Procedure(s)/Report(s)	☒	☐	☒	If Applicable Load Test Procedure(s) and Report(s) to be separate documents To ensure lifting lugs are capable of supporting the weight of the assembly Report(s) to be reviewed and accepted	
Quality Assurance Procedures/Documents						
45.	Quality Program Manual	☒	☐	☒	Prior to PO issue Quality Program Manual is not required to be submitted for review and acceptance if previously accepted previously by another C1 order. Submit the Quality Program Manual for information and include the acceptance page of the previously accepted manual attached and referenced. Quality Program Manual to be included in the documentation package.	

Procedures / Document Checklist

- Proprietary -

Page 10 of 14

CE-505210-FM-002 Rev. 7

Ref. Procedure CE-904026-PRO-210

No.	Description	Document Required for Review and Acceptance by Purchaser	Document part of the HD/HF (Technical Specification 00-00660-TS-001)	Document part of documentation package as per client requirements	Comments	Submission Date Offered by the Supplier
					If there have been revisions to the previously submitted manual, then they are to be submitted for review and approval.	
46.	Inspection and Test Plan	☐	☐	☐		
47.	Dimensional/Visual Inspection Procedure	☒	☐	☒	Dimensional inspection procedures & reports for machined components, valid calibration certificates required for all gauge(s) Report(s) to be reviewed and accepted	
48.	Hydrostatic Test Procedure/Record	☐	☐	☐		
49.	Mechanical Test Procedure/Record	☐	☐	☐		
50.	Mech. Test Spec. Location Illustrations	☐	☐	☐		
51.	Leak Test Procedure	☒	☐	☒	If Applicable Leak Test Procedure and Reports to be separate documents. Report(s) to be reviewed and accepted	
52.	Acceptance Test Procedures/Reports	☒	☐	☒	Acceptance Test Procedure(s) and Acceptance Report(s) will be separate documents Report(s) to be reviewed and accepted	
53.	Ultrasonic Examination/Records	☒	☐	☒	If Applicable Exception: Where NDE applies, test procedure is not required to be submitted for review and acceptance if previously accepted previously by another C1 order. Submit the procedure for information and include the acceptance page of the previously accepted procedure attached and referenced. Procedure to be included in the documentation package.	

Procedures / Document Checklist

- Proprietary -

Page 11 of 14

CE-505210-FM-002 Rev. 7

Ref. Procedure CE-904026-PRO-210

No.	Description	Document Required for Review and Acceptance by Purchaser	Document part of the HD/HF (Technical Specification 00-00660-TS-001)	Document part of documentation package as per client requirements	Comments	Submission Date Offered by the Supplier
					Otherwise, submit procedure for acceptance prior to the test. All NDE records and Qualifications are to provided.	
54.	Eddy Current Examination/Records	☒	☐	☒	If Applicable Exception: Where NDE applies, test procedure is not required to be submitted for review and acceptance if previously accepted previously by another C1 order. Submit the procedure for information and include the acceptance page of the previously accepted procedure attached and referenced. Procedure to be included in the documentation package. Otherwise, submit procedure for acceptance prior to the test. All NDE records and Qualifications are to provided.	
55.	Radiographic Examination/Records	☒	☐	☒	If Applicable Exception: Where NDE applies, test procedure is not required to be submitted for review and acceptance if previously accepted previously by another C1 order. Submit the procedure for information and include the acceptance page of the previously accepted procedure attached and referenced. Procedure to be included in the documentation package.	

Procedures / Document Checklist**- Proprietary -**

Page 12 of 14

CE-505210-FM-002 Rev. 7

Ref. Procedure CE-904026-PRO-210

No.	Description	Document Required for Review and Acceptance by Purchaser	Document part of the HD/HF (Technical Specification 00-00660-TS-001)	Document part of documentation package as per client requirements	Comments	Submission Date Offered by the Supplier
					Otherwise, submit procedure for acceptance prior to the test. All NDE records and Qualifications are to provided.	
56.	Magnetic Particle Examination/Records	☒	☐	☒	If Applicable Exception: Where NDE applies, test procedure is not required to be submitted for review and acceptance if previously accepted previously by another C1 order. Submit the procedure for information and include the acceptance page of the previously accepted procedure attached and referenced. Procedure to be included in the documentation package. Otherwise, submit procedure for acceptance prior to the test. All NDE records and Qualifications are to provided.	
57.	Liquid Penetrant Examination/Records	☒	☐	☒	If Applicable Exception: Where NDE applies, test procedure is not required to be submitted for review and acceptance if previously accepted previously by another C1 order. Submit the procedure for information and include the acceptance page of the previously accepted procedure attached and referenced. Procedure to be included in the documentation package.	

Procedures / Document Checklist**- Proprietary -**

Page 13 of 14

CE-505210-FM-002 Rev. 7

Ref. Procedure CE-904026-PRO-210

No.	Description	Document Required for Review and Acceptance by Purchaser	Document part of the HD/HF (Technical Specification 00-00660-TS-001)	Document part of documentation package as per client requirements	Comments	Submission Date Offered by the Supplier
					Otherwise, submit procedure for acceptance prior to the test. All NDE records and Qualifications are to provided.	
58.	MTR	☐	☐	☒	If applicable Includes Safety Data Sheets for organics	
59.	CMTR	☐	☐	☐		
60.	Deviation Disposition Request /DDR	☒	☐	☒	As applicable When deviation occurs from the design and all documents in the purchased order or tendering documents. Acceptance of such DDRs are required by Candu	
61.	Certificate of Conformance	☒	☐	☒	For each tooling with tooling ID indicated on the purchasing documents, C of C to have the following certification statement with the signature , and include the title of the an authorized rep of the company's quality authority “ The Equipment/tooling listed here in has been inspected by the Company and is in conformance with the contract requirements and approved for shipment”.	
62.	Continuity Test Procedure(s) and Report(s)	☒	☐	☒	As Applicable Continuity Test Procedure(s) and Continuity Test Report(s) will be separate documents	
63.	Megger (HiPot) Test Procedure(s) and Reports(s)	☒	☐	☒	As Applicable Megger (HiPot) Test Procedure(s) and Megger (HiPot) Test Report(s) will be separate documents	

Procedures / Document Checklist

- Proprietary -

Page 14 of 14

CE-505210-FM-002 Rev. 7

Ref. Procedure CE-904026-PRO-210

No.	Description	Document Required for Review and Acceptance by Purchaser	Document part of the HD/HF (Technical Specification 00-00660-TS-001)	Document part of documentation package as per client requirements	Comments	Submission Date Offered by the Supplier
P. O. Line Item Deliverables						
64.	Operation/Maintenance Manual	☐		☐		PO Line item
65.	Quality Records (History Docket)	☐				PO Line item
66.	Quality Records (History File)	☐				PO Line item
67.	Documentation package	☒			QA Documents to be supplied with the shipment for Incoming Inspection and an electronic copy be sent to the buyer. All manufacturing, inspection and test records to be added to the final documentation package	
Generic Comments						

Prepared by:		
	Name:	Date: 2024/05/28
Engineering	R. Ahmad	
Reviewed by:	Name:	Date:
Procurement Quality Assurance Representative (PQAR)	A. Tank	
Discipline Lead	Brian Magda	
Approved by:	Name:	Date:
Discipline Manager	Chris Janeczko	

For the Supplier			
Name	Signature	Date	☐ Submitted with Tender
			☐ Other:



APPENDIX G
Procurement Specifications

142-02120-SP-001	Supplier Procedures – Directions for Format and Technical Content
142-02120-SP-249	Contractors Status Reporting
142-02131-SP-001	Documentation, Packaging, Marking and Shipping Instructions





Specification

SUPPLIER PROCEDURES - DIRECTIONS FOR FORMAT AND TECHNICAL CONTENT

Procurement & Commissioning Procedures and Documents

142-02120-SP-001

Revision 0

Prepared by

A handwritten signature in black ink, appearing to read "M. MacDonald".

MacDonald, Martin M -
CHEMICAL DESIGNER

Reviewed by

A handwritten signature in black ink, appearing to read "Zane Walker".

Walker, Zane - SECTION HEAD -
PROC CHEM & COR

Approved by

A handwritten signature in black ink, appearing to read "Paul Hnatiuk".

Hnatiuk, Paul A. -
MGR, MATERIALS, WELDING &
CHEM ENG

2014/04/09

Candu - Proprietary

Candu Energy Inc.

2285 Speakman Drive
Mississauga, ON
L5K 1B1 Canada



Specification

Supplier Procedures - Directions
for Format and Technical Content

Procurement & Commissioning Procedures and Documents

**142-02120-SP-001
Revision 0**

2014 April

Candu-Proprietary

This document and the information contained in it are the property of Candu Energy Inc. and have been provided subject to the terms and conditions of an agreement with Candu. No use, disclosure or reproduction of this document or the information contained herein is permitted, except in accordance with the applicable agreement or with the prior written consent of Candu.

Candu reserves all intellectual property rights in the document and the information contained herein, including copyright.

Candu Energy Inc.

2285 Speakman Drive
Mississauga, ON
L5K 1B1 Canada



Revision History

Liste de révisions

Candu-Proprietary

CE-511300-FM-168 Rev. 0

Document No. / Numéro de document:	142	02120	SP	001
	Doc. Collection ID ID de la collection de doc.	SI Répertoire du sujet	Section	Serial No. N° de série

Document Details / Détails sur le document

Title Titre	Total no. of pages Nbre total de pages
Supplier Procedures - Directions for Format and Technical Content	61

For Release Information, refer to the Document Transmittal Sheet accompanying this document. / Pour des renseignements portant sur la diffusion, consultez la feuille de transmission de documents ci-jointe.

Revision History / Liste de révisions

Revision / Révision		Details of Rev. / Détails de la rév.	Prepared by Rédigé par	Reviewed by Examiné par	Approved by Approuvé par
No./N°	Date (yyyy/mm/dd)				
D1	2014/02/04	Issued for “Review and Comment”. Sections prepared by the following authors: Brazing - M. Krilic-Andan Cleaning (Chemical or Mechanical) – M. MacDonald Drying – M. MacDonald Forging – M. Krilic Andan Heat Treatment - M. Krilic-Andan Non-Destructive Examination (NDE) – H. McDonald Painting – M. MacDonald Packaging, Shipping, and Storage – M. Krilic-Andan Passivation – M. MacDonald Plating – S. Kim Preservation – M. MacDonald Surface Treatment – S. Kim Welding - M. Krilic-Andan Weld Repair - M. Krilic-Andan	M. MacDonald	Z. Walker K. Farahany	
0	2014/04/09	Issued as “Approved for Use”.	M. MacDonald	Z. Walker	P. Hnatiuk

TABLE OF CONTENTS

SECTION	PAGE
1. PURPOSE.....	1
2. OBJECTIVE	1
3. ACRONYMS.....	1
4. DIRECTIONS FOR MANUFACTURING PROCEDURE FORMAT AND TECHNICAL CONTENT	3
4.1 Brazing	3
4.2 Cleaning (Chemical or Mechanical)	6
4.3 Drying	10
4.4 Forging	14
4.5 Heat Treatment	16
4.6 Non-Destructive Examination (NDE).....	18
4.7 Painting.....	25
4.8 Packaging, Shipping, and Storage	31
4.9 Passivation.....	33
4.10 Plating	39
4.11 Preservation	43
4.12 Surface Treatment	48
4.13 Welding	53
4.14 Weld Repair	56

1. PURPOSE

The purpose of this document is to provide direction/guidelines to suppliers regarding Candu Energy Inc.'s expectations for the format and technical content of their manufacturing procedures.

2. OBJECTIVE

The use of the information contained within this document by the supplier will facilitate the procedure review and acceptance process between them and Candu Energy Inc.

3. ACRONYMS

AMS	Aerospace Material Specification
ASTM	American Society of Testing and Materials
ASME	American Society of Mechanical Engineers
BPS	Brazing Procedure Specification
BPVC	Boiler and Pressure Vessel Code
CANDU	Canada Deuterium Uranium
C of C	Certificate of Conformance
CSA	Canadian Standards Association
CSS	Component Specification Sheet
EQ	Environmentally Qualified
ET	Eddy Current Testing
GMAW	Gas Metal Arc Welding
GTAW	Gas Tungsten Arc Welding
HVOF	High Velocity Oxygen Fuel
ITP	Inspection and Test Plan
MITP	Manufacturing Inspection and Test Plan
MT	Magnetic Particle Testing
NA	Not Applicable
NDE	Non-Destructive Examination
NQA	Nuclear Quality Assurance
PF	Process Flow

PO	Purchase Order
PQR	Procedure Qualification Record
PT	Penetrant Testing
QA	Quality Assurance
QCI	Quality Control Inspector
RT	Radiographic Testing
SAW	Submerged Arc Welding
SI	System International
SMAW	Shielded Metal Arc Welding
SSPC	The Society for Protective Coatings
TS	Technical Specification
UT	Ultrasonic Testing
VSS	Valve Specification Sheet
VT	Visual Testing
WPS	Welding Procedure Specification

4. DIRECTIONS FOR MANUFACTURING PROCEDURE FORMAT AND TECHNICAL CONTENT

The following sub-sections are based on procedure type.

The supplier shall review the content provided in the relevant sub-section(s) of this document and prepare their procedure(s) accordingly so as to be in compliance with the contained procedure content direction and the specified Candu Energy Inc. document/drawing (technical specification, drawing, etc.).

Note that the supplier's methods/processes employed must be in compliance with the requirements of the governing Candu Energy Inc. Technical Specification (TS), drawing(s), Component Specification Sheet (CSS), etc. - the supplier is to reconcile any deviation from these requirements (materials, application, process, etc.).

4.1 Brazing

At a minimum, the content in the following bullets must be provided:

- Supplier/Company Name
- Candu Energy Inc. PO Number
- Supplier Procedure Title
- Supplier Procedure Number and Revision Number
- Supplier Procedure Issue Date
- Candu Energy Inc. Procedure Number and Revision Number
- Name of Item
- Material Type
- ASME or CSA Code Requirements*
- Governing Candu Energy Inc. Technical Specification (TS)*
- Governing Candu Energy Inc. Drawing Number(s)*
- Governing Candu Energy Inc. Component Specification Sheet (CSS)*
- Governing Candu Energy Inc. Valve Specification Sheet (VSS)*

**(if none indicate "NA")*

The procedure shall contain (at a minimum) the following sections:

1) Revision History/Revision Control Sheet:

This section shall (at a minimum) contain the Revision Number, Date, Name of Approver, and Change Description.

2) Purpose/Scope:

This section outlines the procedure objectives, how they are in compliance with the Candu Energy Inc. requirements (e.g. are in compliance with the requirements of the governing Candu Energy Inc. technical specification, drawing(s), CSS, etc.), and defines the limits of the procedure application. A brief description of the brazing method(s) to be used shall be included.

3) Responsibility:

The Quality Control Inspector (QCI), along with the brazing area manager, shall be responsible for ensuring that brazing activities will be performed in accordance with the governing Candu Energy Inc. technical specification, drawing(s), CSS, etc. and that all required documentation will be completed and available for review by Candu Energy Inc.

Personnel performing the brazing activity must be judged competent to perform the task by the Quality Assurance (QA) Manager and shall be trained as per any Candu Energy Inc. requirements. Personnel training documents shall be maintained on file in the supplier's Quality Assurance Department.

4) Methods/Processes:

This section provides the technical aspects of the procedure.

Note that the methods/processes employed must be in compliance with the requirements of the governing Candu Energy Inc. technical specification, drawing(s), CSS, etc. - the supplier is to reconcile any deviation from these requirements (materials, application, process, etc.). Supplementary reference(s) can also be made to Technical Data Sheet(s) for any products suggested for use.

Each manufacture and contactor shall prepare a written Brazing Procedure Specification (BPS). The BPS shall reference the supporting Procedure Qualification Record (PQR).

- a) The brazing procedure specification shall describe all of the essential and nonessential variables for each brazing process used in BPS as per ASME Code IX, listed in QB-250. Each brazing variable shall be defined as per Article XIV, Brazing Data.
- b) The brazing data shall include:
 - Base metal as per QB-402
 - Brazing filler metal as per QB-403
 - Brazing temperature as per QB-404
 - Brazing flux, Fuel gas , or Atmosphere as per QB-406
 - Flow position as per QB-407
 - Joint design as per QB-408
 - Postbrazing heat treatment as per QB-409
 - Technique as per QB-410
- c) Reference shall be made to the following standards:
 - American Society of Mechanical Engineers, Boiler and Pressure Vessel Code:
 - Section II-Part A, Part B, and Part C

- Section III-Division 1
 - Section IX –Brazing Qualification
 - The project specific production drawing and any other engineering specification required by Candu Energy Inc. technical documentation.
- d) The information required to be in the BPS may be in any format to fit the needs of the manufacturer or contractor, as long as every variable outlined in QB-250 through QB-257 and QB-401 is included or referenced.
- e) Form QB-482 ASME IX (Non-mandatory Appendix B) may be used as a guide.
- f) A BPS for Code-production brazing shall be available for reference and review by the Authorized Inspector at the fabrication site.

Procedure Qualification Record(s) (PQR)

- a) A PQR shall document all essential variables of QB-250 for each brazing process used during the brazing of the test coupon. Nonessential or other variables may be recorded at the manufacture's or contractor's request.
- b) The PQR shall be certified by manufacturer or contractor. This certification verifies that the information in the PQR is a true record of the variables used during the brazing of the test coupon and that the test results are in compliance with ASME Section IX.
- c) The type of tests, number of tests, and test results shall be listed in the PQR.
- d) Form QB-483 ASME IX (Non-mandatory Appendix B) may be used as a guide for the PQR. The information required to be in the PQR may be in any format to fit the needs of the manufacturer or contractor, as long as every variable outlined in QB-250 is included or referenced.
- e) The PQR used to support the BPS shall be available, upon request.

4.2 Cleaning (Chemical or Mechanical)

At a minimum, the content in the following bullets must be provided:

- Supplier/Company Name
- Candu Energy Inc. PO Number
- Supplier Procedure Title
- Supplier Procedure Number and Revision Number
- Supplier Procedure Issue Date
- Candu Energy Inc. Procedure Number and Revision Number
- Name of Item
- Material Type
- ASME or CSA Code Requirements*
- Governing Candu Energy Inc. Technical Specification (TS)*
- Governing Candu Energy Inc. Drawing Number(s)*
- Governing Candu Energy Inc. Component Specification Sheet (CSS)*
- Governing Candu Energy Inc. Valve Specification Sheet (VSS)*

**(if none indicate "NA")*

The procedure shall contain (at a minimum) the following sections:

1) Revision History/Revision Control Sheet:

This section shall (at a minimum) contain the Revision Number, Date, Name of Approver, and Change Description.

2) Purpose/Scope:

This section outlines the procedure objectives, how they are in compliance with the Candu Energy Inc. requirements (e.g. are in compliance with the requirements of the governing Candu Energy Inc. technical specification, drawing(s), CSS, etc.), and defines the limits of procedure application (e.g. applies to machined components, castings, etc.). A brief description of the cleaning method(s) to be used shall be included.

3) Responsibility:

The Quality Control Inspector (QCI), along with the cleaning area manager, shall be responsible for ensuring that the cleaning activities will be performed in accordance with the governing Candu Energy Inc. technical specification, drawing(s), CSS, etc. and that all required documentation will be completed and available for review by Candu Energy Inc.

Personnel performing the pre-cleaning/cleaning activity must be judged competent to perform the task by the Quality Assurance (QA) Manager and shall be trained as per any Candu Energy Inc. requirements. Personnel training documents shall be maintained on file in the supplier's Quality Assurance Department.

4) Methods/Processes:

This section provides the technical aspects of the procedure.

Note that the methods/processes employed must be in compliance with the requirements of the governing Candu Energy Inc. technical specification, drawing(s), CSS, etc. - the supplier is to reconcile any deviation from these requirements (materials, application, process, etc.). Supplementary reference can also be made to Technical Data Sheet(s) for any products suggested for use.

- Define the component to be pre-cleaned/cleaned and/or the component area to be pre-cleaned/cleaned (reference the Candu Energy Inc. technical specification, drawing(s), CSS, etc.).
- Pre-cleaning: Describe in detail which method(s) will be employed (wipe, solvent, alkaline, blasting, steam, etc.), reference any Candu Energy Inc. mandated cleaning requirements (example: ASME NQA-1 Subpart 2.1, ASTM E165 Section A1, etc.).
- Cleaning: Describe in detail which method(s) will be employed (wipe, solvent, alkaline, blasting, steam, etc.), reference any Candu Energy Inc. mandated cleaning requirements (example: ASME NQA-1 Subpart 2.1, ASTM E165 Section A1, etc.). Note that prior to application on the component, most powder or liquid cleaning materials (e.g. Alconox) must be first fully dissolved or diluted as per the manufacturer's instructions for use.
- Define which chemical(s) (with concentrations) or materials (define the material) will be used. Technical data sheets for any chemicals proposed for use must be attached to the procedure and must provide sufficient detail so as to demonstrate that any limits provided in the technical specifications are met. Cleaning materials containing deliberate additions of chlorine, fluorine or sulphur based ingredients are not permitted. All other organic cleaning solvents used for cleaning operations shall conform to the requirements for both halogens and sulphur content of paragraph T-641 of Article 6 of Section V of ASME BPVC for Liquid Penetrant Examination materials.
- Define the water quality to be used for chemical make-up and rinsing, reference any Candu Energy Inc. mandated requirement (example: ASME NQA-1 Subpart 2.1, Clause 304, etc.).
- Describe each step that comprises the work activity from start to finish. Description of the process in a flow-sheet or process flow diagram outline is acceptable.
- Describe the inspection method(s) to be employed – provide reference to a supporting inspection procedure or define the method(s) for inspection of the component and the acceptance criteria (see Section 6).
- Describe the steps taken to action non-conforming items found during the inspection process (see Section 6).

5) Performance Tests/Data:

Include any performance test(s) required as per the governing Candu Energy Inc. technical specification or drawing(s). Reference the cleaning product technical data sheet(s) performance data as required.

6) Inspection Criteria:

The method of inspection shall be dictated by the governing Candu Energy Inc. technical specification, drawing(s), CSS, etc. (examples of inspection methods are: Visual, Wipe Test, ASTM Test Method F22 (Water-Break Test)).

Reference other internal/external procedures, standards, or specifications as required.^a It is desirable (but not necessary) that a separate supplier inspection procedure be provided to augment the cleaning procedure (the inspection procedure would be referenced in the cleaning procedure).

Describe how the method(s) of inspection shall be conducted.

Visual inspection method for cleaning shall include the following unless otherwise dictated by the governing Candu Energy Inc. technical specification, drawing(s), CSS, etc., or, by Candu Energy Inc.'s direction to use the manufacturer's cleaning product technical data sheets(s) inspection criteria:

- a) Part is to be inspected from a distance of no more than 2 feet.
- b) Angle of inspection – straight eye line inspection to 45° angle, in any direction.
- c) External surfaces – minimum illumination via shop floor lighting.
- d) Interior surfaces – minimum illumination via shop floor lighting augmented by a portable lighting device (e.g. flash light).

Any instrumentation used for determination of cleanliness quality measurements shall be calibrated prior to use (calibrated as per manufacturer's instructions) and have current calibration documentation available for inspection by Candu Energy Inc.

7) Acceptance Criteria:

Inspection test acceptance criteria shall be dictated by the governing Candu Energy Inc. technical specification, drawing(s), CSS, etc., or, by Candu Energy Inc.'s direction to use the manufacturer's cleaning product technical data sheets(s) inspection test acceptance criteria.

Define how the inspection test results will be interpreted against the test acceptance criteria (i.e. if the inspection test is to determine whether dirt is visible on the component surface and the acceptance criteria is that no dirt shall be visible, then the presence of dirt on the component would be interpreted as a failed cleaning process).

^a An example of a standard is ASTM Test Method F22 (produced by the American Society for Testing and Materials (ASTM)), a specification example is SSPC-SP1 (SSPC (The Society for Protective Coatings (formally known as the Steel Structures Painting Council))).

8) Non-Conforming Material:

The procedure must provide detail on how material found to be in non-conformance with the requirements of the governing Candu Energy Inc. technical specification or drawing(s), CSS, etc., or, by Candu Energy Inc.'s direction to use the manufacturer's cleaning product technical data sheet(s) inspection criteria is dealt with.

Suggested non-conformance mitigation is:

- a) The plant supervisor/manager is notified of the non-conformance.
- b) The non-conforming material is tagged as non-conforming and the details documented (product identified, person rejecting the item, reason for rejection, disposition, and corrective action required).
- c) The corrective action taken to mitigate the non-conformance is documented.
- d) In the event that the corrective action does not meet the requirements of the Candu Energy Inc. technical specification (if provided) or drawing(s), CSS, etc., the non-conformance documentation is to be formally sent to Candu Energy Inc. for review and acceptance.

9) Rework/Repair of Material:

Provide detail on how the rework will be performed: cleaning materials to be used, cleaning material concentrations, cleaning process to be used, inspection process, and acceptance criteria for each stage.

10) Attachments:

The technical data sheets for all cleaning materials used shall be attached to the procedure.

The technical data sheets for any instrumentation used for determination of cleanliness or any other required quality measurement shall be attached to the procedure. If they are not available, the instrument(s) name and model data shall be provided.

Calibration details (calibration method used, date of last calibration, calibration frequency, and calibration standards used) for any instrumentation used for determination of cleanliness or any other required quality measurement shall be attached to the procedure.

Any test reports to support the pre-cleaning/cleaning activity shall be attached to the procedure.

4.3 Drying

At a minimum, the content in the following bullets must be provided:

- Supplier/Company Name
- Candu Energy Inc. PO Number
- Supplier Procedure Title
- Supplier Procedure Number and Revision Number
- Supplier Procedure Issue Date
- Candu Energy Inc. Procedure Number and Revision Number
- Name of Item
- Material Type
- ASME or CSA Code Requirements*
- Governing Candu Energy Inc. Technical Specification (TS)*
- Governing Candu Energy Inc. Drawing Number(s)*
- Governing Candu Energy Inc. Component Specification Sheet (CSS)*
- Governing Candu Energy Inc. Valve Specification Sheet (VSS)*

**(if none indicate "NA")*

The procedure shall contain (at a minimum) the following sections:

1) Revision History/Revision Control Sheet:

This section shall (at a minimum) contain the Revision Number, Date, Name of Approver, and Change Description.

2) Purpose/Scope:

This section outlines the procedure objectives, how they are in compliance with the Candu Energy Inc. requirements (e.g. are in compliance with the requirements of the governing Candu Energy Inc. technical specification, drawing(s), CSS, etc.), and defines the limits of procedure application (e.g. applies to machined components, castings, etc.). A brief description of the drying method(s) to be used shall be included.

3) Responsibility:

The Quality Control Inspector (QCI), along with the drying area manager, shall be responsible for ensuring that drying activities will be performed in accordance with the governing Candu Energy technical specification, drawing(s), CSS, etc. and that all required documentation will be completed and available for review by Candu Energy Inc.

Personnel performing the drying activity must be judged competent to perform the task by the Quality Assurance (QA) Manager and shall be trained as per any Candu Energy Inc. requirements. Personnel training documents shall be maintained on file in the supplier's Quality Assurance Department.

4) Methods/Processes:

This section provides the technical aspects of the procedure.

Note that the methods/processes employed must be in compliance with the requirements of the governing Candu Energy Inc. technical specification, drawing(s), CSS, etc. - the supplier is to reconcile any deviation from it (materials, application, process, etc.). Supplementary reference can also be made to Technical Data Sheet(s) for any products suggested for use.

- Define the component to be dried and/or the component area to be dried (reference the Candu Energy Inc. technical specification, drawing(s), CSS, etc.).
- Describe in detail which drying method(s) will be employed (draining, wipe, compressed air blast, hot air blowing, oven drying, chemical, etc.), reference any Candu Energy Inc. mandated drying requirements.
- Define which chemical(s) (with concentrations) or materials (define the material) will be used. Technical data sheets for any chemicals proposed for use (e.g. acetone) must be attached to the procedure and must provide detail on halogen and sulfur content.
- Describe in detail the process of component disassembly if it is required.
- Describe each step that comprises the drying activity from start to finish. Description of the process in a flow-sheet or process flow diagram outline is acceptable.
- For oven or furnace drying of components, define the temperatures to be employed and the duration of component exposure at these temperatures. Describe how oven/furnace temperature verification is performed (e.g. calibration) and at what frequency. Note: for Environmentally Qualified (EQ) components, drying methods other than oven, furnace, or forced hot-air drying is preferred due to accelerated (heat) aging of any non-removable elastomeric compounds. If oven, furnace, or forced hot-air drying methods need to be performed on EQ sensitive materials, supporting documentation is required to substantiate the temperature(s) and time(s) employed (data acquisition system data, process log sheets, etc.).
- Describe the inspection method(s) to be employed – provide reference to a supporting inspection procedure or define the method(s) for inspection of the component and the acceptance criteria (see Section 6).
- Describe in detail the process of component reassembly if component disassembly had been performed.
- Describe the steps taken to action non-conforming items found during the inspection process (see Section 6).

5) Performance Tests/Data:

Include any performance test(s) required as per the governing Candu Energy Inc. technical specification or drawing(s).

6) Inspection Criteria:

The method of inspection shall be dictated by the governing Candu Energy Inc. technical specification, drawing(s), CSS, etc. (examples of inspection methods are: Visual, Wipe Test, etc.).

Reference other internal/external procedures, standards, or specifications as required.^b It is desirable (but not necessary) that a separate supplier inspection procedure be provided to augment the drying procedure (the inspection procedure would be referenced in the drying procedure).

Describe how the method(s) of inspection shall be conducted.

Visual inspection method for drying shall include the following unless otherwise dictated by the governing Candu Energy Inc. technical specification, drawing(s), CSS, etc., or, by Candu Energy Inc.'s direction to use the manufacturer's drying product technical data sheets(s) inspection criteria:

- a) Part is to be inspected from a distance of no more than 2 feet.
- b) Angle of inspection – straight eye line inspection to 45° angle, in any direction.
- c) External surfaces – minimum illumination via shop floor lighting.
- d) Interior surfaces – minimum illumination via shop floor lighting augmented by a portable lighting device (e.g. flash light).

Any instrumentation used for determination of dryness quality measurements shall be calibrated prior to use (calibrated as per manufacturer's instructions) and have current calibration documentation available for inspection by Candu Energy Inc.

7) Acceptance Criteria:

Inspection test acceptance criteria shall be dictated by the governing Candu Energy Inc. technical specification, drawing(s), CSS, etc., or, by Candu Energy Inc.'s direction to use the manufacturer's drying product technical data sheets(s) inspection test acceptance criteria.

Define how the inspection test results will be interpreted against the test acceptance criteria (i.e. if the inspection test is to determine whether moisture is visible on a previously clean/dry wiper that has just wiped the component surface and the acceptance criteria is that no moisture shall be visible, then the presence of moisture on the wiper would be interpreted as a failed drying process).

8) Non-Conforming Material:

The procedure must provide detail on how material found to be in non-conformance with the requirements of the governing Candu Energy Inc. technical specification or drawing(s), CSS, etc., or, by Candu Energy Inc.'s direction to use the manufacturer's drying product technical data sheet(s) inspection criteria is dealt with.

^b An example of a standard is ASTM Test Method F22 (produced by the American Society for Testing and Materials (ASTM)), a specification example is SSPC-SP1 (SSPC (The Society for Protective Coatings (formally known as the Steel Structures Painting Council))).

Suggested non-conformance mitigation is:

- a) The plant supervisor/manager is notified of the non-conformance.
- b) The non-conforming material is tagged as non-conforming and the details documented (product identified, person rejecting the item, reason for rejection, disposition, and corrective action required).
- c) The corrective action taken to mitigate the non-conformance is documented.
- d) In the event that the corrective action does not meet the requirements of the Candu Energy Inc. technical specification (if provided) or drawing(s), CSS, etc., the non-conformance documentation is to be formally sent to Candu Energy Inc. for review and acceptance.

9) Rework/Repair of Material:

Provide detail on how the rework will be performed: drying materials to be used, concentrations of any chemicals used, drying process to be employed, inspection process, and acceptance criteria for each stage.

10) Attachments:

The technical data sheets for all chemical-based drying materials used shall be attached to the procedure.

The technical data sheets for any instrumentation used for determination of dryness or any other required quality measurement shall be attached to the procedure. If they are not available, the instrument(s) name and model data shall be provided.

Calibration details (calibration method used, date of last calibration, calibration frequency, and calibration standards used) for any instrumentation used for determination of dryness or any other required quality measurement shall be attached to the procedure.

Any test reports to support the drying activity shall be attached to the procedure.

4.4 Forging

At a minimum, the content in the following bullets must be provided:

- Supplier/Company Name
- Candu Energy Inc. PO Number
- Supplier Procedure Title
- Supplier Procedure Number and Revision Number
- Supplier Procedure Issue Date
- Candu Energy Inc. Procedure Number and Revision Number
- Name of Item
- Material Type
- ASME or CSA Code Requirements*
- Governing Candu Energy Inc. Technical Specification (TS)*
- Governing Candu Energy Inc. Drawing Number(s)*
- Governing Candu Energy Inc. Component Specification Sheet (CSS)*
- Governing Candu Energy Inc. Valve Specification Sheet (VSS)*

**(if none indicate "NA")*

The procedure shall contain (at a minimum) the following sections:

1) Revision History/Revision Control Sheet:

This section shall (at a minimum) contain the Revision Number, Date, Name of Approver, and Change Description.

2) Purpose/Scope:

This section outlines the procedure objectives, how they are in compliance with the Candu Energy Inc. requirements (e.g. are in compliance with the requirements of the governing Candu Energy Inc. technical specification, drawing(s), CSS, etc.), and defines the limits of procedure application. The forging procedure shall provide the forge process for both the intermediate billet and the final blanks. A brief description of the forging method shall be included.

3) Responsibility:

The Quality Control Inspector (QCI), along with the forging area manager, shall be responsible for ensuring that forging activities will be performed in accordance with the governing Candu Energy technical specification, drawing(s), CSS, etc. and that all required documentation will be completed and available for review by Candu Energy Inc. Personnel performing the forging activity must be judged competent to perform the task by the Quality Assurance (QA) Manager and shall be trained as per any Candu Energy Inc. requirements. Personnel training documents shall be maintained on file in the supplier's Quality Assurance Department.

4) Methods/Processes:

This section provides the technical aspects of the procedure.

Note that the methods/processes employed must be in compliance with the requirements of the governing Candu Energy Inc. technical specification, drawing(s), CSS, etc. - the supplier is to reconcile any deviation from these requirements (materials, application, process, etc.). Supplementary reference can also be made to Technical Data Sheet(s) for any products suggested for use.

- Procedure shall refer to the drawing number of the part.
- The ASME or ASTM or other applicable material specification shall be referenced.
- a) Intermediate Billet
 - Dimension and number of billet from one ingot shall be presented.
 - Identification of billets for traceability purpose to be specified on steel tag.
 - Marking of billets with PF (process flow), Shop Order #, and grade and heat #.
 - Handling, disposition and cooling regime of billet to be specified.
- b) Final Blank
 - A drawing of a blank to be forged with machined sizes shall be presented.
 - Process specification Forge reference chart will be presented and it shall define:
 - 1) Preheating temperature, soaking time (heating rate) of billet prior forging.
 - 2) Disposition, cooling atmosphere and reference temperature of the blanks after forging.
 - 3) Brief description of the heat treatment (tempering) to provide a stable structure/heat treatment requirements from the drawing.
 - 4) A map showing the location of each piece in the furnace and number per lot to be specified.
 - 5) Interrupted forging, prolonged holding at forging temperature of both, billets and the blanks, due to maintenance or any other unexpected causes shall be covered with detailed production heat cycle procedure for stabilization or any other appropriate instruction for the final resolution of the related cause.
 - 6) Brief description of any pre/post forged process that may be required such as descaling, removal of end defects, porosity, approaches for minimizing thermal shock caused with “winter condition” (ingots stored outside in the cold weather).
 - Reference ASTM standard for any testing on finished part (hardness/tensile, etc.) required.
 - Number of forge parts to be tested per each lot is specified.

4.5 Heat Treatment

The procedure shall contain (at a minimum) the following information:

- Supplier/Company Name
- Candu Energy Inc. PO Number
- Supplier Procedure Title
- Supplier Procedure Number and Revision Number
- Supplier Procedure Issue Date
- Candu Energy Inc. Procedure Number and Revision Number
- Name of Item
- Material Type
- ASME or CSA Code Requirements*
- Governing Candu Energy Inc. Technical Specification (TS)*
- Governing Candu Energy Inc. Drawing Number(s)*
- Governing Candu Energy Inc. Component Specification Sheet (CSS)*
- Governing Candu Energy Inc. Valve Specification Sheet (VSS)*

**(if none indicate "NA")*

The procedure shall contain (at a minimum) the following sections:

1) Revision History/Revision Control Sheet:

This section shall (at a minimum) contain the Revision Number, Date, Name of Approver, and Change Description.

2) Purpose/Scope:

This section outlines the procedure objectives, how they are in compliance with the Candu Energy Inc. requirements (e.g. are in compliance with the requirements of the governing Candu Energy Inc. technical specification, drawing(s), CSS, etc., and defines the limits of procedure application (what material it applies to). A brief description of the heat treatment method(s) and products to be used shall be included.

3) Responsibility:

The Quality Control Inspector (QCI), along with the heat treatment area manager, shall be responsible for ensuring that heat treatment activities will be performed in accordance with the governing Candu Energy technical specification, drawing(s), CSS, etc. and that all required documentation will be completed and available for review by Candu Energy Inc.

Personnel performing the heat treatment activity must be judged competent to perform the task by the Quality Assurance (QA) Manager and shall be trained as per any Candu Energy Inc. requirements. Personnel training documents shall be maintained on file in the supplier's Quality Assurance Department.

4) Methods/Processes:

This section provides the technical aspects of the procedure.

Note that the methods/processes employed must be in compliance with the requirements of the governing Candu Energy Inc. Technical Specification, drawing(s), CSS, etc. - the supplier is to reconcile any deviation from these requirements (materials, application, process, etc.). Supplementary reference can also be made to Technical Data Sheet(s) for any products suggested for use.

- Procedure shall refer to the drawing number of the part being heat treated (dimensions, standard material designator and objectives of heat treatment shall be presented on the drawing).
- Heat treatment temperature shall be presented in both, the imperial and SI-based units.
- Heating and cooling rate shall be presented in terms of degrees of temperature/hour. Holding time at temperature-soaking time shall be presented in terms of time (hr. or min) per in. (mm) of the referenced thickness.
- The tolerance on referenced heat treatment temperature shall be specified in procedure.
- Lot size (number of pieces per lot) min/max shall be specified.
- Number of parts to be tested after heat treatment per each heat treatment lot shall be specified.
- If required, the actual test results for each lot shall be provided.
- Any ASTM standard for any testing on finished part (hardness/tensile, etc.) shall be referenced.
- Heat treatment charts for each lot shall be provided.
- If used, location and number of thermo couple shall be specified and presented in the procedure.
- If part is cleaned prior to heat treatment, the reference cleaning procedure shall be provided.
- If used, protective atmosphere type and composition shall be stated.
- When applicable, type of quenching solutions shall be provided (oil, water, brine, and etcetera.)

4.6 Non-Destructive Examination (NDE)

At a minimum, the content in the following bullets must be provided:

- Supplier/Company Name
- Candu Energy Inc. PO Number
- Supplier Procedure Title
- Supplier Procedure Number and Revision Number
- Supplier Procedure Issue Date
- Candu Energy Inc. Procedure Number and Revision Number
- Name of Item
- Material Type
- ASME or CSA Code Requirements*
- Governing Candu Energy Inc. Technical Specification (TS)*
- Governing Candu Energy Inc. Drawing Number(s)*
- Governing Candu Energy Inc. Component Specification Sheet (CSS)*
- Governing Candu Energy Inc. Valve Specification Sheet (VSS)*

**(if none indicate "NA")*

The procedure shall contain (at a minimum) the following sections:

1) Revision History/Revision Control Sheet:

This section shall (at a minimum) contain the Revision Number, Date, Name of Approver, and Change Description.

2) Purpose/Scope:

This section outlines the procedure objectives and how they are in compliance with the Candu Energy Inc. requirements (i.e. applicable Candu Energy Inc. technical specification, drawing(s), CSS, codes & standards, etc.). In addition the purpose defines the limits of procedure application (e.g. applies to machined components, castings, etc.). A brief description of the NDE method shall also be included.

3) Responsibility:

The Quality Control Inspector (QCI), along with the NDE manager, shall be responsible for ensuring that NDE activities will be performed in accordance with the governing Candu Energy technical specification, drawing(s), CSS, codes & standards, etc. and that all required documentation will be completed and available for review by Candu Energy Inc.

Personnel performing the NDE activity must be judged competent to perform the task by the Quality Assurance (QA) Manager and shall be trained as per any Candu requirements. Personnel training documents shall be maintained on file in the supplier's Quality Assurance Department.

4) General:

a) Personnel Qualification

All personnel performing NDE shall be qualified in accordance with the appropriate standards identified in the applicable governing Candu Energy Inc. documents.

b) Equipment/Calibration

The appropriate equipment to be used shall be identified in the procedure. In addition, as part of the procedure, the calibration should be verified and recorded in the report.

c) Cleanliness

When required by the applicable Candu Energy Inc. technical specification, drawing(s), CSS, codes & standards, etc. the NDE technique shall ensure that the procedure and applicable materials shall meet the applicable sulphur and halogen restrictions. A Certificate of Conformance (C of C) for all NDE materials, including cleaning solvents, shall be supplied.

5) Procedure:

This section provides the required technical content aspects of the procedure.

Note that the methods/processes employed must be in compliance with the requirements of the governing Candu Energy Inc. technical specification, drawing(s), CSS, codes & standards, etc.

a) Radiographic Examination/Radiographic Testing (RT)

- material type and thickness range
- isotope or maximum X-ray voltage used
- source-to-object distance
- distance from source side of object to film
- source size
- film brand and designation
- screens used
- image quality indicator
- C of C for marking materials

b) Ultrasonic Examination of Welds/Ultrasonic Testing (UT) of Welds

- Weld configurations to be examined, including thickness dimensions and base material product form (pipe, plate, etc.)
- The surfaces from which the examination shall be performed

- Technique(s) (straight beam, angle beam, contact, and/or immersion)
- Angle(s) and mode(s) of wave propagation in the material
- Search unit type(s), frequency(ies), and element size(s)/shape(s)
- Special search units, wedges, shoes, or saddles, when used
- Ultrasonic instrument(s)
- Calibration [calibration block(s) and technique(s)]
- Directions and extent of scanning
- Scanning (manual vs. automatic)
- Method for discriminating geometric from flaw indications
- Method for sizing indications
- Computer enhanced data acquisition, when used
- Scan overlap
- Personnel performance requirements, when required
- Personnel qualification requirements
- Surface condition (examination surface, calibration block)
- Couplant: brand name or type and C of C
- Post-examination cleaning technique
- Automatic alarm and/or recording equipment, when applicable
- Records, including minimum calibration data to be recorded (e.g., instrument settings)

c) Ultrasonic Examination of Material/Ultrasonic Testing (UT) of Material

- Material types and configurations to be examined, including thickness dimensions and product form (castings, forgings, plate, etc.)
- The surfaces from which the examination shall be performed
- Technique(s) (straight beam, angle beam, contact, and/or immersion)
- Angle(s) and mode(s) of wave propagation in the material
- Search unit type(s), frequency(ies), and element size(s)/shape(s)
- Special search units, wedges, shoes, or saddles, when used
- Ultrasonic instrument(s)
- Calibration [calibration block(s) and technique(s)]
- Directions and extent of scanning
- Scanning (manual vs. automatic)
- Method for sizing indications
- Computer enhanced data acquisition, when used
- Scan overlap (decrease only)
- Personnel performance requirements, when required
- Personnel qualification requirements
- Surface condition (examination surface, calibration block)
- Couplant: brand name or type and C of C
- Post-examination cleaning technique
- Automatic alarm and/or recording equipment, when applicable

- Records, including minimum calibration data to be recorded (e.g., instrument settings)
- d) Liquid Penetrant Examination/Penetrant Testing (PT)
- Identification of type and family group of penetrant materials including developers, emulsifiers, etc.
 - Surface preparation (finishing and cleaning, including type of cleaning solvent)
 - Method of applying penetrant
 - Method of removing excess surface penetrant
 - Hydrophilic or lipophilic emulsifier concentration and dwell time in dip tanks and agitation time for hydrophilic emulsifiers
 - Hydrophilic emulsifier concentration in spray applications
 - Method of applying developer
 - Minimum and maximum time periods between steps and drying aids
 - Penetrant dwell time
 - Developer dwell time (Interpretation Time)
 - Light intensity range
 - Surface temperature range
 - Performance demonstration, when required
 - Personnel qualification requirements
 - Materials, shapes, or sizes to be examined and the extent of examination
 - Post-examination cleaning technique
 - Penetrant materials C of C for sulfur and halogen content
- e) Magnetic Particle Examination/Magnetic Particle Testing (MT)
- Magnetizing technique
 - Magnetizing current type and amperage
 - Surface preparation
 - Magnetic particles (fluorescent/visible, color, particle size, wet/dry)
 - Method of particle application
 - Method of excess particle removal
 - Minimum light intensity
 - Existing coatings, coating type and thickness demonstrated
 - Nonmagnetic surface contrast enhancement, when utilized
 - Performance demonstration, when required
 - Qualified examination part surface temperature range
 - Shape or size of the examination object
 - Equipment type
 - Demagnetizing technique and specified residual magnetization
 - Post-examination cleaning technique
 - Personnel qualification requirements
 - Materials C of C for sulfur and halogen content

f) Eddy Current Examination/Eddy Current Testing (ET)

- Tube material
- Tube diameter and wall thickness
- Mode of inspection — differential or absolute
- Probe type and size
- Length of probe cable and probe extension cables
- Probe manufacturer, part number, and description
- Examination frequencies, drive voltage, and gain settings
- Manufacturer and model of eddy current equipment
- Scanning direction during data recording, i.e., push or pull
- Scanning mode — manual, mechanized probe driver, remote controlled fixture
- Fixture location verification
- Identity of calibration reference standard(s)
- Minimum digitization rate
- Maximum scanning speed during data recording
- Personnel requirements
- Data recording equipment manufacturer and model
- Scanning speed during insertion or retraction, no data recording
- Side of application — inlet or outlet
- Data analysis parameters
- Tube numbering
- Tube examination surface preparation

g) Visual Examination/Visual Testing (VT)

- Technique used (i.e. direct, remote)
- Visual aids
- Personnel performance requirements, when required
- Lighting intensity
- Configurations to be examined and base material product forms (pipe, plate, forgings, etc.)
- Lighting equipment
- Methods or tools used for surface preparation
- Equipment or devices used for a direct technique
- Sequence of examination
- Personnel qualifications

6) Acceptance Criteria:

Inspection test acceptance criteria shall be dictated by the governing Candu Energy Inc. technical specification, drawing(s), CSS, codes & standards, etc.

Define how the inspection test results will be interpreted against the test acceptance criteria.

7) Non-Conforming Material:

The procedure must provide detail on how material found to be in non-conformance with the requirements of the governing Candu Energy Inc. technical specification or drawing(s), CSS, codes & standards, etc., is dealt with.

Suggested non-conformance action(s) are:

- a) The plant supervisor/manager is notified of the non-conformance,
- b) The non-conforming material is tagged as non-conforming and quarantined. The details documented (product identified, person rejecting the item, reason for rejection, disposition, and corrective action required).
- c) The non-conformance shall be brought to the attention of the customer for resolution and/or acceptance.
- d) The corrective action taken to resolve the non-conformance shall be documented.

8) Rework/Repair of Material:

Reference appropriate contingency procedure, alternatively the procedure may be identified in applicable ITP or MITP or shop traveler (as appropriate).

9) Attachments:

- a) Calibration details: The calibration method used, date of last calibration, calibration frequency, and calibration standards used for any instrumentation used.
- b) C of C: When applicable, C of Cs for materials coming into contact with the item(s) being tested.
- c) NDE Report Template

10) Records/Documentation:

- a) Documentation and records shall be prepared as specified by the referencing Code Section and the applicable requirements of this Section. Examination records shall include the following information as a minimum:
 - (1) Date of the examination.
 - (2) Name and/or identity and certification level (if applicable) for personnel performing the examination.
 - (3) Identification of any NDE equipment used and the calibration date and date due.
 - (4) Identification of the weld, part, or component examined including weld number, serial number, or other identifier.
 - (5) Examination method, technique, procedure Identification, and revision.
 - (6) Results of the examination.
- b) Personnel qualifications and procedure performance demonstrations shall be documented as specified by the referencing Code Section.

- c) When documentation requirements for personnel qualification and procedure performance demonstrations performed are not specified by the referencing Code Section, the following information shall be recorded as a minimum:
 - (1) Name of organization responsible for preparation and approval of the examination procedure.
 - (2) Examination method applied
 - (3) Procedure number or designation
 - (4) Number and date of most recent revision.
 - (5) Date of the demonstration.
 - (6) Name and/or identity and certification level (if applicable) of personnel performing demonstration.
- d) Retention of examination records and related documentation (e.g., radiographs and review forms, ultrasonic scan files, etc.) shall be as specified by the referencing Code Section.
- e) Digital images and reviewing software shall be retained under an appropriate record retention system that is capable of securely storing and retrieving data for the time period specified by the referencing Code Section.

4.7 Painting

The procedure shall contain (at a minimum) the following information:

- Supplier/Company Name
- Candu Energy Inc. PO Number
- Supplier Procedure Title
- Supplier Procedure Number and Revision Number
- Supplier Procedure Issue Date
- Candu Energy Inc. Procedure Number and Revision Number
- Name of Item
- Material Type
- ASME or CSA Code Requirements*
- Governing Candu Energy Inc. Technical Specification (TS)*
- Governing Candu Energy Inc. Drawing Number(s)*
- Governing Candu Energy Inc. Component Specification Sheet (CSS)*
- Governing Candu Energy Inc. Valve Specification Sheet (VSS)*

**(if none indicate "NA")*

The procedure shall contain (at a minimum) the following sections:

1) Revision History/Revision Control Sheet:

This section shall (at a minimum) contain the Revision Number, Date, Name of Approver, and Change Description.

2) Purpose/Scope:

This section outlines the procedure objectives, how they are in compliance with the Candu Energy Inc. requirements (e.g. are in compliance with the requirements of the governing Candu Energy Inc. technical specification, drawing(s), CSS, etc., and defines the limits of procedure application (e.g. applies to machined components, castings, etc.). A brief description of the painting method(s) and products to be used shall be included.

3) Responsibility:

The Quality Control Inspector (QCI), along with the painting area manager, shall be responsible for ensuring that painting activities will be performed in accordance with the governing Candu Energy technical specification, drawing(s), CSS, etc. and that all required documentation will be completed and available for review by Candu Energy Inc.

Personnel performing the painting activity must be judged competent to perform the task by the Quality Assurance (QA) Manager and shall be trained as per any Candu Energy Inc. requirements. Personnel training documents shall be maintained on file in the supplier's Quality Assurance Department.

4) Methods/Processes:

This section provides the technical aspects of the procedure.

Note that the methods/processes employed must be in compliance with the requirements of the governing Candu Energy Inc. technical specification, drawing(s), CSS, etc. - the supplier is to reconcile any deviation from these requirements (materials, application, process, etc.). Supplementary reference can also be made to Technical Data Sheet(s) for any products suggested for use.

a) Pretreatment Section^c

- Cleaning method(s) to be employed - describe in detail the process for component cleaning^d.
- Surface preparation level required (brushing, blasting, etc.) - describe in detail the process for component surface preparation².
- If a corrosion preventative application step is to be performed (e.g. phosphating), the corrosion preventative must be identified and the process conditions for its application described in detail (concentration, mode of contact, temperature, duration, rinsing, etc.).
- Areas to be cleaned and/or treated with a corrosion preventative must be defined (reference the Candu Energy Inc. drawing or other).
- Inspection methods to be employed – describe in detail the method(s) for inspection of components after each process activity performed (e.g. after solvent cleaning, after wire brushing, after applying corrosion preventative product, etc.) and the acceptance criteria (see Section 5).
- Describe the steps taken to action non-conforming items found during the inspection process (see Section 6).

b) Paint Reduction, Mixing, and Application Equipment Section

- Reduction of paints and requirements for mixing shall be governed by the associated Technical Data Sheets for the products to be applied.
- All paint products, containers, and application equipment must be kept free of contaminants.

c) Primer Coat

- The primer material to be applied shall be identified.

^c The methods/processes employed must be in compliance with the requirements of the governing Candu Energy Inc. Technical Specification (if provided) and/or the coating product Technical Data Sheet(s).

^d Reference other internal/external procedures or standards if necessary (an example of a standard is a surface preparation standard produced by the SSPC (The Society for Protective Coatings (formally known as the Steel Structures Painting Council)).

- The required number of primer coats shall be identified (as per the governing Candu Energy Inc. technical specification (if provided) and/or the coating product Technical Data Sheet(s)).
- The required primer color shall be identified (as per Candu Energy Inc. requirements).
- The required primer thickness shall be identified (as per the governing Candu Energy Inc. technical specification (if provided) and/or the coating product Technical Data Sheet(s)).
- Areas to be cleaned and/or treated with a corrosion preventative must be defined (reference the Candu Energy Inc. drawing or other Candu Energy requirements).
- Areas to be primed must be defined (reference the Candu Energy Inc. drawing or other Candu Energy requirements).
- The method of primer material application shall be identified.
- Environmental requirements at time of primer material application shall be identified (temperature, humidity, etc.).
- Any baking, dry/cure-time, or recoating requirements shall be identified and described (time, temperature, humidity, etc.).
- The requirements for coating thickness measurements shall be defined (number of measurements per unit area, number of duplicate measurements, etc.).
- Coating inspection methods to be employed – describe in detail the method(s) for inspection of components after coating and the acceptance criteria (as per Section 5).
- Describe the steps taken to action non-conforming items found during the inspection process (see Section 6).

d) Top Coat

- The top coat material to be applied shall be identified.
- The required number of top coat applications shall be identified (as per the governing Candu Energy Inc. technical specification (if provided) and/or the coating product Technical Data Sheet(s)).
- The required top coat color shall be identified (as per Candu Energy Inc. requirements).
- The required top coat thickness shall be identified (as per the governing Candu Energy Inc. technical specification (if provided) and/or the coating product Technical Data Sheet(s)).
- The method of top coat material application shall be identified.
- Environmental requirements at time of top coat material application shall be identified (temperature, humidity, etc.).
- Any baking, dry/cure-time, or recoating requirements shall be identified and described (time, temperature, humidity, etc.).

- The requirements for coating thickness measurements shall be defined (number of measurements per unit area, number of duplicate measurements, etc.).
- Coating inspection methods to be employed – describe in detail the method(s) for inspection of components after coating and the acceptance criteria (as per Section 5).
- Describe the steps taken to action non-conforming items found during the inspection process (see Section 6).

e) Powder Paint

- The powder coat material to be applied shall be identified.
- The method of powder coat application shall be identified.
- The application equipment requirements shall be identified (grounding requirements, etc.).
- The required powder coat color shall be identified (as per Candu Energy Inc. requirements).
- The required powder coat thickness shall be identified (as per the governing Candu Energy Inc. technical specification (if provided) and/or the coating product Technical Data Sheet(s)).
- Environmental requirements for time of powder coating material application shall be identified (temperature, humidity, etc.).
- Areas to be powder coated must be defined (reference the Candu Energy Inc. drawing or other).
- Any baking requirements shall be identified and described (time, temperature, etc.).
- The requirements for coating thickness measurements shall be defined (number of measurements per unit area, number of duplicate measurements, etc.).
- Powder coating material reapplication requirements shall be provided.
- Coating inspection methods to be employed – describe in detail the method(s) for inspection of components after coating and the acceptance criteria (as per Section 5).
- Describe the steps taken to action non-conforming items found during the inspection process (see Section 6).

5) Performance Tests/Data:

Include any performance test(s) as required as per the governing Candu Energy Inc. technical specification (if provided) and/or the coating product Technical Data Sheet(s) (e.g. gloss reading, cure test, adhesion test, etc.).

6) Inspection Criteria:

The method of inspection shall be dictated by the governing Candu Energy Inc. technical specification, drawing(s), CSS, etc.

Reference other internal/external procedures, standards, or specifications as required. It is desirable (but not necessary) that a separate supplier inspection procedure be provided to augment the painting procedure (the inspection procedure would be referenced in the painting procedure).

Describe how the method(s) of inspection shall be conducted.

Visual inspection should include the following unless otherwise dictated by the governing Candu Energy Inc. technical specification (if provided) and/or the coating product Technical Data Sheet(s):

- a) Part is to be inspected from a distance of no more than 2 feet.
- b) Angle of inspection – straight eye line inspection to 45° angle, in any direction.
- c) External surfaces – minimum illumination via shop floor lighting.
- d) Interior surfaces – minimum illumination via shop floor lighting augmented by a portable lighting device (e.g. flash light).

Inspection method is typically performed via visual examination, acceptance criteria would be “no visible imperfections with a uniformity of appearance, color, finish, and coverage” and to include a statement similar to: “a surface void of runs, sags, blisters, blotches, excess over-spray, flaking, stains, fish eye, smears, pin holes, craters, dust, bubbles, scratches, or chipped coating.”

Instrumentation used for determination of coating thickness, gloss measurements, etc. shall be calibrated prior to use (calibrated as per manufacturer’s instructions) and have current calibration documentation available for inspection by Candu Energy Inc.

7) Acceptance Criteria:

Inspection test acceptance criteria shall be dictated by the governing Candu Energy Inc. technical specification, drawing(s), CSS, etc., or, by Candu Energy Inc.’s direction to use the manufacturer’s cleaning, reduction, and painting product technical data sheet inspection test acceptance criteria.

Define how the inspection test results will be interpreted against the test acceptance criteria (i.e. if the inspection test is to determine whether pin holes are visible on the component surface and the acceptance criteria is that no pin holes shall be visible, then the presence of pin holes on the component would be interpreted as a failed painting process).

8) Non-Conforming Material:

The procedure must provide detail on how material found to be in non-conformance with the requirements of the governing Candu Energy Inc. technical specification (if provided) and/or the coating product Technical Data Sheet(s) is dealt with.

Suggested non-conformance mitigation is:

- a) The plant supervisor/manager is notified of the non-conformance.
- b) The non-conforming material is tagged as non-conforming and the details documented (product identified, person rejecting the item, reason for rejection, disposition, and corrective action required).
- c) The corrective action taken to mitigate the non-conformance is documented.
- d) In the event that the corrective action does not meet the requirements of the Candu Energy Inc. technical specification (if provided) and/or the coating product Technical Data Sheet(s), the non-conformance is to be formally sent to Candu Energy Inc. for review and acceptance.

9) Rework/Repair of Material:

The procedure must provide detail on how the repair or touch-up of a material coating will be performed: surface preparation, coating preparation and application, inspection, and acceptance requirements.

10) Attachments:

The technical data sheets for all cleaning, corrosion prevention, primer, reduction, and coating materials used on the parts/equipment shall be attached to the procedure.

The technical data sheets for any instrumentation used for determination of cleanliness, coating thickness, or any other required quality measurement shall be attached to the procedure. If they are not available, the instrument(s) name and model data shall be provided.

Calibration details (calibration method used, date of last calibration, calibration frequency, and calibration standards used) for any instrumentation used for determination of cleanliness, coating thickness, or any other required quality measurement shall be attached to the procedure.

Any test reports to support the cleaning and coating activity shall be attached to the procedure.

4.8 Packaging, Shipping, and Storage

At a minimum, the content in the following bullets must be provided:

- Supplier/Company Name
- Candu Energy Inc. PO Number
- Supplier Procedure Title
- Supplier Procedure Number and Revision Number
- Supplier Procedure Issue Date
- Candu Energy Inc. Procedure Number and Revision Number
- Name of Item
- Material Type
- ASME or CSA Code Requirements*
- Governing Candu Energy Inc. Technical Specification (TS)*
- Governing Candu Energy Inc. Drawing Number(s)*
- Governing Candu Energy Inc. Component Specification Sheet (CSS)*
- Governing Candu Energy Inc. Valve Specification Sheet (VSS)*

**(if none indicate "NA")*

The procedure shall contain (at a minimum) the following sections:

1) Revision History/Revision Control Sheet:

This section shall (at a minimum) contain the Revision Number, Date, Name of Approver, and Change Description.

2) Purpose/Scope:

This section outlines the procedure objectives, how they are in compliance with the Candu Energy Inc. requirements (e.g. are in compliance with the requirements of the governing Candu Energy Inc. technical specification, drawing(s), CSS, etc. It is designed to serve as a guide for a minimum acceptable level of packaging, shipping, and storage by the supplier.

3) Responsibility:

The Quality Control Inspector (QCI), along with the packaging/shipping/storage area manager, shall be responsible for ensuring that all packing, shipping, and storage activities will be performed in accordance with the governing Candu Energy technical specification, drawing(s), CSS, etc. and that all required documentation will be completed and available for review by Candu Energy Inc.

Personnel performing the packaging/shipping/storage activities must be judged competent to perform the tasks by the Quality Assurance (QA) Manager and shall be trained as per any Candu Energy Inc. requirements. Personnel training documents shall be maintained on file in the supplier's Quality Assurance Department.

4) General Requirements:

(Note that the Candu Energy technical specification always takes precedent – the supplier is to reconcile any differences (material, application, process, etc.).

- Level A, B, C or D as applicable will be specified as per ASME NQA-1, Subpart 2.2.
- The supplier will describe in detail the process for goods packaging that will satisfy the requirements that are necessary during shipping, handling and storage outlined in responding requirements for that Level, requirements listed in the TS and in the Purchase Contract.
- The supplier shall describe measures for packaging, shipping, receiving, storage and handling of specified “goods” assuring conformance to the distinct packaging /shipping requirements (e.g. by air, sea container or break-bulk) set out in the tender/purchase contract.
- The supplier shall describe how goods are to be preserved in a manner that shall ensure protection of loss, entry of dust, sand, water, mechanical and chemical damage, rust, etc. during shipment. This includes all measures that may be taken during handling and transport by carrier to ensure safe arrival of the Goods at their ultimate destination. The packaging shall provide in all instances means of handling by forklift trucks, slings and other conventional means of lifting and moving. This should include: contact preservative; type of dessicants; type of cushioning; type of blocking, bracing or anchoring.

6) Marking Requirements

All packages, boxes crates, bundles, and unpacked goods shall be clearly marked to show necessary shipping information. The marking shall be by means of waterproof paint or ink protected by clear shellac, and shall be applied by a method suitable to the type of product and packaging involved and shall be applied on two sides of the package as follows:

- Shipment number (as provided by Candu Energy Inc.’s order administrator or logistics representative)
- Destination
- Purchase Contract Number
- Package number and local number of packages
- QA Release Number
- Weight and Dimension of Package
- Storage Level
- Handling Instructions, i.e. Fragile, Keep Dry, This Side Up, Centre of Gravity, Sign Here, Stacking Limitations, etc. as required
- Special Instructions, etc. Desiccant Inside, Unpacking Instructions, etc.

4.9 Passivation

At a minimum, the content in the following bullets must be provided:

- Supplier/Company Name
- Candu Energy Inc. PO Number
- Supplier Procedure Title
- Supplier Procedure Number and Revision Number
- Supplier Procedure Issue Date
- Candu Energy Inc. Procedure Number and Revision Number
- Name of Item
- Material Type
- ASME or CSA Code Requirements*
- Governing Candu Energy Inc. Technical Specification (TS)*
- Governing Candu Energy Inc. Drawing Number(s)*
- Governing Candu Energy Inc. Component Specification Sheet (CSS)*
- Governing Candu Energy Inc. Valve Specification Sheet (VSS)*

**(if none indicate "NA")*

The procedure shall contain (at a minimum) the following sections:

1) Revision History/Revision Control Sheet:

This section shall (at a minimum) contain the Revision Number, Date, Name of Approver, and Change Description.

2) Purpose/Scope:

This section outlines the procedure objectives, how they are in compliance with the Candu Energy Inc. requirements (e.g. are in compliance with the requirements of the governing Candu Energy Inc. technical specification, drawing(s), CSS, etc.), and defines the limits of procedure application (e.g. applies to machined components, weldment, etc.). A brief description of the passivation method to be used shall be included.

3) Responsibility:

The Quality Control Inspector (QCI), along with the passivation area manager, shall be responsible for ensuring that cleaning, descaling, and passivation activities will be performed in accordance with the governing Candu Energy technical specification, drawing(s), CSS, etc. and that all required documentation will be completed and available for review by Candu Energy Inc.

Personnel performing the passivation activity must be judged competent to perform the task by the Quality Assurance (QA) Manager and shall be trained as per any Candu requirements. Personnel training documents shall be maintained on file in the supplier's Quality Assurance Department.

4) Methods/Processes:

This section provides the technical aspects of the procedure.

Note that the methods/processes employed must be in compliance with the requirements of the governing Candu Energy Inc. technical specification, drawing(s), CSS, etc. - the supplier is to reconcile any deviation from these requirements (materials, application, process, etc.). Supplementary reference can also be made to Technical Data Sheet(s) for any products suggested for use.

a) Pre-Cleaning and/or Cleaning Process Section

- Define the component to be pre-cleaned/cleaned and/or the component area to be pre-cleaned/cleaned (reference the Candu Energy Inc. technical specification, drawing(s), CSS, etc.).
- Pre-cleaning: Describe in detail which method(s) will be employed (wipe, solvent, alkaline, blasting, steam, etc.), reference any Candu Energy Inc. mandated cleaning requirements (example: ASME NQA-1 Subpart 2.1, ASTM E165 Section A1, etc.).
- Cleaning: Describe in detail which method(s) will be employed (wipe, solvent, alkaline, blasting, steam, etc.), reference any Candu Energy Inc. mandated cleaning requirements (example: ASME NQA-1 Subpart 2.1, ASTM E165 Section A1, etc.).
- Describe which chemical(s) (with concentrations) or materials (define the material) will be used. Technical data sheets for any chemicals proposed for use must be attached to the procedure and must provide detail on halogen and sulfur content. Cleaning materials containing deliberate additions of chlorine, fluorine or sulphur based ingredients are not permitted. All other organic cleaning solvents used for cleaning operations shall conform to the requirements for both halogens and sulphur content of paragraph T-641 of Article 6 of Section V of ASME BPVC for Liquid Penetrant Examination materials.
- Define the water quality to be used for chemical make-up and rinsing, reference any Candu Energy Inc. mandated requirement (example: ASME NQA-1 Subpart 2.1, Clause 304, etc.).
- Describe each step that comprises the work activity from start to finish, including relevant process temperatures and duration. Description of the process in a flow-sheet or process flow diagram outline is acceptable.
- Inspection method(s) to be employed – provide reference to a supporting inspection procedure or define the method(s) for inspection of the component and the acceptance criteria (see Section 6).
- Describe the steps taken to action non-conforming items found during the inspection process (see Section 6).

b) Descaling Process Section

- Define the component area to be descaled (reference the Candu Energy Inc. technical specification, drawing(s), CSS, etc.).

- Describe in detail which method(s) will be employed: mechanical (e.g. abrasive blasting, power brushing, sanding, grinding, etc.) and/or chemical (e.g. Pickling Solution Code A, B, C of Table A1.1 ASTM A380), reference any Candu Energy Inc. mandated descaling requirements.
- Describe which materials (define the material and provide iron concentration) and/or chemical(s) (with concentrations) will be used. Technical data sheets for any chemicals proposed for use must be attached to the procedure and must provide detail on halogen and sulfur content.
- Define the water quality to be used for chemical make-up and rinsing, reference any Candu Energy Inc. mandated requirement (example: ASME NQA-1 Subpart 2.1, Clause 304, etc.).
- Describe each step that comprises the work activity from start to finish, including relevant process temperatures and duration. Description of the process in a flow-sheet or process flow diagram outline is acceptable.
- Inspection method(s) to be employed – provide reference to a supporting inspection procedure or define the method(s) for inspection of the component and the acceptance criteria (see Section 6).
- Describe the steps taken to action non-conforming items found during the inspection process (see Section 6).

c) Passivation Process Section

- Define the component to be passivated and/or the component area to be passivated (reference the Candu Energy Inc. technical specification, drawing(s), CSS, etc.).
- Define the Candu Energy Inc. mandated standard practice or standard specification that will be followed for conducting the passivation process.^e Reference the Candu Energy Inc. technical specification, drawing(s), CSS, etc.
- Define which chemical(s) (with concentrations) will be used. Technical data sheets for all chemicals proposed for use must be attached to the procedure and must provide detail on halogen and sulfur content.
- Define the water quality to be used for chemical make-up and rinsing, reference any Candu Energy Inc. mandated requirement (example: ASME NQA-1 Subpart 2.1, Clause 304, etc.).
- Describe each step that comprises the work activity from start to finish, including relevant process temperatures and durations. Description of the process in a flow-sheet or process flow diagram outline is acceptable.
- Describe the inspection method(s) to be employed – provide reference to a supporting inspection procedure or define the method(s) for inspection of the component and the acceptance criteria (see Section 6).
- Describe the steps taken to action non-conforming items found during the inspection process (see Section 6).

^e Example: ASTM A967-05 “Standard Specification for Chemical Passivation Treatments for Stainless Steel Parts”

5) Performance Tests/Data:

Include any performance test(s) as required as per the governing Candu Energy Inc. technical specification or other supporting documentation. Reference the cleaning, descaling, or passivation product technical data sheet(s) performance data as required.

6) Inspection Criteria:

(a) Pre-Cleaning/Cleaning Process Section

The method of inspection shall be dictated by the governing Candu Energy Inc. technical specification, drawing(s), CSS, etc., examples of inspection methods are: Visual, Wipe Test, ASTM Test Method F22 (Water-Break Test), etc.

Reference other internal/external procedures, standards, or specifications as required.^f It is desirable (but not necessary) that a separate supplier inspection procedure be provided to augment the cleaning procedure (inspection procedures would be referenced in the cleaning section).

Describe how the method(s) of inspection shall be conducted.

Visual inspection method for cleaning shall include the following unless otherwise dictated by the governing Candu Energy Inc. technical specification, drawing(s), CSS, etc., or, by Candu Energy Inc.'s direction to use the manufacturer's cleaning product technical data sheet(s) inspection criteria:

- a) Part is to be inspected from a distance of no more than 2 feet.
- b) Angle of inspection – straight eye line inspection to 45° angle, in any direction.
- c) External surfaces – minimum illumination via shop floor lighting.
- d) Interior surfaces – minimum illumination via shop floor lighting augmented by a portable lighting device (e.g. flash light).

Any instrumentation used for determination of cleanliness quality measurements shall be calibrated prior to use (calibrated as per manufacturer's instructions) and have current calibration documentation available for inspection by Candu Energy Inc.

(b) Descaling Process Section

The method of inspection shall be dictated by the governing Candu Energy Inc. technical specification, drawing(s), CSS, etc., example of the inspection method is: Visual, etc.

Reference other internal/external procedures, standards, or specifications as required.

Describe how the method(s) of inspection shall be conducted.

Visual inspection method for descaling shall include the following unless otherwise dictated by the governing Candu Energy Inc. technical specification, drawing(s), CSS,

^f An example of a standard is ASTM Test Method F22 (produced by the American Society for Testing and Materials (ASTM)), a specification example is SSPC-SP1 (SSPC (The Society for Protective Coatings (formally known as the Steel Structures Painting Council))).

etc., or, by Candu Energy Inc.'s direction to use the manufacturer's descaling product technical data sheet(s) inspection criteria:

- a) Part is to be inspected from a distance of no more than 2 feet.
- b) Angle of inspection – straight eye line inspection to 45° angle, in any direction.
- c) External surfaces – minimum illumination via shop floor lighting.
- d) Interior surfaces – minimum illumination via shop floor lighting augmented by a portable lighting device (e.g. flash light).

Any instrumentation used for determination of descaling quality measurements shall be calibrated prior to use (calibrated as per manufacturer's instructions) and have current calibration documentation available for inspection by Candu Energy Inc.

(c) Passivation

The method of inspection shall be dictated by the governing Candu Energy Inc. technical specification, drawing(s), CSS, etc., examples of inspection methods are: ASTM A380-06 Clauses 7.2.5 and 7.3 (Water-Wetting and Drying, Copper Sulfate Test, Ferroxyl Test for Free Iron, etc.).

Reference other internal/external procedures, standards, or specifications as required. It is desirable (but not necessary) that a separate supplier inspection procedure be provided to augment the passivation procedure (inspection procedures would be referenced in the passivation section).

Describe how the method(s) of inspection shall be conducted.

Define which chemical(s) (with concentrations) will be used in the inspection process. Technical data sheets for all chemicals proposed for use must be attached to the procedure and must provide detail on halogen and sulfur content.

Any visual inspection method for passivation shall include the following unless otherwise dictated by the governing Candu Energy Inc. technical specification, drawing(s), CSS, etc., or, by Candu Energy Inc.'s direction to use the manufacturer's passivation product technical data sheet(s) inspection criteria:

- a) Part is to be inspected from a distance of no more than 2 feet.
- b) Angle of inspection – straight eye line inspection to 45° angle, in any direction.
- c) External surfaces – minimum illumination via shop floor lighting.
- d) Interior surfaces – minimum illumination via shop floor lighting augmented by a portable lighting device (e.g. flash light).

Any instrumentation used for determination of passivation quality measurements shall be calibrated prior to use (calibrated as per manufacturer's instructions) and have current calibration documentation available for inspection by Candu Energy Inc.

7) Acceptance Criteria:

Inspection test acceptance criteria shall be dictated by the governing Candu Energy Inc.

technical specification, drawing(s), CSS, etc., or, by Candu Energy Inc.'s direction to use the manufacturer's cleaning, descaling, and/or passivation product technical data sheets(s) inspection test acceptance criteria.

Define how the inspection test results will be interpreted against the test acceptance criteria (i.e. if the passivation effectiveness test is to determine whether rust appears after a wetting-drying test, the presence of rust would be non-compliant with the ASTM A380 residual surface iron requirements and be interpreted as a failed passivation process).

8) Non-Conforming Material:

The procedure must provide detail on how material found to be in non-conformance with the requirements of the governing Candu Energy Inc. technical specification or drawing(s), CSS, etc., or, by Candu Energy Inc.'s direction to use the manufacturer's cleaning, descaling, or passivation product technical data sheet(s) inspection criteria is dealt with.

Suggested non-conformance mitigation is:

- a) The plant supervisor/manager is notified of the non-conformance.
- b) The non-conforming material is tagged as non-conforming and the details documented (product identified, person rejecting the item, reason for rejection, disposition, and corrective action required).
- c) The corrective action taken to mitigate the non-conformance is documented.
- d) In the event that the corrective action does not meet the requirements of the Candu Energy Inc. Technical Specification (if provided) and/or the manufacturer's product Technical Data Sheet(s), the non-conformance is to be formally sent to Candu Energy Inc. for review and acceptance.

9) Reprocessing of Material:

The procedure must provide detail on how reprocessing of the non-conforming material will be performed: surface re-cleaning, re-descaling, re-passivation, re-inspection, and acceptance criteria for each stage.

10) Attachments:

The technical data sheets for all cleaning, descaling, and passivation solutions/materials used on the parts/equipment shall be attached to the procedure.

The technical data sheets for any instrumentation used for determination of cleanliness, descaling, or passivation effectiveness, or any other required quality measurement shall be attached to the procedure. If they are not available, the instrument(s) name and model data shall be provided.

Calibration details (calibration method used, date of last calibration, calibration frequency, and calibration standards used) for any instrumentation used for determination of cleanliness, descaling, or passivation effectiveness, or any other required quality measurement shall be attached to the procedure.

Any test reports to support the passivation activity shall be attached to the procedure.

4.10 Plating

At a minimum, the content in the following bullets must be provided:

- Supplier/Company Name
- Candu Energy Inc. PO Number
- Supplier Procedure Title
- Supplier Procedure Number and Revision Number
- Supplier Procedure Issue Date
- Candu Energy Inc. Procedure Number and Revision Number
- Name of Item
- Material Type
- ASME or CSA Code Requirements*
- Governing Candu Energy Inc. Technical Specification (TS)*
- Governing Candu Energy Inc. Drawing Number(s)*
- Governing Candu Energy Inc. Component Specification Sheet (CSS)*
- Governing Candu Energy Inc. Valve Specification Sheet (VSS)*

**(if none indicate "NA")*

The procedure shall contain (at a minimum) the following sections:

1) Revision History/Revision Control Sheet:

This section shall (at a minimum) contain the Revision Number, Date, Name of Approver, and Change Description.

2) Purpose/Scope:

This section outlines the procedure objectives, how they are in compliance with the Candu Energy Inc. requirements (e.g. are in compliance with the requirements of the governing Candu Energy Inc. technical specification, drawing(s), etc.), and defines the limits of procedure application (e.g. applies to machined components, castings, etc.). A brief description of the plating method(s) to be used shall be included.

3) Responsibility:

The Quality Control Inspection (QCI), along with the plating area manager, shall be responsible for ensuring that the plating activities will be performed in accordance with the governing Candu Energy Technical Specification, drawing(s), CSS, etc. and that all required documentation will be completed and available for review by Candu Energy Inc.

Personnel performing the plating activity must be judged competent to perform the task by the Quality Assurance (QA) Manager and shall be trained as per any Candu Energy Inc. requirements. Personnel training documents shall be maintained on file in the supplier's Quality Assurance Department.

4) Methods/Processes:

This section provides the technical aspects of the procedure.

Note that the methods/processes employed must be in compliance with the requirements of the governing Candu Energy Inc. technical specification, drawing(s), CSS etc. – the supplier is to reconcile any deviation from it (materials, application, process, etc.). Supplementary reference can also be made to Technical Data Sheet(s) for any products suggested for use.

- Define the component and/or the component area to be plated (reference the Candu Energy Inc. technical specification, drawing(s), or other internal/external procedure(s) or standard(s)^g if necessary).
- Define minimum plating thickness (or range) as per drawing, TS, etc.
- Describe which plating methods will be employed (tank plating, brush plating, flow-plating, etc.), reference any Candu Energy Inc. mandated plating requirements.
- Describe which chemical(s) (with concentrations) or materials (define the material) will be used. Technical data sheets for any chemical proposed for use as the final cleaning solution (e.g., acetone) must be attached to the procedure and must provide detail on halogen and sulfur content if mandated.
- Describe (or show in a sketch) location of any electrical connections to the parts being plated.
- Describe in detail the process of component disassembly if it is required.
- Describe each step that comprises the work activity from start to finish including pre-treatment of the surface, racking/masking and activation process to obtain acceptably sound adhesion of the plating. If the hydrogen baking post plating is specified, the baking temperature and duration must be provided, together with the transfer/standby time from the plating tank to the oven/furnace. Description of the process in form of flow-sheet or process flow outline is also acceptable.
- Describe the inspection methods to be employed – provide reference to a supporting inspection procedure or define the method(s) for inspection of the component and the acceptance criteria (see Section 6).
- Describe in detail the process of component reassembly if component disassembly had been performed.
- Describe the steps taken to action non-conforming items found during the inspection process (see Section 6). Also describe the stripping and re-plating processes if required.

5) Performance Tests/Data:

^g An example of a standard is ASTM B183 “Standard Practice for Preparation of Low-Carbon Steel for Electroplating” produced by the ASTM International (American Society for Testing and Materials).

Include any performance test(s) required as per the governing Candu Energy Inc., technical specification, or drawing(s).

6) Inspection Criteria:

The method of inspection shall be dictated by the governing Candu Energy Inc. technical specification, drawing(s), CSS, etc. (examples of inspection methods are: visual, hardness, bend or tape tests for evaluating the plating adhesion, etc.).

The inspection criteria should reference other internal/external procedures, standards, or specifications as required^h. It is desirable (but not necessary) that a separate supplier inspection procedure be provided to augment the plating procedure (the inspection procedure would be referenced in the plating procedure).

Describe how the method(s) of inspection shall be conducted.

Any instrumentation used for determination of cleanliness quality measurements shall be calibrated prior to use (calibrated as per manufacturer's instructions) and have current calibration documentation available for inspection by Candu Energy Inc.

7) Acceptance Criteria:

Inspection test acceptance criteria shall be dictated by the governing Candu Energy Inc. technical specification, drawing(s), CSS, etc., or, by Candu Energy Inc.'s direction to use associated standard inspection criteria.

Define how the inspection test results will be interpreted against the test acceptance criteria (i.e., if the inspection test is to determine the coating thickness and the use of non-destructive testing is not the option of the supplier, destructive methods such as microscopic measurements may be performed for thickness measurement on a separately plated sample processed at the same time as the production parts. The separate specimen shall be of the same basis metal as that of the production component.)

8) Non-Conforming Material:

The procedure must provide detail on how material found to be in non-conformance with the requirements of the governing Candu Energy Inc. technical specification or drawing(s), etc. is dealt with.

Suggested non-conformance mitigation is:

- a) The plant supervisor/manager is notified of the non-conformance.
- b) The non-conforming material is tagged as non-conforming and the details documented (product identified, person rejecting the item, reason for rejection, disposition, and corrective action required).

^h An example of a standard is ASTM Test Method B499 "Measurement of Coating Thicknesses by the Magnetic Method: Nonmagnetic Coatings on Magnetic Basis Metal (produced by the American Society for Testing and Materials (ASTM)).

- c) The corrective action taken to mitigate the non-conformance is documented.
- d) In the event that the corrective action does not meet the requirements of the Candu Energy Inc. Technical Specification (if provided), the non-conformance is to be formally sent to Candu Energy Inc. for review and acceptance.

9) Rework/Repair of Material:

Provide detail on how the rework will be performed: plating removal (stripping) process to be used, concentrations of any chemicals to be used, and inspection and acceptance criteria.

10) Attachments:

The technical data sheets for any instrumentation used for any required quality measurement shall be attached to the procedure. If they are not available, the instrument(s) name and model data shall be provided.

Calibration details (calibration method used, date of last calibration, calibration frequency, and calibration standards used) for any instrumentation used for determination of plating thickness or any other required quality measurement shall be attached to the procedure.

Any test reports to support the plating activity shall be attached to the procedure.

- C of C with the requirements of the governing Candu Energy Inc. technical specification (if provided) and/or other plating specification/standards shall be attached to the procedure.
- For non-destructive measurement of plating thickness, the coating thickness instrument Technical Data Sheet shall be provided. If not available, the instrument name and model shall be provided.
- The coating thickness instrument calibration details shall be provided (calibration method used, date of last calibration, calibration frequency, and calibration standards used).

4.11 Preservation

At a minimum, the content in the following bullets must be provided:

- Supplier/Company Name
- Candu Energy Inc. PO Number
- Supplier Procedure Title
- Supplier Procedure Number and Revision Number
- Supplier Procedure Issue Date
- Candu Energy Inc. Procedure Number and Revision Number
- Name of Item
- Material Type
- ASME or CSA Code Requirements*
- Governing Candu Energy Inc. Technical Specification (TS)*
- Governing Candu Energy Inc. Drawing Number(s)*
- Governing Candu Energy Inc. Component Specification Sheet (CSS)*
- Governing Candu Energy Inc. Valve Specification Sheet (VSS)*

**(if none indicate "NA")*

The procedure shall contain (at a minimum) the following sections:

1) Revision History/Revision Control Sheet:

This section shall (at a minimum) contain the Revision Number, Date, Name of Approver, and Change Description.

2) Purpose/Scope:

This section outlines the procedure objectives, how they are in compliance with the Candu Energy Inc. requirements (e.g. are in compliance with the requirements of the governing Candu Energy Inc. technical specification, drawing(s), CSS, etc., and defines the limits of procedure application (e.g. applies to machined components, castings, etc.). A brief description of the preservation method(s) and products to be used shall be included. Note that preservation requirements for a component may or may not extend to include its storageⁱ. Direction on storage procedure content direction is provided in the Packaging, Shipping, and Storage section.

3) Responsibility:

The Quality Control Inspector (QCI), along with the preservation area manager, shall be responsible for ensuring that preservation activities will be performed in accordance with the governing Candu Energy technical specification, drawing(s), CSS, etc. and that all required documentation will be completed and available for review by Candu Energy Inc.

ⁱ Per ASME NQA-1-2008 Subpart 2.2, Clause 101; Storage is defined as the act of holding items in storage facilities, which are defined as a warehouse, yard, or other areas designated and prepared for holding of items.

Personnel performing the preservation activity must be judged competent to perform the task by the Quality Assurance (QA) Manager and shall be trained as per any Candu Energy Inc. requirements. Personnel training documents shall be maintained on file in the supplier's Quality Assurance Department.

4) Methods/Processes:

This section provides the technical aspects of the procedure.

Note that the methods/processes employed must be in compliance with the requirements of the governing Candu Energy Inc. technical specification, drawing(s), CSS, etc. - the supplier is to reconcile any deviation from these requirements (materials, application, process, etc.). Supplementary reference can also be made to Technical Data Sheet(s) for any products suggested for use.

- Define the component to be preserved and/or the component area to be preserved (reference the Candu Energy Inc. technical specification, drawing(s), CSS, etc.).
- Describe in detail the process of component disassembly (if required).
- Describe in detail which preservation method(s) will be employed (preservation coating, inert gas, silica gel, vapor corrosion inhibitors, wraps, etc.), and reference any Candu Energy Inc. mandated preservation requirements.
- Define the component cleanliness requirements prior to application of the preservation material(s) and reference any Candu Energy Inc. mandated preservation requirements.
- Describe any cleaning method(s) to be employed and provide reference to a supporting cleaning procedure or describe in detail the process for component cleaning. Reference other internal/external procedures, standards, or specifications as required. It is desirable that a separate supplier inspection procedure be provided to augment the cleaning procedure (the inspection procedure would be referenced in the cleaning procedure).
- If a separate cleaning procedure is not being referenced, describe in detail which method(s) will be employed (wipe, solvent, alkaline, blasting, steam, etc.) and reference any Candu Energy Inc. mandated cleaning requirements (example: ASME NQA-1 Subpart 2.1, ASTM E165 Section A1, etc.).
- Define which chemical(s) (with concentrations) or materials (define the material) will be used for any cleaning activity. Technical data sheets for any chemicals proposed for use must be attached to the procedure and must provide detail on halogen and sulfur content. Cleaning materials containing deliberate additions of chlorine, fluorine or sulphur based ingredients are not permitted. All other organic cleaning solvents used for cleaning operations shall conform to the requirements for both halogens and sulphur content of paragraph T-641 of Article 6 of Section V of ASME BPVC for Liquid Penetrant Examination materials.
- Define the water quality to be used for cleaning chemical make-up and rinsing, reference any Candu Energy Inc. mandated requirement (example: ASME NQA-1 Subpart 2.1, Clause 304, etc.).

- Define the preservation product(s)/material(s) that will be used, the technical data sheets for each must be attached to the procedure and must provide detail on halogen and sulfur content.
- Describe each step that comprises the preservation activity from start to finish. Description of the process in a flow-sheet or process flow diagram outline is acceptable.
- Describe the inspection method(s) to be employed to verify preservation material(s) placement, thickness, concentration, etc. Provide reference to a supporting inspection procedure or define the method(s) for inspection of the component and the acceptance criteria (see Section 6).
- Describe in detail the process of component reassembly (if required).
- Define the storage requirements (if any) for the preserved component (environmental conditions, structural supports, etc.); reference the Candu Energy Inc. technical specification, drawing(s), CSS, etc.
- Describe in detail how the preservation product(s)/material(s) are to be removed from the components, provide specifics on how this will be done for internal cavities (if applicable). Define which chemical(s) (with concentrations) or materials (define the material) would be used for the preservation material removal activity. Technical data sheets for any chemicals proposed for use must be attached to the procedure and must provide detail on halogen and sulfur content.
- Describe the steps taken to action non-conforming items found during the inspection process (see Section 6).

5) Performance Tests/Data:

Include any performance test(s) required as per the governing Candu Energy Inc. technical specification or drawing(s) and/or the coating product Technical Data Sheet(s).

6) Inspection Criteria:

The method of inspection shall be dictated by the governing Candu Energy Inc. technical specification, drawing(s), CSS, etc. (examples of inspection methods are: visual, thickness, etc.).

Reference other internal/external procedures, standards, or specifications as required.^j It is desirable (but not necessary) that a separate supplier inspection procedure be provided to augment the preservation procedure (the inspection procedure would be referenced in the preservation procedure).

Describe how the method(s) of inspection shall be conducted.

Visual inspection method for preservation shall include the following unless otherwise dictated by the governing Candu Energy Inc. technical specification, drawing(s), CSS, etc., or, by Candu Energy Inc.'s direction to use the manufacturer's preservation product technical data sheets(s) inspection criteria:

^j An example of a standard is ASME NQA-1, Quality Assurance Requirements for Nuclear Facility Applications.

- a) Part is to be inspected from a distance of no more than 2 feet.
- b) Angle of inspection – straight eye line inspection to 45° angle, in any direction.
- c) External surfaces – minimum illumination via shop floor lighting.
- d) Interior surfaces – minimum illumination via shop floor lighting augmented by a portable lighting device (e.g. flash light).

Any instrumentation used for determination of preservation quality measurements shall be calibrated prior to use and have documentation of calibration available for inspection by Candu Energy Inc.

7) Acceptance Criteria:

Inspection test acceptance criteria shall be dictated by the governing Candu Energy Inc. technical specification, drawing(s), CSS, etc., or, by Candu Energy Inc.'s direction to use the manufacturer's preservation product technical data sheets(s) inspection test acceptance criteria.

Define how the inspection test results will be interpreted against the test acceptance criteria (i.e. if the inspection test is to determine the preservation product coating thickness on the component at 6 different locations, and the acceptance criteria is that the coating thickness shall be minimum 1.5 mm, then a measured thickness of minimum 1.5 mm at all tested locations would be interpreted as an adequate preservation coating thickness).

8) Non-Conforming Material:

The procedure must provide detail on how material found to be in non-conformance with the requirements of the governing Candu Energy Inc. technical specification or drawing(s), CSS, etc., or, by Candu Energy Inc.'s direction to use the manufacturer's preservation product technical data sheet(s) inspection criteria is dealt with.

Suggested non-conformance mitigation is:

- a) The plant supervisor/manager is notified of the non-conformance.
- b) The non-conforming material is tagged as non-conforming and the details documented (product identified, person rejecting the item, reason for rejection, disposition, and corrective action required).
- c) The corrective action taken to mitigate the non-conformance is documented.
- d) In the event that the corrective action does not meet the requirements of the Candu Energy Inc. technical specification (if provided) or drawing(s), CSS, etc., the non-conformance documentation is to be formally sent to Candu Energy Inc. for review and acceptance.

9) Rework/Repair of Material:

Provide detail on how the rework will be performed: cleaning/preservation/removal chemicals/materials to be used, concentrations of any chemicals used, removal/cleaning/re-application of product, inspection process, and acceptance criteria for each stage.

10) Attachments:

The technical data sheets for all preservation product(s)/material(s) to be used shall be attached to the procedure.

The technical data sheets for any instrumentation used for determination of a preservation product/material parameter or any other required quality measurement shall be attached to the procedure. If they are not available, the instrument(s) name and model data shall be provided.

Calibration details (calibration method used, date of last calibration, calibration frequency, and calibration standards used) for any instrumentation used for determination of a preservation product/material parameter or any other required quality measurement shall be attached to the procedure. Any test reports to support the preservation activity shall be attached to the procedure.

4.12 Surface Treatment

The surface treatment processes covered in this article are intended for metallic components to enhance their surface properties (wear/corrosion) and may include:

- Hardfacing
- Thermal spraying
- Carburization (or carburizing)
- Other case hardening (nitriding/malcomizing)
- Anodizing (or other chemical conversion coatings, e.g., black oxide treatment)

These techniques can be grouped into two broad categories:

- Techniques to build-up an inorganic surface layer of different composition and/or structure (hardfacing and thermal spraying).
- Techniques to modify an existing surface topographically, chemically, and/or microstructurally without significantly changing its dimensions (anodizing, and case hardening such as carburizing and nitriding/malcomizing).

Note the technical contents expected from other forms of common surface finishing procedures, plating (also known as electroplating or electrodeposition) and painting, are covered in separate sections of this document.

At a minimum, the content in the following bullets must be provided:

- Supplier Company Name
- Candu Energy Inc. PO Number
- Supplier Procedure Title
- Supplier Procedure Number and Revision Number
- Supplier Procedure Issue Date
- Candu Energy Inc. Procedure Number and Revision Number
- Name of Item
- Material Type
- ASME or CSA Code Requirements*
- Governing Candu Energy Inc. Technical Specification (TS)*
- Governing Candu Energy Inc. Drawing Number(s)*
- Governing Candu Energy Inc. Component Specification Sheet (CSS)*
- Other industry standard(s) referenced in the supplier surface treatment procedure (for example, the Aerospace Material Specification, AMS 2469, "Hard Anodic Coating Treatment of Aluminum and Aluminum Alloys Processing and Performance Requirements" used as a reference for anodizing requirements)

**(if none indicate "NA")*

The procedure shall contain (at a minimum) the following sections:

1) Revision History/Revision Control Sheet:

This section shall (at a minimum) contain the Revision Number, Date, Name of Approver, and Change Description.

2) Purpose/Scope:

This section outlines the procedure objectives, how they are in compliance with the Candu Energy Inc. requirements (e.g. are in compliance with the requirements of the Candu Energy Inc. Technical Specification, drawing(s), etc.), and defines the limits of procedure application (e.g., applies to machined components, castings, etc.). A brief description of the main surface treatment method employed shall be included.

3) Responsibility:

The Quality Control Inspection (QCI), along with the production area manager, shall be responsible for ensuring that the surface treatment activities will be performed in accordance with the governing Candu Energy Technical Specification, drawing(s), CSS, etc. and that all required documentation will be completed and available for review by Candu Energy Inc.

Personnel performing the surface treatment activity must be judged competent to perform the task by the Quality Assurance (QA) Manager and shall be trained as per any Candu Energy Inc. requirements. Personnel training documents shall be maintained on file in the supplier's Quality Assurance Department.

4) Methods/Processes:

This section provides the technical aspects of the procedure.

Note that the methods/processes employed must be in compliance with the requirements of the governing Candu Energy Inc. technical specification, drawing(s), CSS etc. - the supplier is to reconcile any deviation from these requirements (material, application, process, etc.). Supplementary reference can also be made to Technical Data Sheet(s) for any products suggested for use.

- Define the component and/or the component area to be treated for surface hardening (reference the Candu Energy Inc. technical specification, drawing(s), or other internal/external procedure(s) or standard(s) if necessary)
- Define the minimum surface layer thickness (or range) as specified in drawing, TS, etc.
- Define coating properties, such as case depth, hardness, etc. as specified in the drawings or TS.
- Describe which surface treatment method will be employed (see below examples), reference any Candu Energy Inc. mandated requirements.
 - Examples of thermal spray processes include flame spray, electric-arc spray, plasma spray, and HVOF.

- Examples of hardfacing processes include oxyfuel welding, powder torch welding, arc welding and laser welding.
- Examples of carburizing processes include (gas-, pack-, plasma- and vacuum carburizing).
- Examples of nitriding processes include (gas-, salt bath- and plasma carburizing).
- Examples of anodizing processes include chromic acid, sulfuric acid, and hard anodizing.
- Describe which chemical(s) (with concentrations) or materials (define the material) will be used. Technical data sheets for any chemical proposed for use as cleaning solution (e.g., acetone) must be attached to the procedure and must provide detail on halogen and sulfur content if mandated.
- Describe in detail the process of component disassembly if it is required.
- Describe each step that comprises the work activity from start to finish including pretreatment of the surface (vapor degreasing, alkaline soak cleaning, de-smutting etc.) and racking/masking for selective surface hardening treatment. Description of process in form of flow-sheet or process flow outline is also acceptable.
- Describe the inspection methods to be employed – provide reference to a supporting inspection procedure or define the method(s) for inspection of the component and the acceptance criteria (see Section 6).
- Describe in detail the process of component reassembly if component disassembly had been performed.
- Describe the steps taken to action non-conforming items found during the inspection process (see Section 6).

5) Performance Tests/Data:

Include any performance test(s) required as per the governing Candu Energy Inc., Technical Specification or drawing(s).

6) Inspection Criteria:

The method of inspection shall be dictated by the governing Candu Energy Inc. technical specification, drawing(s), CSS, etc. (Examples of inspection methods are: visual, hardness, bend or tape tests for evaluating the plating adhesion, etc.).

Reference other internal/external procedures, standards, or specifications as required^k. It is desirable (but not necessary) that a separate supplier inspection procedure be provided to augment the surface treatment procedure (the inspection procedure would be referenced in the surface treatment procedure).

^k An example of a testing standard is ASTM Test Method B499 “Measurement of Coating Thicknesses by the Magnetic Method: Nonmagnetic Coatings on Magnetic Basis Metal (produced by the American Society for Testing and Materials (ASTM)).

Describe how the method(s) of inspection shall be conducted. Any instrumentation used for determination of processed surface layer quality measurements shall be calibrated prior to use and have documentation of calibration available for inspection by Candu Energy Inc.

7) Acceptance Criteria:

Inspection test acceptance criteria shall be dictated by the governing Candu Energy Inc. technical specification, drawing(s), SCC, etc., or, by Candu Energy Inc.'s direction to use associated standard inspection criteria.

Define how the inspection test results will be interpreted against the test acceptance criteria (i.e., if the inspection test is to determine the hardened layer thickness and the use of non-destructive testing is not the option of the supplier, destructive methods such as microscopic measurements may be performed for thickness measurement on a separately sample processed at the same time as the production parts. The separate specimen shall be of the same base metal as the production component.

8) Non-Conforming Material:

The procedure must provide detail on how material found to be in non-conformance with the requirements of the governing Candu Energy Inc. technical specification or drawing(s), etc. is dealt with.

Suggested non-conformance mitigation is:

- a) The plant supervisor/manager is notified of the non-conformance.
- b) The non-conforming material is tagged as non-conforming and the details documented (product identified, person rejecting the item, reason for rejection, disposition, and corrective action required).
- c) The corrective action taken to mitigate the non-conformance is documented.
- d) In the event that the corrective action does not meet the requirements of the Candu Energy Inc. Technical Specification (if provided) or drawing(s), CSS, etc., the non-conformance documentation is to be formally sent to Candu Energy Inc. for review and acceptance.

9) Rework/Repair of Material:

Provide detail on how the rework will be performed.

10) Attachments:

The technical data sheets for any instrumentation used for determination of hardened layer thickness or any other required quality measurement shall be attached to the procedure. If the technical data sheets are not available, the instrument(s) name and model information shall be provided.

Calibration details (the method used, date of last calibration, calibration frequency, and calibration standard(s) used) for any instrumentation used for determination of hardened layer thickness or any other required quality measurement shall be attached to the procedure.

Any test reports to support the surface treatment activity shall be attached to the procedure such as Certificate of compliance with the requirements of the governing Candu Energy Inc. technical specification (if provided) and/or other surface treatment specification/standards shall be attached to the procedure.

4.13 Welding

At a minimum, the content in the following bullets must be provided:

- Supplier/Company Name
- Candu Energy Inc. PO Number
- Supplier Procedure Title
- Supplier Procedure Number and Revision Number
- Supplier Procedure Issue Date
- Candu Energy Inc. Procedure Number and Revision Number
- Name of Item
- Material Type
- ASME or CSA Code Requirements*
- Governing Candu Energy Inc. Technical Specification (TS)*
- Governing Candu Energy Inc. Drawing Number(s)*
- Governing Candu Energy Inc. Component Specification Sheet (CSS)*
- Governing Candu Energy Inc. Valve Specification Sheet (VSS)*

**(if none indicate "NA")*

The procedure shall contain (at a minimum) the following sections:

1) Revision History/Revision Control Sheet:

This section shall (at a minimum) contain the Revision Number, Date, Name of Approver, and Change Description.

2) Purpose/Scope:

This section outlines the procedure objectives, how they are in compliance with the Candu Energy Inc. requirements (e.g. are in compliance with the requirements of the governing Candu Energy Inc. technical specification, drawing(s), CSS, etc.), and defines the limits of procedure application. A brief description of the welding method(s) to be used shall be included.

3) Responsibility:

The Quality Control Inspector (QCI), along with the welding area manager, shall be responsible for ensuring that welding activities will be performed in accordance with the governing Candu Energy technical specification, drawing(s), CSS, etc. and that all required documentation will be completed and available for review by Candu Energy Inc.

Personnel performing the welding activity must be judged competent to perform the task by the Quality Assurance (QA) Manager and shall be trained as per any Candu requirements. Personnel training documents shall be maintained on file in the supplier's Quality Assurance Department.

4) Methods/Processes:

This section provides the technical aspects of the procedure.

Note that the methods/processes employed must be in compliance with the requirements of the governing Candu Energy Inc. technical specification, drawing(s), CSS, etc. - the supplier is to reconcile any deviation from these requirements (materials, application, process, etc.). Supplementary reference can also be made to Technical Data Sheet(s) for any products suggested for use.

Welding Procedure Specification (WPS)

Each manufacture and contractor shall prepare written Welding Procedure Specification (WPS). The WPS shall reference the supporting Procedure Qualification Record(s) PQR.

- a) The welding procedure specification shall describe, within the limits defined in paragraph on essential variables, nonessential, and when required, supplementary essential variables as per ASME Code IX, listed in QW-250 through QW-280. Each welding variable shall be defined as per Article IV, Welding Data.
 - Joints as per QW-402
 - Base metals as per QW-403
 - Filler metals as per QW-404
 - Position as per QW-405
 - Preheat as per QW-405
 - Post weld heat treatment as per QW-407
 - Shielding/purging gas as per QW-408
 - Electrical characteristics as per QW--409
 - Welding technic as per QW-410
- b) Reference shall be made to the following standards:
American Society of Mechanical Engineers, Boiler and Pressure Vessel Code:
 - Section II-Part A and Part C
 - Section III-Division 1
 - Section IX -Welding Qualification
 - Any other engineering specification required by Candu Energy Inc. technical documentation.
- c) The information required to be in the WPS may be in any format to fit the needs of manufacturer or contractor, as long as every variable outlined in QW-250 through QW-280 is included or referenced.
- d) Form QW-482 ASME IX (Non-mandatory Appendix B) may be used for the SMAW, SAW, GMAW and GTAW, for the rest of welding processes this form may be adjusted accordingly.
- e) A welding procedure for Code production welding shall be available for reference and review by the Authorized Inspector at the fabrication site.

Procedure Qualification Record(s) (PQR)

- a) A PQR shall document all essential and, when required, supplementary essential welding data described in the responding WPS that are used to weld a test coupon. Welding variables shall normally be recorded in a small range of actual variables that are used in production welding.
- b) Nonessential or other variables may be recorded at the manufacture's or contractor's request.
- c) The PQR shall be certified by manufacturer or contractor. This certification verifies that the information in the PQR is true record of the variables used during the welding of the test coupon and that the resulting test resulting tensile, bend, or macro test results are in compliance with ASME Section IX.
- d) Form QW-483 ASME IX (Non-mandatory Appendix B) may be used as a guide for the PQR.
- e) The information required to be in the PQR may be in any format to fit the needs of manufacturer or contractor, as long as every variable outlined in QW-250 through QW-280 is included or referenced.
- f) The type of tests, number of tests, and test results shall be listed in the PQR.
- g) PQR used to support WPS shall be available, upon request

4.14 Weld Repair

At a minimum, the content in the following bullets must be provided:

- Supplier/Company Name
- Candu Energy Inc. PO Number
- Supplier Procedure Title
- Supplier Procedure Number and Revision Number
- Supplier Procedure Issue Date
- Candu Energy Inc. Procedure Number and Revision Number
- Name of Item
- Material Type
- ASME or CSA Code Requirements*
- Governing Candu Energy Inc. Technical Specification (TS)*
- Governing Candu Energy Inc. Drawing Number(s)*
- Governing Candu Energy Inc. Component Specification Sheet (CSS)*
- Governing Candu Energy Inc. Valve Specification Sheet (VSS)*

**(if none indicate "NA")*

The procedure shall contain (at a minimum) the following sections:

1) Revision History/Revision Control Sheet:

This section shall (at a minimum) contain the Revision Number, Date, Name of Approver, and Change Description.

2) Purpose/Scope:

This section outlines the procedure objectives, how they are in compliance with the Candu Energy Inc. requirements (e.g. are in compliance with the requirements of the governing Candu Energy Inc. technical specification, drawing(s), CSS, etc.), and defines the limits of procedure application. A brief description of the weld repair method(s) to be used shall be included.

3) Responsibility:

The Quality Control Inspector (QCI), along with the welding area manager, shall be responsible for ensuring that weld repair activities will be performed in accordance with the governing Candu Energy technical specification, drawing(s), CSS, etc. and that all required documentation will be completed and available for review by Candu Energy Inc.

Personnel performing the weld repair activity must be judged competent to perform the task by the Quality Assurance (QA) Manager and shall be trained as per any Candu requirements. Personnel training documents shall be maintained on file in the supplier's Quality Assurance Department.

4) Methods/Processes:

This section provides the technical aspects of the procedure.

Note that the methods/processes employed must be in compliance with the requirements of the governing Candu Energy Inc. technical specification, drawing(s), CSS, etc. - the supplier is to reconcile any deviation from these requirements (materials, application, process, etc.). Supplementary reference can also be made to Technical Data Sheet(s) for any products suggested for use.

Welding Repair Procedure Specification (WPS)

Each manufacture and contactor shall prepare written Welding Procedure Specification (WPS). The WPS shall reference the supporting Procedure Qualification Record(s) PQR.

- a) The welding procedure specification shall describe, within the limits defined in paragraph on essential variables, nonessential, and when required, supplementary essential variables as per ASME Code IX, listed in QW-250 through QW-280. Each welding variable shall be defined as per ASME Article IV, Welding Data.
- b) The welding data shall include:
 - Joints as per QW-402 with drawing for detailed joint repair design.
 - Base metals as per QW-403
 - Filler metals as per QW-404
 - Position as per QW-405
 - Preheat as per QW-405
 - Post weld heat treatment as per QW-407
 - Shielding/purging gas as per QW-408
 - Electrical characteristics as per QW-409
 - Welding technic as per QW-410
- c) Reference shall be made to the following standards: American Society of Mechanical Engineers, Boiler and Pressure Vessel Code:
 - Section II-Part A and Part C
 - Section III-Division 1
 - Section IX -Welding Qualification
 - The project specific production drawing and any other engineering specification required by Candu Energy Inc. technical documentation.
- d) The information required to be in the WPS may be in any format to fit the needs of the manufacturer or contractor, as long as every variable outlined in QW-250 through QW-280 is included or referenced.
- e) Form QW-482 ASME IX (Non-mandatory Appendix B) may be used for the SMAV, SAW, GMAW and GTAW, for the rest of welding processes this form may be adjusted accordingly.

- f) A welding procedure for Code production welding shall be available for reference and review by the Authorized Inspector at the fabrication site.

Procedure Qualification Record(s) (PQR)

- a) A PQR shall document all essential and, when required, supplementary essential welding data described in the responding WPS that are used to weld a test coupon. Welding variables shall normally be recorded in a small range of actual variables that are used in production welding.
- b) Nonessential or other variables may be recorded at the manufacture's or contractor's request.
- c) The PQR shall be certified by manufacturer or contractor. This certification verifies that the information in the PQR is true record of the variables used during the welding of the test coupon and that the resulting test resulting tensile, bend, or macro test results are in compliance with ASME Section IX.
- d) Form QW-483 ASME IX (Non-mandatory Appendix B) may be used as a guide for the PQR.
- e) The information required to be in the PQR may be in any format to fit the needs of the manufacturer or contractor, as long as every variable outlined in QW-250 through QW-280 is included or referenced.
- f) The type of tests, number of tests, and test results shall be listed in the PQR as per provision of the ASME IX and responding requirements stipulated in the ASME III Division 1, Subsection NB, NC, ND, NE, NF, NG and NH.
- g) PQR used to support WPS shall be available, upon request.

Additional requirements for repair welding shall be provided:

- Weld examination and weld defect removal will be completed according to the provision of the ASME Section III, Division 1. Detail instructions for the elimination of any defect and subsequent repair weld examination shall be in compliance with the requirements stipulated in the relevant Subsection of the ASME III, Division 1, such as: NB NC, ND, NE, NF, NG and NH.
- Supplementary information containing specific requirements that are relevant to the each repair scenario (such as, instruction for placing layers, their thickness, cleaning instruction, girding of reinforcement layer, etc.).



Specification

Contractor's Status Reporting

Procurement

142-02120-SP-249

Revision 0

2011 November

novembre 2011

Candu - Proprietary

This document and the information contained in it is the property of Candu Energy Inc. (Candu) and must be safeguarded in accordance with Candu requirements and in a manner generally consistent with the Treasury Board requirements. These requirements include but may not be limited to keeping the document secure when not in use, accessibility of the information to pre-screened personnel, and special handling, storage and destruction measures. No use, disclosure, exploitation or transfer of any information contained herein is permitted except in accordance with Candu requirements or Candu's prior written consent.

Le présent document et l'information qu'il contient sont la propriété d'Énergie atomique du Canada limitée (Candu) et doivent être protégés conformément aux exigences Candu et, de façon générale, à celles du Conseil du Trésor. Ces exigences comprennent, entre autres, la garantie de sécurité du document lorsqu'il n'est pas utilisé, la restriction de l'accès aux renseignements aux personnes qui ont obtenu la cote de sécurité appropriée et les mesures de manipulation, d'entreposage et de destruction spéciales. Les renseignements qu'il contient ne peuvent être utilisés, ni divulgués, ni exploités ni communiqués sauf conformément aux exigences Candu L ou à l'autorisation préalable écrite Candu .

© Candu Energy Inc.

2285 Speakman Drive
Mississauga, Ontario
Canada L5K 1B1

© Candu Energy Inc.

2285, rue Speakman
Mississauga (Ontario)
Canada L5K 1B1



Revision History

Liste de révisions

Candu - Proprietary

Page 1 of / de 1

Document No. / Numéro de document:

142	02120	SP	249
Doc. Collection ID ID de la collection de doc.	SI Répertoire du sujet	Section	Serial No. N° de série

Document Details / Détails sur le document

Title Titre	Total no. of pages N ^{bre} total de pages
Contractor's Status Reporting	6

For Release Information, refer to the Document Transmittal Sheet accompanying this document. / Pour des renseignements portant sur la diffusion, consultez la feuille de transmission de documents ci-jointe.

Revision History / Liste de révisions

Revision / Révision		Details of Rev. / Détails de la rév.	Prepared by Rédigé par	Reviewed by Examiné par	Approved by Approuvé par
No./N°	Date				
0	2011/11/11	Issued as "Approved for Use".	L. Costantini		D. Vella

TABLE OF CONTENTS

SECTION		PAGE
1.	SCOPE	1
2.	REPORTING SEQUENCE AND CONTENTS.....	1
2.1	Title Page	1
2.2	Summary	1
2.3	Material	2
2.4	Tooling.....	2
2.5	Manufacturing Status	2
2.6	Photographs.....	2
2.7	Testing.....	2
2.8	One Month Look Ahead	2
3.	FREQUENCY OF REPORTING	2

1. SCOPE

Monthly Contract Status Reports are a requirement of this tender and are identified in the General Contract Conditions. Monthly Contract Status Reports are considered part of the Equipment Documentation for the purposes of assessing "Liquidated Damages".

2. REPORTING SEQUENCE AND CONTENTS

2.1 Title Page

The Title Page shall include the following:

- a) Project Name
- b) Name of the Equipment
- c) Contract Number
- d) Date of the Status Report

2.2 Summary

The summary will be divided into two sections.

- a) The "Technical Status Summary".
- b) The "Schedule Status Summary".

The Technical Status Summary shall be a summary of the design, development, procedure status, and drawing status.

The procedure and drawing status should include:

- a) A dialogue on both the design and development status as well as any problem areas in submittal or acceptance of the Contractor's documents. Include a statement indicating percent of Drawings and Procedures submitted and percent accepted.
- b) The Candu Software Schedule must be kept current by the Contractor and submitted monthly as part of the Status Report.
- c) A list, description and status of all DDR's (Deviation Disposition Requests).
- d) A Quality trend report summarizing current or projected Quality issues including, but not limited to as applicable;
 - 1) QA Organization changes, if any,
 - 2) critical QA, inspection and test issues,
 - 3) forthcoming major Witness, Hold and QA release points,
 - 4) status of major inspections and tests.

The Schedule Status Summary shall include:

- a) A double bar chart, one line showing the original barchart schedule and the line directly below that one showing the Contract status at the time of publishing and:
- b) A dialogue giving the status of the overall Contract with special notes explaining any delays to the schedule shown in a) above.

2.3 Material

This section will include:

- a) a list of all material required to be purchased for the Contract with predict and actual dates for the receipt of the material;
- b) a list of the Free Issue Material (if applicable) required for the Contract with required receipt dates and actual receipt dates for such material.

2.4 Tooling

This section will include the status of any special tooling required to be made for the manufacturing of the equipment in the Contract.

2.5 Manufacturing Status

This section will consist of a brief dialogue describing the manufacturing status along with any problems in manufacturing that may affect the schedule.

2.6 Photographs

The Purchaser requires photographs of Equipment at various stages of manufacturing up to and including preparation and packaging for shipment. [Photographs should be approximately 15 cm (6 inch) x 10 cm (4 inch) and in colour.] Photographs may be taken when the Equipment is away from machines, jigs, fixtures etc. so that only the Equipment is shown. With the photographs, include a very brief description of the Equipment and its present stage in the manufacturing process. The Contractor shall discuss with the Purchaser the frequency and format of submitting such photographs.

2.7 Testing

This section shall include a description and schedule for the design and procurement of any special testing equipment required for the Contract as well as a dialogue of the status of testing of the equipment in the Contract.

2.8 One Month Look Ahead

This section shall include a summary of Work to be done on the Contract in the following month.

3. FREQUENCY OF REPORTING

Status Reports including photographs are required monthly and shall be submitted within the first 10 days of each month and shall cover the preceding month.

NOTE: Invoices will not be paid if Monthly Contract Status Reports including photographs are not submitted on time and in sufficient detail to enable the Purchaser to assess the status and progress of the work.



DOCUMENTATION, PACKAGING, MARKING AND SHIPPING INSTRUCTIONS

Procurement & Commissioning Procedures and Documents

142-02131-SP-001

Revision 0

Prepared by
Rédigé par

A handwritten signature in black ink, appearing to read "Bruce Morrison".

Morrison, Bruce F - MANAGER,
LOGISTICS

Reviewed by
Vérifié par

A handwritten signature in black ink, appearing to read "Chris Mlinaric".

Mlinaric, Chris - MANAGER,
SUPPORT SERVICES

Approved by
Approuvé par

A handwritten signature in black ink, appearing to read "Ed Ruiter".

Ruiter, Ed -

2012/01/05

**OFFICIAL
USE ONLY**

2012/01/05

**À USAGE
EXCLUSIF**

Candu Energy Inc.

2285 Speakman Drive
Mississauga, ON
L5K 1B1 Canada

Candu Energy Inc.

2285 Speakman Drive
Mississauga, ON
L5K 1B1 Canada



Specification

Documentation, Packaging,
Marking and Shipping
Instructions

Procurement & Commissioning Procedures and Documents

**142-02131-SP-001
Revision 0**

2012 January

janvier 2012

**OFFICIAL USE
ONLY**

**À USAGE
EXCLUSIF**

This document and the information contained in it is the property of Candu Energy Inc. No use, disclosure, exploitation or transfer of any information contained herein is permitted in the absence of an agreement with Candu Energy Inc., and the document may not be released without the written consent of Candu Energy Inc.

Le présent document et l'information qu'il contient sont la propriété de Candu Energy Inc. Le présent document ne peut être transmis sans l'autorisation écrite de Candu Energy Inc. et les renseignements qu'il contient ne peuvent être utilisés, ni divulgués, ni exploités ni communiqués sans entente à cet effet avec Candu Energy Inc.

Candu Energy Inc.
2285 Speakman Drive
Mississauga, ON
L5K 1B1 Canada

Candu Energy Inc.
2285 Speakman Drive
Mississauga, ON
L5K 1B1 Canada



Revision History

Liste de révisions

OFFICIAL USE ONLY À USAGE EXCLUSIF

Page 1 of /de 1

CE-511300-FM-168 Rev. 0

Ref. Procedure CE-511300-PRO-161

Document No. / Numéro de document:

142	02131	SP	001
Doc. Collection ID ID de la collection de doc.	SI Répertoire du sujet	Section	Serial No. N° de série

Document Details / Détails sur le document

Title Titre	Total no. of pages Nbre total de pages
Documentation, Packaging, Marking and Shipping Instructions	56

For Release Information, refer to the Document Transmittal Sheet accompanying this document. / Pour des renseignements portant sur la diffusion, consultez la feuille de transmission de documents ci-jointe.

Revision History / Liste de révisions

Revision / Révision		Details of Rev. / Détails de la rév.	Prepared by Rédigé par	Reviewed by Examiné par	Approved by Approuvé par
No./N°	Date (yyyy/mm/dd)				
D1	2011/11/11	Issued for Review & Comment	B. Morrison	R. Prosser C. Mlinaric	
0	2012/01/05	Issued as “Approved for Use”.	B. Morrison	E. Ruiter C. Mlinaric	E. Ruiter

TABLE OF CONTENTS

SECTION	PAGE
1. PURPOSE.....	1
1.1 Introduction	1
1.2 Definitions	2
2. CONFLICTING REQUIREMENTS	3
3. GUIDELINES FOR PACKAGING	4
3.1 Minimum Preparation Requirements by the Supplier/Project.....	4
3.2 General Packaging Instructions	4
3.3 Types of Packaging	5
3.4 Wooden Case	5
3.5 Wooden Crates.....	7
3.5.1 Metal and PVC Barrels.....	10
3.5.2 Wooden Drums (Wooden Reels)	10
3.5.3 Equipment on Saddles and Skids	10
3.5.4 Twenty (20) and Forty (40) feet not-Returnable Shippers Owned Containers	11
3.5.5 Bundle and Loose GOODS	11
3.6 Supply Categories	11
3.6.1 Machinery (pumps, compressors, turbines, engines, ventilators including blowers, reduction gears, bridge cranes).....	11
3.6.2 Cables and any Kind of Material Wound on Drum	12
3.6.3 Pipes and Tubes	13
3.6.4 Fittings	14
3.6.5 Valves	15
3.6.6 Gaskets	15
3.6.7 Equipment (vessels, columns, boilers, etc.) and Skids (complete plant package).	15
3.6.8 Steel Structures	16
3.6.9 Linkage, Nuts and Bolts	16
3.6.10 Plates	16
3.6.11 Electrodes	17
3.6.12 Tiles and Cements	17
3.6.13 Air Coolers.....	17
3.6.14 Electrical and Telecommunications Goods (electric motors, transformers, power-battery, telephone equipment, electrical equipment in general).	17
3.6.15 Instruments (Control and safety valves, instrument and control panels, analysers, digital indicators, automotive valves, electronic computer systems, instrumentation in general, etc.)	18

TABLE OF CONTENTS

SECTION	PAGE
3.7 Packaging Inspection	19
3.8 Packaging & Packaging Waste (European Union only)	19
4. MARKING INSTRUCTIONS	20
5. PACKING LIST GENERAL REQUIREMENTS.....	24
6. STORAGE RECOMMENDATIONS.....	26
6.1 Storage Levels	26
6.2 Vapour Barrier Recommendations	26
7. OCEAN FREIGHT TRANSPORTATION.....	28
7.1 Ocean Containers	28
7.2 Break-Bulk Cargo (Conventional)	29
8. AIR FREIGHT.....	30
9. HAZARDOUS MATERIALS & DANGEROUS GOODS.....	33
10. HEAVY LIFTS / OVERSIZED CARGOES (ABNORMAL LOADS)	35
11. DOMESTIC PACKAGED SHIPMENTS	36
12. INTERNATIONAL SHIPMENTS.....	37
12.1 For shipments destined to Canada:	37
12.2 For other International shipments	38
13. EXPORT PERMITS	39
14. ATTACHMENTS.....	40

FIGURES

Figure 1 AWS type container	32
Figure 2 AVE type container.....	32

APPENDICES

Attachment A Case for Weights up to 500 kg	41
Attachment B Case for Weights up to 4,000 kg.....	42
Attachment C Case for Weights up to 20,000 kg.....	43
Attachment D International Handling Symbols	44

TABLE OF CONTENTS

SECTION	PAGE
Attachment E International Symbols	45
Attachment F Hazardous International Symbols SAFETY MARKS	46
Attachment G Aircraft Pallet Sizes	48

1. PURPOSE

1.1 Introduction

This instruction covers the general requirements for standard Packaging & Marking for all types of Goods (equipment, materials & tooling) - for international shipment as well as for Goods to be stored for an extended period. It is designed to serve as a guide to a minimum acceptable level of Packaging by our suppliers, and by our Projects as they return goods upon completion of Project use.

Supplier or Project published specifications may supersede these instructions as long as the minimum Candu Energy Inc. requirements are met and they have been approved by Candu Energy Inc. for each project.

The following instructions are intended as minimum requirements, and adherence to these instructions in no way absolves or relieves the Supplier/Project of any responsibility or obligation outlined elsewhere (i.e. in a Purchase Order).

Methods used shall be such as to warrant safe preservation to protect goods against environmental induced corrosion, damage and or degradation during the storage, handling and delivery of the commodity to its ultimate destination at the lowest reasonable cost.

Responsibilities will be defined for the Supplier/Project, throughout this Documentation, Packaging, Marking and Shipping Instruction (GDPMSI), providing the minimum requirements for material preparation, preservation and seaworthy packing.

This GDPMSI will become an integral part of the conditions of all Purchase Orders generated by the Contractor.

The acceptance of the Purchase Order implies that the Supplier agrees to accept all conditions and will follow all instructions, set forth therein, and that the Supplier will reimburse the Contractor for any demurrage, penalties and fines, due to the Supplier's negligence to comply with these instructions. Aforementioned includes losses or damages due to use of inadequate protective materials, faulty or insufficient packing, improper marking and/or incorrect documentation.

Note that this instruction does not govern shipments of contaminated goods. Other procedures document the process for shipment of contaminated goods.

1.2 Definitions

Contractor	Candu Energy Inc. for the execution of the Project.
Forwarder	Supplier of Transportation services acting for and on behalf of Contractor. Forwarder may also provide Customs Clearance services.
Goods	All material and equipment, consumable or non-consumable, hazardous or non-hazardous, described within or not, or as part of the sub-items of the purchase order.
Hazardous Cargo	Dangerous cargo marked by Supplier such as explosives, flammable, corrosives, poisons, precisely and strictly in accordance with rules and regulations of the country of origin, as well as the International Maritime Dangerous Goods Code (IMDG) for ocean freight, or International Air Transport Association Regulations (IATA) for air freight.
Job-Site	The area where the Goods are to be delivered and if not the job-site then to a destination to be determined by the Contractor.
Package	Any outside enclosure such as a case, crate, barrel, drum, bundle or other such facility used to enclose or protect the items in transit.
Packing List	The shipping document detailing the description of Goods, shipping configuration of items, number of items contained within any Package, dimensions & weights for each Package and any other specific requirement (i.e. P.O. line item weight, etc.) if required.

2. CONFLICTING REQUIREMENTS

In case of conflicting requirements, the following items govern in descending order of precedence.

Agreements outlined in mutually approved minutes of meetings subsequent to the issuance of the purchase order.

Purchase Order and relevant revisions or supplements.

Specifications detailed herein.

3. GUIDELINES FOR PACKAGING

Packaging means to efficiently and effectively protect Goods, from the moment they leave the factory until they are installed on Job-Site or from the moment they leave a Project until they are required for future Project work.

Packaging shall be sufficient to withstand without limitation, rough handling during loading/unloading, sea, air and inland transport, and exposure to extreme temperature, salt, and precipitation during transit, and outdoor storage. Packaging includes pre-packaging treatment and protection activities such as application of anti-corrosion, coating, vacuum, peeling, desiccants, solvents, etc. Depending on their type, Goods are to be protected against both mechanical damage (shocks, rupture, breakage, loss) and corrosion (rain, salty atmosphere, sand, etc.).

In every instance the desiccant/moisture absorbent supplied within the Packaging must be suitable for outdoor storage during extreme temperature variances (+40 to -40 C) at the Contractor jobsite for a minimum of 6 (six) months. The Supplier/Project will ensure that locations and expiry date of the desiccant is clearly identified on each package to ensure change out of desiccant occurs.

3.1 Minimum Preparation Requirements by the Supplier/Project

Regardless of General Purchase Conditions, the Supplier/Project has ultimate responsibility to assure that Goods shall be adequately prepared to provide against entry of dust, sand, water, damage, theft, rust, etc. with inner Packaging and wrapping, prior to export boxing. This constitutes preparation and Packaging of the Goods which includes all pre-crating protection activities, such as application of anti-corrosion coatings, vacuum packing, desiccants, solvents, end caps, etc. (Specific lubricants types, inhibitors and other internal detail preservation requirements are the responsibility of the Supplier.

The sealing barrier, if any, shall be such that the Goods can be identified (e.g., by customs authorities) without breaking the sealing (This will not apply to foil bag).

3.2 General Packaging Instructions

In article below (3.3) is reported a brief summary on different allowable types of packing and materials to be used.

The metric system for weights and dimensional measurements shall be used where ever possible and depending on normal customs and practices of the shipping location.

Please note that "special" handling for certain types of equipment may be included in the Purchase Order. These specific packaging instructions will take precedence over requirements stated below. It remains the Supplier/Project's responsibility to supply this minimum or better and more suitable protection and/or packaging than as specified below, to ensure safe arrival of the Goods at their ultimate destination. Any variances from this must be mutually agreed upon by both the Supplier/Project and Contractor.

The Supplier will be responsible to identify products that will be harmed by the extreme weather and temperature changes, as well as those products require inside temperature controlled storage.

Goods of more than one Purchase Order shall not be packed together in one package.

Outside packaging MUST take into consideration risk of exposure to a variety of extreme climatic conditions for inland transport and short & long term storage at the Job-Site. This protection must be effective for a 12-month period after delivery on Job-Site.

Spare Parts shall be packed separately where feasible.

3.3 Types of Packaging

In accordance with the nature of the contents, the following types of Packaging shall be considered.

Inner or outer Packaging will not be returned, unless otherwise specified in the order.

The structural characteristic will be explained below, and details of each type of Packaging will then be listed.

Under no circumstances shall Fibreboard, Cardboard, or similar cartons be used as outside Packaging.

3.4 Wooden Case

The cases must always allow for handling by fork lift truck; they must therefore be provided with battens which allow the forks to pass at right angles beneath the longest side.

Base boards must be simply fined. The base must not be covered with tarred paper or polythene to allow for ventilation and water drain from the case.

In some instances, tarred paper may be allowed to avoid moisture from getting in through the bottom of the case.

The base structure (flats and boards) must be such as to equally distribute the weight of the apparatus.

The cases and crates are subdivided into three categories:

- weight of contents up to 500 kg
- weight of contents up to 2,000 kg
- weight of contents up to 20,000 kg (*)
- (*) In the case of exceptional and particularly cumbersome equipment (e.g. apparatus weighing over 20,000 kg), Supplier will consult Contractor to agree to the most appropriate type of Packaging.
- Cases up to 500 kg (reference: attachment A)

- | | |
|------------------------------|--|
| Carrying built crossed bars: | Where the contents permit cases may be without base slat, these will be replaced by carrying crossbars to permit movement with a fork lift truck and will also serve as a batten for lifting ropes. Minimum dimensions of the base bars: 6 x 10 cm. The carrying crossbars on the boarding and on the cover must have minimum dimensions of 10 x 2.5 cm. |
| Planking: | Planking must be 2 to 2.3 cm thick. |
| Base: | Base planks must be 3 cm thick. |
| Bands: | Each case must be bound with at least two bands. |
| Cover: | The cover must be lined internally with polyethylene sheet (top) and a sheet of tarred paper (bottom). |
- Cases up to 2,000 kg (reference: attachment B)

Base Blocks:	Cases must be provided with battens on the bottom parallel to the longitudinal access. The dimensions of the battens must be 12 x 12 cm, spacing approx. 60 cm. The section and the space between must be proportionate to the structural characteristics, dimensions and weight of the product. The ends of the slats must be bevelled at 45° to allow for the positioning of ropes.
Slats:	Slats 8 cm thick should be placed crossways beneath the beams, raising the Package for lifting ropes or to allow movement with a fork lift truck. The position and number of the slats will be determined in accordance with the length of the case and the relative centre of gravity. The slats must be about twice as wide as they are thick and may be fixed by screw (preferred), nails or staples.
Base:	The base boarding must be 4 cm thick and be nailed or bolted or screwed to the base crossbars.
Planks:	Planks must be laid vertically so as to achieve improved resistance to superimposed weights.

Stiffening struts: Stiffening struts must be placed immediately beneath the cover (10 x 4 cm minimum and 80 cm minimum spacing).

Since their objective is to protect the case against the pressure imposed by the lifting ropes, the dimension of the struts must be proportionate to the weight of the Package and to its size.

Cover: Horizontal bracing may be external or internal. The dimensions and spacing will be proportionate to the dimensions and weight of the case.

The cover must be lined internally with polyethylene sheet (top) and a sheet of tarred paper (bottom).

- Cases up to 20.000 kg (reference: attachment C).
They are similar to the previous category. Obviously they must be sturdier.

Planking: Planking must be 3 cm thick.

Bottom boards: Closed tight and of the following dimensions:

12 x 12 cm for weights up to 6,000 kg

15 x 15 cm for weights up to 9,000 kg

20 x 20 cm for weights up to 20,000 kg

Base: The base boarding must be at least 5 cm thick and be bolted or screwed to the base beams.

Cover: Horizontal bracing may be external or internal. Section 20 x 4 cm (80 cm max spacing) the cover of these cases must be supported by a lattice frame (section approx. 15 x 3 cm). The squares or rectangles in the lattice may not exceed 40/50 cm at the side, to obtain a sufficiently resistant cover.

The cover must be lined internally with polyethylene sheet (top) and a sheet of tarred paper (bottom).

3.5 Wooden Crates

General

All wood and/or lumber used in Packaging, pallets or for securing within containers must be heat treated and/or fumigated so that they meet with the requirements of the country of export and/or destination . The treated Packaging materials must be marked with a universally recognized symbol or another approved by the Canadian Food Inspection Agency (CFIA) or other internationally recognized organization, prior to any shipments being released, to demonstrate they meet with the requirements of the recipient country.

Recycled, remanufactured or repaired wood Packaging material should be re-certified and re-marked.

Plywood made from veneer which has been rotary cut, sliced or sawed, must be well seasoned, commercially dry and free from defects that would materially lessen the strength. Plywood must be at least 3 ply for boxes when cleated and at least 5 ply for boxes when nailed. Nailed boxes must be assembled with grain of outer plywood face in the direction of the longest faces of the box and securely nailed or fastened to corner posts or ends. Boxes exceeding 17 kg (35 pounds) must have 4 vertical corner posts or other equally suitable devices or fasteners.

All components of such material should be treated. Any and all costs associated as a result of non-compliance to phytosanitary regulations including the *ISPM 15 regulation* shall be to the seller's account.

Export and International

Supplier/Project shall abide by and be fully compliant with the International Standards for Phytosanitary Measures (ISPM #15) as described in The Guidelines for Regulating Packaging Material in International Trade set out by the Secretariat of the International Plant Protection Convention administered by the Food and Agriculture Organization (FAO) of the United Nations.

As an exporter, the Canadian law requires that importing countries phytosanitary import requirements be met as provided in policy D-01-5 of CFIA.

Import into Canada

Supplier/Project shall abide by and be fully compliant with The Entry Requirements for Wood Packaging Materials Produced in All Areas Other Than the United States and China (D-98-08) as set out by the CFIA. In the case where wood Packaging materials are produced in China the CFIA directive (D-98-10) is applicable. Canada has adopted the Regulatory Guidelines Established in International Standard for Phytosanitary Measures as set out by the FAO.

These regulations apply to all non-remanufactured and non-treated wood utilized for crates, pallets, skids, case, bracing, spacers, bearers, dunnage etc. except between Canada and the United States of America. Participating countries at the time include Canada, United States, Mexico, European Union, Korea, China Australia and New Zealand. Other countries must be verified by project.

The wood Packaging must be heat treated or fumigated using methyl bromide at rates specified in policy D-98-08 (CFIA).

The treatment or a process of treatment must be officially endorsed by the National Plant Protection Organization (NPPO) of the exporting seller.

The treated Packaging must be marked with a universally recognized symbol or another approved by the CFIA prior to any shipments being shipped by the seller.

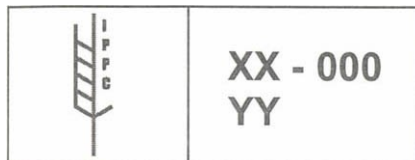
Recycled, remanufactured or repaired wood Packaging material should be re-certified and re-marked. All components of such material should be treated.

Marking

Seller/Project must ensure that the internationally approved mark to certify that the wood Packaging material that bears the mark has been subjected to an approved measure and is clearly visible on the crate, dunnage, case, pallet, etc... for approval of entry. A sample of the logo is available from CFIA and the FAO.

The mark shown below is to certify that the wood Packaging material that bears the mark has been subjected to an approved measure.

The mark should at minimum include the:



- Symbol
- ISO two letter country code followed by a unique number assigned by the NPPO to the producer of the wood Packaging material, who is responsible for ensuring appropriate wood is used and properly marked
- IPPC abbreviation for approved measure used:
 - HT-Heat Treated
 - MB- Methyl Bromide
 - KD-Kiln Dried
 - CPI-Chemical Pressure Impregnation
- Marking should be:
 - according to model shown above
 - legible
 - permanent and not transferable
 - placed in a visible location, at least two opposite sides of the article being certified.

Non-Compliance

Any costs associated with the inspection or with bringing a shipment into compliance is the responsibility of the Supplier. Failure to meet foreign import requirements may result in the refusal of entry of the shipment, destruction of the wood Packaging, suspension of trade and domestic penalties or fines. Any such costs will be to the seller's account.

Reference Material

Details of these regulations are available on the internet:

- a) Canada: www.inspection.gc.ca

- b) International: www.fao.org
- c) Other site: www.dcmnr.ie

3.5.1 Metal and PVC Barrels

Barrels with a capacity of less than 100 litres will be packaged in crates or arranged on pallets protected with a heat-shrunk cover and cross hoops.

Barrels with a capacity of/or greater than 100 litres will be sent individually.

Metal Strapping shall be of unannealed steel and applied with stretching tool and secured with crimped steel seals. Sizes of strapping utilized should follow latest practices developed by manufacturers depending on weight of package being strapped.

Nails should be cement coated or spiral type and should comply with latest issue of CSA Standards.

3.5.2 Wooden Drums (Wooden Reels)

Drums should be made of robust construction to afford effective protection during transport and handling. Wooden reels must be complete, in good condition and have been recently painted with waterproof paint.

Drums must be closed by staves nailed to the flanges with strengthening boards, 8 x 8 cm, placed laterally below the staves at distance of 50 cm.

For drums over 1.5 m in diameter, these reinforcing boards must be replaced by stays bolted to the flanges.

A sheet of tarred paper and a polystyrene sheet must be placed beneath the staves.

Drums must have at least 2 complete turns of strapping.

3.5.3 Equipment on Saddles and Skids

Openings in equipment shall be closed by metal sheets or wooden covers and painted a different colour. Sheets and covers shall be firmly attached to equipment or material, i.e. NO tape but preferably bolting.

All skid and saddle locations shall be indicated by the Supplier on the transport and lifting drawing (reference point 9) and approved by Contractor engineering department. Surface areas which may come into contact with slings during lifting are to be covered with wooden hurdles.

The equipment shall be fitted with suitable lifting lugs, or be properly marked as where to place the slings. The Centre of Gravity (CG) shall be indicated by the Supplier.

(For cargoes originating from the European Union, all lifting devices supplied, shall be officially Certified by a recognized authority. Subject certificate shall be sent to Contractor Inspection Department before the effective transport date).

3.5.4 Twenty (20) and Forty (40) feet not-Returnable Shippers Owned Containers

The use of ocean freight containers as Packaging (for sea transportation) shall be formally forbidden, except by Contractor's written authorization.

40' metal containers may be used for the consolidation and packaging of equipment. The contents of the containers must be properly blocked and braced in order to prevent any movement inside the container. Weight will be properly distributed in container.

Containers are to be new, or if previously used then they are to be free from dents and perforations and require a marine survey certificate certifying they are wind and water tight and conform to CSC standards. 20' containers are not generally recommended and will only be accepted if the Supplier requests in writing and receives Contractor's authorization to use twenty-foot containers.

The Contractor reserves the right to request also RINA approval.

3.5.5 Bundle and Loose GOODS

Only pipes, steel structures and plates can be packed in bundles or loose.

Each length of pipe of 150 mm or larger diameter shall be treated as an individual package and marked accordingly. Pipe smaller than 150 mm shall be bundled by strapping cleats or cradles of 50 x 100 mm lumber above and below the load. Ends of pipe must be capped if not in crates.

The Packaging detail will be shown in the point 3.4.

3.6 Supply Categories

Main supply categories listed below shall be packed in accordance with the types of Packaging shown in point 3.3.

3.6.1 Machinery (pumps, compressors, turbines, engines, ventilators including blowers, reduction gears, bridge cranes)

The type of Packaging used for machinery will be a wooden case.

In order to protect the machinery, heat-insulated bag must be used and, to maintain the humidity at 35%, dehydrating agents must be placed inside the bag.

All auxiliary threaded connections must always be hermetically sealed by means of threaded metal plug, and greased.

Machine finished surfaces shall be coated for protection against water and humidity for ocean shipment. All unfinished surfaces of machinery are to be given one coat of rust protecting paint.

All unprotected metal surfaces subject to corrosion or deterioration, shall be coated with a soft drying petroleum base preservation compound. Care shall be exercised in

applying preservation coatings so as to ensure that items fabricated of leather, mica, rubber and similar materials are not coated with the preservative compound.

All critical working surfaces such as close tolerance shafts, bearings, seals, and spindles, shall be protected from the entrance of dirt and moisture by wrapping with a greaseproof waterproof paper having a neutral pH factor. This wrapping shall be secured firmly in place with pressure sensitive tape or wax coated twine. All openings of electric motors, generators and other electrical equipment shall be sealed with pressure sensitive tape. Brushes and armatures of motors and generators shall be also wrapped with a piece of Grade A greaseproof paper.

VPI, VCI, and Vapour Barrier papers to be utilized where critical moving parts of machinery, are subject to rusting due to condensation. These uses are limited by the various metal content of certain products. (see section

Equipment such as electronics and other instrumentation shall be packed in an interior moisture-vapour-proof barrier with silica gel, or a comparable desiccant, to absorb moisture within the package and marked fragile. Where required equipment will be vacuum packed to ensure materiel is not damaged by elements during transport or storage period in tropical climates.

Openings in electric motors, generators and other electrical equipment shall be sealed with waterproof tape or in some equally efficient manner. A protective paper (Grade A - Greaseproof or equivalent) shall be inserted between the brushes and armature of motors and generators.

Fragile items shall be wrapped in crepe cellulose wadding or some equally efficient cushioning material and/or floated in excelsior. Fragile articles must be packed in substantial wooden cases with special precautions against risk of breakage.

3.6.2 Cables and any Kind of Material Wound on Drum

Drums: Cables and any kind of material wound on drum (flexible tubing i.e. hose, multiple tubing, copper cables and tubes) will normally be supplied on wooden drums (wooden reels).

Sizes must be those specified in the Data Sheet, and the tolerances allowed under the relevant standards must be observed.

Cases: Power and instrument cables and copper tubing wound in skein (small dimensions) should be packed in cases.

3.6.3 Pipes and Tubes

Steel pipes: Pipes of carbon steel, of low-alloy steel, galvanized, in diameter up to 1/2" and less than 2 mm in thickness will be packed in crates, the gross weight of each crate not to exceed 1,500 kg.

Crates must contain, so far as possible pipes of the same diameter, length, composition and thickness. Pipes are to be wrapped in polyethylene sheet and packed tightly inside the crate.

Pipes over 1/2" and up to 6" in diameter are to be made up into bundles that can easily be handled and stacked, with fastenings of twisted steel rope (4 strands) of a minimum thickness of 5 mm, positioned 1 m apart.

Internal pipe surfaces must be protected through closure of the two ends by plastic caps secured by adhesive tape. The bundles should contain pipes of the same diameter, thickness, composition and length.

Pipes over 6" in diameter will be despatched loose.

If necessary these pipes can be strengthened to prevent ovalization or deformation. The ends will be protected by pressure caps and taped or bolted to flange

Pipes in stainless steel, copper and aluminium must be packed in cases.

Pipes in PVC and glass fibre must be packed in wooden case protected against deformation during transport or ambient temperatures while stacked in the open.

Tubing for exchangers must be closed at the ends with wooden or plastic plugs, secured in bundles and packed in cases. The equipment must be wrapped in polyethylene sheet and held firmly in position inside the case.

Tubing for instruments will be packed in wooden cases secured by metal bands. Tubing with threaded ends must be provided with metal or plastic plugs to protect the thread.

3.6.4

Fittings

- Flanges:** All flanges must be protected against oxidation (corrosion for the stainless steel types) by means of special semi-permanent grease.
- Carbon steel flanges of small dimensions (up to 6") will be threaded in necklace fashion, placed in strong polyethylene or similar bags and packed in wooden cases (not exceed 1,000 kg in weight).
- Carbon steel flanges over 6" must be assembled in piles held together by at least tie-bolts, and packed in wooden crates.
- Discs of soft material must be interposed between the flanges.
- Stainless steel flanges must be pre-packed in the manner described for carbon steel flanges and packed in cases. The flanges will be wrapped in polyethylene sheet inside the cases.
- PVC and glass fibre flanges up to 4" will be packed in wooden cases and those of larger dimensions in wooden crates.
- General fittings:** All fittings made of carbon steel, drawn, forged and, in particular, threaded, must be protected against oxidation (corrosion for the stainless steel types) by means of special long lasting grease and bevels should be provided of appropriate protections.
- Fittings up to 6" will be threaded in necklace fashion, placed in strong polyethylene or similar bags and packed in wooden cases (not exceed 1,000 kg in weight). Each bag will contain items of the same type.
- Fittings over 6" will be arranged, unassembled, in wooden crates.
- All fittings in stainless steel, copper and aluminium alloy pre-packed as described for fittings in carbon steel and packed in wooden cases.
- Fittings in PVC and glass fibre, up to 6", will be packed in wooden cases and, those over 6" in wooden crates.
- Accessories for tubing:** All accessories for tubing (blinds, sealing, steam traps, hoses, spring supports, flow spy-hole, sprayers, etc. will be properly protected against corrosion with the application of a preservative, placed in polyethylene bags and packed in cases

3.6.5 Valves

- Steel Valves:** All valves must be protected against oxidization by means of special long lasting grease and flanges will be closed with laminated discs and rubber gaskets.
- Gate valves will be dispatched in the closed position, but plug cocks must be open.
- Valves of up to 4", as well as threaded necklace fashion and placed in polyethylene bags, will be in wooden cases.
- Valves of over 4" will be packed in wooden crates after hand wheels and moving parts have been protected by means of polyethylene hoods and cushioning materials. Valves will be packed vertically to prevent damage to hand wheels or the hand wheels can be removed
- Stainless steel valves:** Stainless steel valves will be protected in the manner prescribed for carbon steel valves and packed in wooden cases.

3.6.6 Gaskets

Gaskets will be grouped by type, material and size.

Metal gaskets and ring-joints will be treated with long-lasting grease and individually wrapped in waxed, passivating crepe paper. Small-diameter gaskets will be placed in ventilated polyethylene bags, duly identified, then in waterproofed boxes and finally packed in wooden cases. Large-diameter gaskets will be attached to wooden frames and packed in wooden cases. The cases must contain instructions regarding unpacking of contents.

3.6.7 Equipment (vessels, columns, boilers, etc.) and Skids (complete plant package).

If the package is transported already assembled there is no need for Packaging as such. It will be sufficient for any projecting of especially vulnerable parts which cannot be dismantled to be protected by means of boxing and binding. All threaded connections must be fitted with plug, greased and bound with tape and flanges must also be closed using steel discs and rubber gaskets bolted to the flanges.

For prefabricated sections consisting solely of metal structures, suitable strengtheners in wood or iron must be provided to prevent permanent warping and/or deformation during the various stages of handling and transport. Stays, beams, crosspieces and lifting hooks must be applied at the most suitable positions for prevention of damage.

Any equipment required for transport and/or to place on board or unload the above packages, such as spreaders, saddles, sized ropes, stiffening structures etc., is included in the supply and the above mentioned equipments will follow the Goods to the final destination.

3.6.8 Steel Structures

Steel structures involving structures over 5 mm thickness will be tied in bundles that can easily be handled and stacked, with fastenings of steel rope of a minimum thickness of 5 mm. Each bundle should be provided with eyebolts for lifting purposes.

Structural work, unless otherwise indicated, must be appropriately grouped, with individual pieces being pre-welded to ensure they are sufficiently robust.

Welding operations must not impair in any way the soundness and usability of the items being dispatched, and not be undertaken for structural items that have been galvanized, treated with preservatives or any other surface preparation.

Structural work of a thickness under 5 mm should be packed in wooden crates.

Small items, such as brackets, tacking plates, rings, hooks, etc., should be packed in wooden cases.

Steps iron will be dispatched as follows: the step elements will be tied into bundles and the safety caging will be assembled at the installation Job-Site, and the calendared elements should be tied into bundles separately from the uprights, which should also be tied in bundles.

3.6.9 Linkage, Nuts and Bolts

Stays, nuts, bolts, metal inserts, etc., unless protected by other stable agents (galvanized, etc.) will be greased, sorted into types, enclosed in polypropylene bags and packed in wooden cases. Items must be identified by means of a label tied onto the outside of the sack, and another label placed inside.

Small sized nuts and bolts should be placed in cartons and packed in wooden cases.

3.6.10 Plates

Raw plate and shell-plate:

Plates and shell-plates comprising a single package do not require any special Packaging. Planking or boards should be inserted as spacers between each plate to facilitate hoisting.

In stacking shell-plates, the first plate must rest on saddles or wedges to avoid any deformation.

Plates assembled in bundles must be held together with bands of stainless steel. If plates have no chamfers for welding, it is expedient to weld vertically pieces or straps onto the sides of the bundles to make them more compact. Wooden beams should be interposed between each bundle to facilitate lifting.

Stainless steel plates: Plates of this type must be packaged, secured with stainless steel bands and packed in wooden cases. A sheet of passivity or waxed paper must be laid between the plates.

3.6.11 Electrodes

Electrodes will be packed in wooden cases and pre-packed as follows:

Individual Packaging of electrodes will be marked with the denomination of the electrode they contain and its technical characteristics, wrapped in heat-shrunk polyethylene.

Packaging will then be placed inside heat-sealed bags into which moisture-absorptive agents have first been placed.

3.6.12 Tiles and Cements

Tiles: Ordinary and refractory tiles will be packed in wooden cases in a criss-cross pattern. Tiles will be protected by coverings of polyethylene, which will allow for the circulation of air. Cases must be strapped.

Refractory items in general must be packed in cases.

Cements: Ordinary and refractory cements must be placed in multilayer paper bags and then in a double bag of polyethylene and packed in wooden cases.

The volume of each Packaging should be one (1) m³ approx.

3.6.13 Air Coolers

Fin surfaces must be protected on both sides with a layer of galvanized sheet, suitably attached to the cleat of the bale. Finned tubes must be suitably separated from one another by spacers to protect them during loading and unloading operations.

The detached parts of refrigerating units, such as hubs, driving shafts, bearings, etc., must be packed in wooden cases. Before being placed in the cases, parts must be carefully greased and wrapped in anti-corrosive waxed paper.

3.6.14 Electrical and Telecommunications Goods (electric motors, transformers, power-battery, telephone equipment, electrical equipment in general).

All Goods should be packed in wooden cases. MIL-B-131-F Foil bags and moisture-absorptive agents will be used in all cases. To prevent the protective bag from being damaged, any sharp edges on the equipment must be padded. In case of vulnerable parts of equipment that cannot be treated with protective substances, self-adhesive waxed paper impregnated with passivity substances can be used.

The main instruments on boards, panels and control desks, which are likely to suffer damage during carriage, handling or storage, should be dismantled and packed in cases. Panel coupling elements for medium and low voltage distribution boards and motor control panels will be packed in separate cases.

Accumulator batteries, appliances and instruments made of brittle materials must be given a form of protection inside the cases.

Switchboards and control panels must be packed as follows:

- a) No case should contain more than three coupled panels or boards.
- b) Arcing contact parts, if positioned in the contact-breakers and isolators, must be firmly secured; if they are being dispatched separately, each must be packed individually inside the case to avoid breakage of the contact.

Switchboards, control panels and control desks should be packed in wooden cases, together with all their accessories.

Motors will normally be supplied without the couplings. Motors will be firmly bolted down to the bottom of the case, and suitably protected. Junction boxes must be placed with the connection and coupling output channel towards the bottom. The drive shaft of the motor must be greased and protected.

Fluorescent lamps should be packed separate from the frames. They must be protected against possible breakage during loading, unloading, and transport. Frames must be packed in wooden crates separate from the lamps.

Mercury-vapour lamps and incandescent lamps must be provided with protection inside the case by means of spacers of poly-sponge or other suitable material.

Shunt and connection boxes must be packed in cases.

Large transformers must be packed in crates. It must be possible to fasten a sling through the transformer eyebolts even after Packaging. Any dismantled parts must be empty of oil and packed in cases. Measuring and control instruments must be dismantled and packed in wooden cases. The transformer must be bolted down to the base and held securely inside the crate to prevent it from moving.

Small transformers up to 2000 kVA, must be packed in wooden cases.

3.6.15 Instruments (Control and safety valves, instrument and control panels, analysers, digital indicators, automotive valves, electronic computer systems, instrumentation in general, etc.)

Instruments equipment must be suitably packed in wooden cases.

Individual appliances must be packed in cases and enclosed in MIL-B-131-F Foil Bag. Apparatus must be cushioned before being placed in the bags.

Before the bags are sealed, moisture-absorptive agents must be inserted, and a partial vacuum must be created inside the bag.

Precision instruments must be wrapped in anti-corrosive waxed paper, padded and enclosed in protective bags in which moisture- absorptive agents have been placed; they must then be put inside a Packaging of foamed polystyrene of a suitable shape. The whole will then be placed in a water-resistant carton long-fibre natural craft board. The cartons must all be stowed next to one another inside a wooden case.

The Supplier of Goods is ultimately responsible to adhere to ALL regulatory requirements for Health, Safety and Environmental concerns of the Country of exportation and of importation (GSPLAJ), in regards all Goods used in export Packaging.

3.7 Packaging Inspection

Irrespective of the Delivery Term agreed in the Purchase Order, the preparation for export shipping (including Packaging) may be subject to inspection by Contractor or Contractor's representative.

The Supplier shall give the Contractor written notice of no less than ten (10) days of the time and place at which the Goods are to be packaged for shipment.

The client or appointed representative may be present at Supplier's shop/factory prior to shipment and at dockside prior to loading aboard ship. Inspection shall not relieve Supplier of any responsibilities or obligations under the terms and conditions of the Purchase Order. Contractor reserves the right to reject the Packaging found not adequate during inspection. In such cases, Contractor will ask Supplier to immediately repack the Goods in acceptable Packaging for account of Supplier. If the Packaging rejection occurs at the port of loading, the repackaging activities must be carried out at dockside in the notified working day and all costs to obtain entry authorization by Port Authorities will be the responsibility of Supplier.

To obtain such authorization Supplier may contact the Contractor's appointed Forwarder.

If any part of the Goods for any reason whatsoever has not been inspected during the fabrication or before Packaging, the Contractor shall have the right to inspect the Goods at the agreed delivery point, provided that Contractor notifies Supplier of its intent to do so by their written order to proceed with the Packaging.

3.8 Packaging & Packaging Waste (European Union only)

All parties involved will comply with the European Packaging & Packaging Waste legislation in force (directive number 94/62/EC of December 20, 1994) or when situated outside the European Union the local legislation in force will apply.

4. MARKING INSTRUCTIONS

Each case, bundle, crate or container which is larger than two cubic meters shall be stencilled with indelible ink over coated with clear preservative on two opposite sides other than the side on which it rests or it is intended to be stacked, with the following marks in five centimetres letters and numbers, all in the English language. Packages smaller than two cubic meters may use labels reduced in size, and if necessary, the label may be affixed using a tag.

When an individual shipment consists of multiple cases, skids, crates, containers or other then each shall be individually and sequentially numbered.

All boxes and crates shall bear the following markings:

- a) Name of Contractor
- b) Project Name, Contract and P.O. numbers
- c) Port of Destination
- d) Equipment Tag Number (to be specified by the Supplier)
- e) *Name of Supplier
- f) Seller's Ref. Number (to be specified by the Supplier)
- g) Case Number _____ of _____
(Continuous series)
- h) Dimensions in meters (width, length, height, to be specified by the Supplier)
- i) Volume in Cubic Meters (to be specified by the Supplier)
- j) Gross Weight in Kilograms (to be specified by the Supplier)
- k) Net Weight in Kilograms (to be specified by the Supplier)
- l) Colour Codes, when required (to be specified by the Supplier)

In accordance with the characteristics and different requirements in loading, unloading and shipping different Contract Equipment, the package shall be conspicuously marked with "Handle With Care", "Right Side Up", "Keep Dry", etc., in English and with appropriate international trade practice marks and illustrative marks, such as (Please refer to attachment D):

- a) Chains
- b) Indicates the place through which lifting chains are to be passed for loading and unloading. The chain shall be painted red.
- c) Cross Sign
- d) Indicates the centre of gravity of the case.
- e) Umbrella
- f) Indicates that equipment must be stored in a warehouse with extreme care.
- g) Glass Cup

- h) Indicates that the shipment is fragile and should be handled with extreme care.
- i) Slings Points
- j) Seller shall indicate sling points and/or furnish slinging diagram

To facilitate identification of the contents, in case of breakage or deterioration of packaging, the packed equipment or machine should be also marked on one of its sides, and, if this should not be possible, marks shall be stamped on corrosion resistant sheet metal or other suitable corrosion resistant material, which shall be fixed to the equipment in such a way, as to avoid being detached during handling.

In the case of goods which, because of their nature, are customarily supplied and accepted by the Contractor without packaging, marks and indications, shall be stamped on two sides of the equipment as described above.

All unpacked equipment, cases, parcels or bundles weighing more than 4 (four) metric tons, shall bear indications of their center of gravity using international symbol and of the appropriate places through which lifting braces are to be fixed for the adequate handling during loading, transfer and unloading operations.

All cases, crates and bundles weighing a metric ton or more shall have a mark, indicating the maximum load that can be placed on top of them.

One set of packing lists shall be in a waterproof bag affixed to the case, bundle, crate, or container and covered with a metal plate. This metal plate must be placed in the left top quarter of one of the largest surfaces and firmly affixed. One additional copy of the packing list shall be included inside the case, crate, or container.

International symbols are to be stencilled using indelible red ink over coated with clear preservative to prevent erosion. Any crate(s) identified as containing dangerous goods are to be prepared as a shipment separate from the main consignment and shall contain a separate set of packing lists. Appropriate markings identifying the contents, the product identification numbers, and any other markings required for the transport of dangerous goods shall be clearly affixed to or stencilled on the crate(s) on the two opposite sides other than the side on which the crate rests or is intended to be stacked. On cylinders the safety marks must appear on shoulder.

The colour to be used shall contrast with the package surface (never use yellow or red paint).

Black or white paint are recommended.

Packages/equipment of five tons or more must be marked with slinging points and also clearly show on all sides the centre of gravity (CG). This is of paramount importance for Pressure Vessels and Heat Exchangers.

Packages containing Hazardous Material / Dangerous Goods are to be marked on at least two sides with appropriate labels in conformity with IMDG/IATA regulations.

Chemicals/ Dangerous Goods shall be identified on a separate invoice, packing list and Bill of Lading. All hazardous Goods shall be identified by appropriate hazard class and technical/proper shipping name (Refer to attachment F).

All spare parts orders must be clearly marked as spares and shall not be included with main equipment (with exception).

Shipping marks must appear on all packages and all shipping documents such as packing lists, invoices.

Bundles and loose items which are not suitable for application of shipping marks directly onto bundles or pieces, must be supplied with at least 2 (two) indestructible labels, 1 (one) at each end, bearing shipping marks. Labels are tied firmly to bundles with pieces of steel wire.

Metal or other weatherproof tags shall be clearly legible and must invariably be affixed to bundles and single units at both ends. Do not use linen, paper or light cardboard tags.

Hand lettering with felt tip markers is not acceptable.

Precautionary statements will be stencilled on each package to indicate proper handling and storage.

MINIMUM REQUIREMENTS

-----60 cm x 40 cm-----

Marks:

Consignee	:	AAAA
Project Number	:	Job No. XXXX
Final Destination	:	BBBB
P.O. No.	:	
Supplier's Name	:	
Country of Origin	:	
Item No.	:	
Description of Material	:	
Package No.	:	1 of ..., 2 of ..., etc.
Gross Weight	:	Kg
Net Weight	:	Kg
Dimensions	:	cms (L x W x H)
Volume	:	CBM

Storage Recommendations:

Cargo Handling Symbol:
(Identify / describe)

Special Instructions:

Stencil on two sides and one end in clear characters at least 5 centimetres high (where case size permits, otherwise use optimum size for each package dimension).

For the identification of Goods subject to special handling, the following international symbols shall be used to ensure proper handling and warehousing (see Attachment 0):

Symbol	Description	Remarks (symbols to be shown on)
1	THIS SIDE UP	All Top corners
2	FRAGILE GOODS – HANDLE WITH CARE	All Top corners
3	KEEP DRY	All Top corners
4	PROTECT FROM SOLAR RADIATION	All Top corners
5	USE NO HOOKS	All sides
6	CENTRE OF GRAVITY	Two sides of packages over 2,000 kg
7	SLING HERE	All slinging points

Special remarks: symbols 15 cm high to be painted in black, on bright surface.

All boxes and crates are to be marked clearly on all four sides:

"HANDLE WITH CARE"

"RIGHT SIDE UP" and

"KEEP DRY"

5. PACKING LIST GENERAL REQUIREMENTS

The Supplier will not ship Goods and materials from the point of origin, regardless of the delivery terms included in the Purchase Order, until the final release for shipment is issued to the Supplier by the Contractor's Representative.

90 (Ninety) days prior to the Contractual shipping date, Supplier must supply the Contractor with 4 (four) copies of shipping drawings for each large piece exceeding 10,000 kilograms or 12.2m L x 3m W x 2.7m H meters.

The shipping drawing must show at least the following information;

- Purchase Order No.
- Item No.
- Overall Dimensions (length x width x height)
- Total Weight
- Centre of Gravity
- Lift Points
- Any Special Lifting Requirements (i.e., use of lifting beam)

Supplier must submit a "provisional packing list" (provisional packing list is to include estimated weight and dimensions of all packages) 30 days prior to scheduled shipping date which must include the following information:

- Confirmation of Purchase Order shipping date.
- Quantity
- Description
- Reference Tag Number (where applicable)
- Supplier's reference part number and serial number if applicable.
- Estimated gross weight (kilograms) of each shipping item (crate, skid)
- Length, width and height (meters) of each shipping item (crate, skid)

30 (Thirty) days prior to the Contractual shipping date, Supplier must furnish Contractor a copy of safety information for all easily combustible Goods, dangerous Goods and poisonous Goods.

These safety instructions must include the following information:

- Name of Goods
- Nature of Goods
- Measure of protection
- Emergency treatment
- Material Safety Data Sheets

- IMCO Code
- UN Product Code

30 (Thirty) days prior to the Purchase Order shipping date Supplier must furnish Contractor with a description of the "special precautions" to be taken for the equipment/material with special requirements for temperature, shock, etc., during the transportation and storage.

Upon the receipt of the release for shipment from the Contractor Inspection group, the Supplier will be obligated to submit the final packing list (c/w weight and dimensions) to the Contractor.

Any shipment that proceeds without an authorized order (release for shipment) will be subjected to charges not exceeding the freight charges.

Item numbers shall be listed accurately on the packing list to allow proper identification.

The packing list shall indicate details of the items packed, piece by piece, accessory by accessory (Purchase Order line item level). Indication of sets, lots or the like is not allowed.

Whenever packages are stuffed into containers, details of their contents shall be listed.

Monetary value shall never be indicated on any packing list.

One copy of the packing list shall be placed inside the box and two copies of it shall be fixed to the outside of the packing in a waterproof envelope protected by a plastic or aluminium cover, marked "packing list".

6. STORAGE RECOMMENDATIONS

6.1 Storage Levels

The Supplier's storage recommendations shall be marked on each Package and indicated on the packing list.

The symbols for the type of storage are:

- X or "A" = storage in heated/ventilated warehouse and with special precautions (air conditioning)
- XX or "B" = storage in closed warehouse
- XXX or "C" = storage in open shed
- XXXX or "D" = storage in open air (covered by tarpaulins)

If for any reason, Contractor is unable to accept dispatch of the Goods at the time when the Goods are due and ready for dispatch, Contractor shall so inform Supplier in writing and Supplier shall store the Goods, safeguard it, and take all steps to prevent its deterioration during such storage and Contractor shall be liable to Seller:

- a) if applicable, for that instalment of the Contract Price that would have been paid in accordance with the Contract had the Goods been dispatched and
- b) for the reasonable justified costs, including insurance of so doing, after a period of grace of 2 (two) months free storage (unless otherwise specifically provided).

6.2 Vapour Barrier Recommendations

a) For Six (6) Months Transit and Storage, the following products – or equivalent - shall be used:

- VAPPRO V-5 & V-10 VCI Foam Emitters (Product Codes V5E-6-3-24 & V10E-9-7-12):
 - Place in all enclosed areas of equipment
- VAPPRO 9868 RUST INHIBITOR (Product Code V9868-12-400):
 - Spray on all bare metal parts
- VAPPRO 826 VCI 6MIL FILM (Product Code V826-6-20-100):
 - Fully enclose all equipment, folding and stapling or heat sealing all edges.
- MIL-PRF-131K (Product Code TMF-275-48-600):
 - Wrap and hermetically seal all equipment

b) For Two (2) Years Transit and Storage, the following products – or equivalent - shall be used:

- VAPPRO V-5 & V-10 VCI Foam Emitters (Product Codes V5E-6-3-24 & V10E-9-7-12):
 - Place in all enclosed areas of equipment

- VAPPRO VCI POUCHES (Product Code VP-9-15-10):
 - Place inside VAPPRO 826 VCI 6MIL FILM, next to equipment
 - VAPPRO 9868 RUST INHIBITOR (Product Code V9868-12-400):
 - Spray on all bare metal parts
 - VAPPRO 870CB ELECTRO SPRAY (Product Code V870CB-12-500):
 - Spray all electrical and electronic components
 - VAPPRO 826 VCI 6MIL FILM (Product Code V826-6-20-100):
 - Fully enclose all equipment, folding and stapling or heat sealing all edges.
 - MIL-PRF-131K (Product Code TMF-275-48-600):
 - Wrap and hermetically seal all equipment
- c) The MIL-B-131 Type 1, Class1, vapour barrier shroud is a silver poly-propylene and aluminium foil which encapsulates the crated goods. This material is used in conjunction with humidity absorbing desiccant within the bag, then heat sealed and vacuum packed. The vacuum bag is used as well with a Humidity Indicator Plug that gauges moisture build up within the bag. These materials are used in both military and commercial applications where transit and storage conditions are long term and severe.
- d) Other effective preservation materials are Cortec products. These preservation materials are used in the prevention of rust, corrosion or oxidation. Recommended Cortec Products – or equivalent - are:
- VCI 105 Foam emitters for electronic cabinets,
 - VCI 126 blue or transparent film with VpCI for multimetal protection,
 - VCI 132 VpCI impregnated foam pads, multimetal, non-toxic and nitrite free, and
 - Cor-Pak VpCI Pipe Strips protecting ferrous or non-ferrous tubular products.

7. OCEAN FREIGHT TRANSPORTATION

7.1 Ocean Containers

Contractor will position the Shipping Company containers c/o Supplier's shop/factory. Unless otherwise agreed, to all costs for containerization of the Goods (stowing, securing, stuffing, etc.) will be for account of Supplier.

The material to be loaded into containers must be packed according to point 3.5.4.

Container inspection, loading, securing and limitations will be in accordance with International Standards and Maritime Law & Regulations, regardless of the country of export.

Material is to be bundled, boxed or crated restricting weight to a gross maximum of 4 metric tons, where feasible. After cleaning and preserving, all equipment shall be packed in waterproof paper lined export type shipping containers.

Large pumps, compressors, and other heavy machinery that will not be damaged by water or exposure may be packed in open crates or semi-boxed. All openings of pipe, large valves, and machinery must be protected by wooden or PVC covers or plugs, and the machined threads must be covered by caps or wrapping to protect against damage in transit.

Up to 225 kg

Boxes for weight of contents up to 225 kg shall be constructed with minimum 2.54 centimetres cleats and of nominal 2.54 centimetres lumber or equivalent plywood, and shall be strapped with two 1.90 centimetres unannealed flat steel strapping secured with crimped steel seals.

225 kg to 500 kg

Boxes for weight over 225 kg to 500 kg shall be constructed of nominal 2.54 centimetres lumber or equivalent and be modified to include a skid base of nominal 2.54 centimetres deck, nominal 5 centimetres x 10 centimetres lumber skids and minimum 5 centimetres x 10 centimetres frame members. Boxes shall be strapped with a minimum of two x 3 centimetres unannealed flat steel strapping secured with crimped steel seals.

501 kg to 4000 kg

Boxes for weight in excess of 500 kg shall have skid type bases. These skids shall be minimum nominal 7.5 centimetres x 10 centimetres or 10 centimetres x 10 centimetres lumber secured to headers with 1.25 centimetres machine bolts. The deck shall be of nominal 2.54 centimetres lumber or heavier. Framing members of the sides and ends shall be of nominal 5 centimetres x 7.5 centimetres or heavier. The sheathing shall be of nominal 2.54 centimetres lumber or plywood equivalent.

Top construction shall be of nominal 2.54 centimetres lumber with waterproof paper liner supported by minimum 5 centimetres x 10 centimetres lumber joists on edge, where the width of the case so requires, and cleared to the sides. The corners of

prefabricated panels shall be reinforced by the addition of 3 centimetres steel straps where necessary.

Over 4000 kg

Where the weight exceeds 4000 Kilos, all construction will be in accordance with above specification however material used is to be altered as the weight is increased.

Ventilation shall be provided in all shipping containers to permit free circulation of air by spacing of floor boards 0.63 centimetres to 0.48 centimetres apart and by drilling a cluster of holes in the sides of large cases upward from the outside at 45°. These should be a minimum of five such holes 1.25 centimetres in diameter.

Heavy machinery shall be mounted and bolted to the skid which shall be of sufficient strength to provide support and to prevent distortion.

TABLE OF SKID SIZES

Gross load Weight (kg)	Max. skid Length (m)	Runner size (centimetres)
225-450	2.44	5 x 10
451-1,000	4.88	7.5 x 10
1,001-5,000	6.10	10 x 10
5,001-10,000	9.15	10 x 15.25
10,001-15,000	9.15	15.25 x 15.25
15,001-25,000	9.15	15.25 x 20
25,001-36,000	10.67	20 x 20
36,001-45,500	10.67	20 x 25

All skids will be spaced no greater than 1.22 metre apart, centre to centre. The headers will be the same size as the skid runners and shall be bolted into each skid runner. The deck of the skid shall be a minimum of 5 centimetres lumber. All internal frame work shall be a minimum of 10 centimetres lumber and shall have diagonal bracing.

If the length exceeds the limit shown in the table above the next larger size is to be used.

7.2 Break-Bulk Cargo (Conventional)

In general, break-bulk cargo is any ocean cargo which is not containerized, exceeds the inside dimensions and weight limitations set for specific types of containers or otherwise designated as such, as per the delivery requirements specific to the project.

8. AIR FREIGHT

Cardboard Containers

Cardboard containers may only be used for airfreight, if they meet IATA packaging requirements. Otherwise, any material packaged in cardboard containers must be repackaged or enclosed in another crate or bundle.

Machinery Boxes

Material is to be bundled, boxed or crated restricting weight to a gross maximum of 2,720 kg. All openings of pipe, large valves, and machinery must be protected by wooden or PVC covers or plugs, and the machined threads must be covered by caps or wrapping to protect against damage in transit. Maximum size for boxes or crates is not to exceed 152.4 cm D x 274.3 cm W x 152.4 cm H.

Up to 225 kg

Boxes for weight of contents up to 225 kg shall be constructed with minimum 2.54 cm cleats and of nominal 2.54 cm lumber or equivalent plywood, and shall be strapped with two 1.9 cm unannealed flat steel strapping secured with crimped steel seals.

225 kg to 500 kg

Boxes for weight over 225 kg to 500 kg shall be constructed of nominal 2.54 cm lumber or equivalent and be modified to include a skid base of nominal 2.54 cm deck, nominal 5.08 cm x 10.16 cm lumber skids and minimum 5.08 cm x 10.16 cm frame members. Boxes shall be strapped with a minimum of two – 3.18 cm unannealed flat steel strapping secured with crimped steel seals.

501 kg to 2,720 kg

Boxes for weight in excess of 500 kg shall have skid type base. These skids shall be minimum nominal 7.5 centimetres x 10 centimetres lumber secured to headers with 1.25 centimetres machine bolts. The deck shall be of nominal 2.54 centimetres lumber or heavier.

Framing members of the sides and ends shall be of nominal 5 centimetres x 7.54 centimetres or heavier. The sheathing shall be of nominal 2.54 centimetres lumber or plywood equivalent.

Top construction shall be of nominal 2.54 centimetres lumber supported by minimum 5 centimetres x 10 centimetres joists on edge, where the width of the case so requires, and cleated to the sides. The corners of prefabricated panels shall be reinforced by the addition of 3 centimetres steel straps where necessary.

SKIDS

Heavy machinery shall be mounted and bolted to a skid which shall be of sufficient strength to provide support and to prevent distortion.

TABLE OF SKID SIZES

Gross load Weight (kg)	Max. skid Length (m)	Runner size (centimetres)
225-500	12.70 x 24.13	5 x 10
501-2,720	12.70 x 24.13	7.54 x 10

All skids will be spaced no greater than 1.22 metre apart, centre to centre. The headers will be the same size as the skid runners and shall be bolted into each skid runner.

The deck of the skid shall be a minimum of 2.54 centimetres lumber. All internal frameworks shall be a minimum of 5 centimetres lumber and shall have diagonal bracing.

Contoured Structural Container / Igloo

Contoured containers/Igloos shall be constructed as described below:

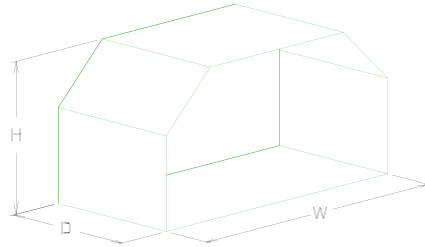


Figure 1: AWS type container

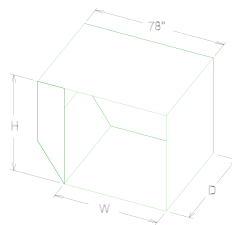


Figure 2: AVE type container

Class	Size	IATA
		Description
AWS	Full	Maximum gross weight (including tare) 2,720kg 1.55 m L x 3.17 m W x 1.62 m H Vol: 7 cubic meters
AVE	Half	Maximum gross weight (including tare) 1,590kg. 1.55 m L x 1.58 m W x 1.63 m H Vol: 4 cubic meters

Containers shall have skid type bases. These skids shall be a minimum nominal of 7.54 centimetres x 10 centimetres lumber secured to headers with 1.25 centimetres machine bolts. The deck shall be of nominal 2.54 centimetres lumber or heavier. Framing members of the sides and ends shall be of nominal 5 centimetres x 7.54 centimetres or heavier. The sheathing shall be of nominal 2.54 centimetres lumber or plywood equivalent. Top construction shall be of nominal 2.54 centimetres lumber supported by minimum 5 centimetres x 10 centimetres joists on edge and cleared to the sides. The corners of prefabricated panels shall be reinforced by the addition of 3.17 centimetres steel straps where necessary.

9. HAZARDOUS MATERIALS & DANGEROUS GOODS

Transportation of these good must comply with all International, Federal, Provincial, State and municipal regulations.

For any materials or residues which are deemed by the appropriate transport regulatory authorities to be hazardous or regulated cargo for transportation, the Supplier must provide the Contractor in writing, with advance notice, the following safety instructions:

- I. Name of goods
- II. Nature of goods
- III. Measure of protection
- IV. Emergency treatment
- V. UN Number
- VI. IMCO code
- VII. Safety Data Sheets
- VIII. Spreader Bar details c/w Certified Calculations

All hazardous goods must be packaged in accordance with appropriate transportation mode rules and regulations. Supplier must ensure all shipping units comply with the packing/packaging provisions of the International Maritime Dangerous Goods (IMDG) or International Air Transport Association (IATA) code, as appropriate to the mode of transportation. Supplier must ensure that hazardous materials, especially liquids, are in suitable containers that will allow use without spillage and can be satisfactorily resealed after a portion of the material has been extracted. (Re-closed containers should allow normal handling and transport without significant escape of liquids or vapours. Containers of 5 litres or more of hazardous, low-viscosity liquids such as paint thinners, chlorinated solvents, and insecticides should have a retractable metal or plastic spout or a similar design that will meet the stated objective).

Ensure all packages are properly and adequately marked and labelled in compliance with provisions of the IMDG or IATA code and properly display safety marks, product identification number(s), and any other information required by the regulations governing the transport of dangerous goods.

The Supplier must assist the Contractor in obtaining all necessary certificates and permits.

These requirements also affect the transport of solvents, pre-mixed paint, equipment with batteries, oil filled equipment, magnetized materials, pressurized equipment or containers, etc. The specific mode of transport being employed and the quantities involved may determine if the cargo is classified as dangerous or not.

Thirty (30) days before the consignment is due to be dispatched, the Supplier must send to the Contractor the following documents for each product:

- Multimodal dangerous Goods form for sea freight transport correctly completed and signed
- Shipper's declaration for airfreight transport correctly completed and signed
- Material safety data sheet

Hazardous material shall be identified on a **separate invoice and packing list** to those for non-hazardous material. All hazardous material shall be identified by the appropriate hazard class and technical or proper shipping name. All invoices and packing lists for hazardous material shall contain the following statement:

"This is to certify that the above-named material is properly classified, described, packaged, marked and labelled, and is in proper condition for transportation according to the appropriate Government or International Transport regulations."

_____ NAME	_____ COMPANY NAME
_____ TITLE	_____ SIGNATURE
_____ HAZARD CLASS NO	_____ TECHNICAL SHIPPING NAME

The Supplier shall produce any necessary hazardous cargo certificates, in accordance with the appropriate Government and International Transportation regulations for all shipments.

10. HEAVY LIFTS / OVERSIZED CARGOES (ABNORMAL LOADS)

Goods exceeding one of the following dimensions and/or weight are to be considered to be heavy lifts and/or oversized cargoes.

Loads

Length : 2 meter
Width : 2.50 meter
Height : 2.50 meter
Weight : 25 Tons

If the above-mentioned figures will be exceeded, Supplier must give advance notice to Contractor and shall present to Contractor transport and lifting drawing(s).

Plus showing, other information below:

- Purchase Order/Tag number.
- Weight and dimensions of each piece.
- Centre of Gravity (CG).
- Lifting lugs and/or sling locations.
- Size of lugs.
- Saddle locations (incl. width, height and weight).
- Any specific transport requirements.

Drawing packages are to be provided by Supplier (numbers to be determined on a case-by-case basis by the Contractor), a minimum of 90 (ninety) days prior to "Ready for Shipment" date.

Heavy-lifts and/or over-sized Goods shall be accompanied with "spreader bars" whenever required by the Contractor or the Job-Site. The Supplier will specify each item with lifting and sling points that will distribute the load equally and keep the item in a stable, horizontal position when lifted by one hook. All items up to 125 tons are to be designated for a single point lifting method.

11. DOMESTIC PACKAGED SHIPMENTS

10 (Ten) days prior to the contractual shipping date, Supplier must provide (via Courier Express or email) two copies of the shipping documents to the Contractor addressed as follows:

- Contractor (As agents or as representatives for – when applicable)
- Contractor's Buyer, (project office address)

Attention: Document Controller, Project: (project number and project name).

At time of shipment, a final packing list shall be prepared by Supplier for each package, showing the equipment / material description and appropriate tag number where applicable. **1 (One) copy** of the packing list shall be placed inside the shipping container / crate affixed to the equipment / material, **1 (One) copy** shall be secured in a weatherproof pouch firmly attached to the outside of each crate/pallet etc.

12. INTERNATIONAL SHIPMENTS

12.1 For shipments destined to Canada:

- a) 2 (Two) original copies of a Commercial Invoice with Harmonized System code per item,
- b) 1 (One) original copy of the NAFTA Certificate with HS code per item,
- c) If a NAFTA certificate is not available or not applicable, then two (2) original copies of a Certificate of Origin (if applicable).
- d) Packing List and/or a separate packing list for each shipment (in this latter case the Commercial Invoice must indicate 'partial shipment' and be issued for each individual shipment, the total of which must equal the total shipment quantity/value).
- e) Copy of truck bill, airway bill or bill of lading.
- f) 1 (One) copy of each of the above to be sent to the Logistics Lead of the project or to the Buyer at the given address via email or courier, if no Logistics Lead is assigned to the project.

The 'Importer of Record' should clearly show: *****

It is the responsibility of the equipment Supplier to define and supply the HS Code description on the Commercial invoice and on the NAFTA Certificate. If the Supplier is unable to supply or is unsure of the HS code to be used, they must then contact the Project appointed Customs Broker for assistance.

In order to comply with Canada Customs Non-Compliance regulations, the Commercial invoice must imperatively show the following elements:

- Supplier name and details.
- Invoicing address.
- Purchaser / Consignee's name and details (XXX location purchasing the Goods).
- Delivery address, if different from consignee's address.
- Other references such as invoice and purchase order numbers.
- Country of Origin (a blanket certificate of origin or an affidavit might be asked upon request to prove the origin of the Goods). The terms "Manufactured in" and "Made in" are the only acceptable terms.
- Mode of transportation.
- Currency of settlement.
- Number of packages
- Specification of commodities (kind of package, marks and numbers, general description and characteristics such as proper identifying description in commercial terms).

- Unit and total price paid or payable in the currency of settlement.
- Invoice total
- Total weight
- Originator: name of the person completing the invoice.
- Customs Broker details and contact

12.2 For other International shipments

The following minimum documentation will be required for the clearance of all commercial shipments irrespective of the mode of shipment. This documentation is to be prepared and supplied by the SUPPLIER:

- a) Commercial Invoice
 - 1 (One) original and 2 (two) copies destined to Customs Broker.
 - 1 (One)) copy destined to Contractor's Representative.
- b) Certificate of Origin
 - 1 (One) original and 2 (two) copies destined to Customs Broker.
 - 1 (One) copy to Contractor's Representative

These documents must be sent to the following parties' indicating the Project name and Project number:

- Contractor (As agents or as representatives for – when applicable)
- Contractor's Buyer, (project office address)
- Document Controller
- Contractor's appointed Freight Forwarder

All documents must be received by the Customs Broker and Contractor Representative 15 (fifteen) days prior to the schedule date of arrival of the shipment. All documentation shall be addressed according to instructions issued by the Contractor.

Documentation for international shipments will vary depending on the importing Country requirements and existence of special trade arrangements between Country Nations.

Specific or additional documentation needs and completion instructions will be provided to the Supplier prior to any shipment being agreed upon and will be included as part of the Purchase Order (Special) Terms and Conditions.

In order to ensure that the above documents are verified properly in time for shipment, the original copies should be couriered. If the documents are not correctly completed the Supplier will be advised that new documents are needed. Shipments will not be expedited until all Import related documents are completed and in good order.

13. EXPORT PERMITS

The following requirements on Export Permits are applicable to all Projects where goods are being purchased.

The Supplier must advise the Contractor of all Export Permits required for the export of the Goods covered by the Purchase Order.

This is a condition of the validity of the Purchase Order that the required Export Permits for the Goods be issued in time for the scheduled delivery, failing which the Purchase Order shall irrevocably be deemed null and void, and Contractor shall in such event not be liable to the Supplier for any damages nor compensation.

The Supplier shall at its own cost obtain the required Export Permits.

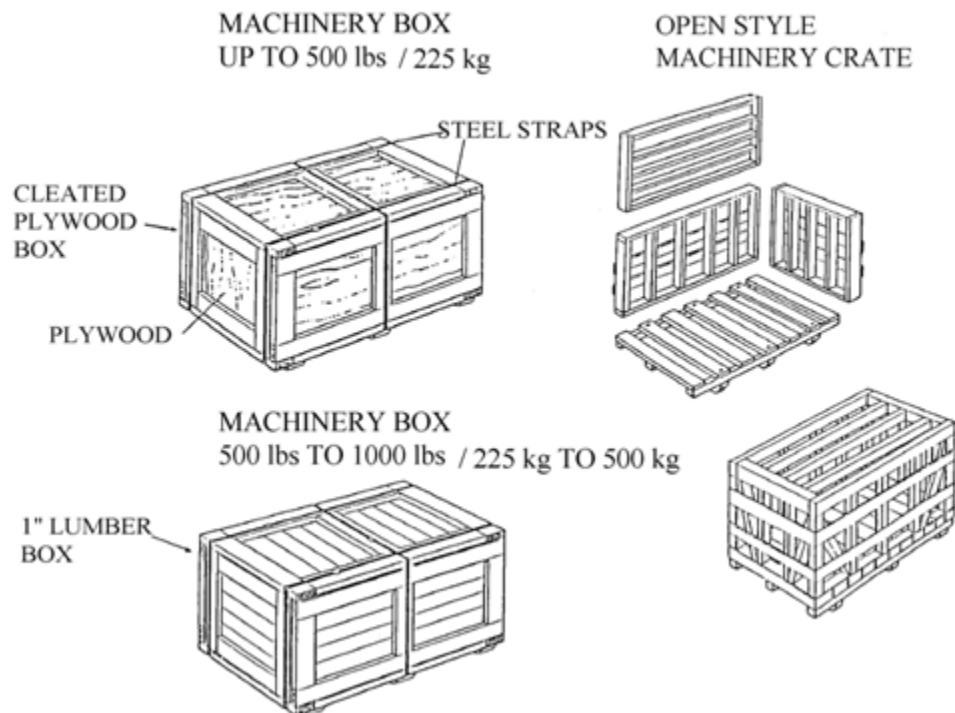
Should the Contractor be the Exporter of Record, the Supplier shall, at its own cost, assist the Contractor in all aspects in preparing applications for the export permits and in obtaining the required approvals. Where, under the delivery term, the Supplier is responsible for obtaining the required export permits, the Supplier shall do so at its own cost.

14. ATTACHMENTS

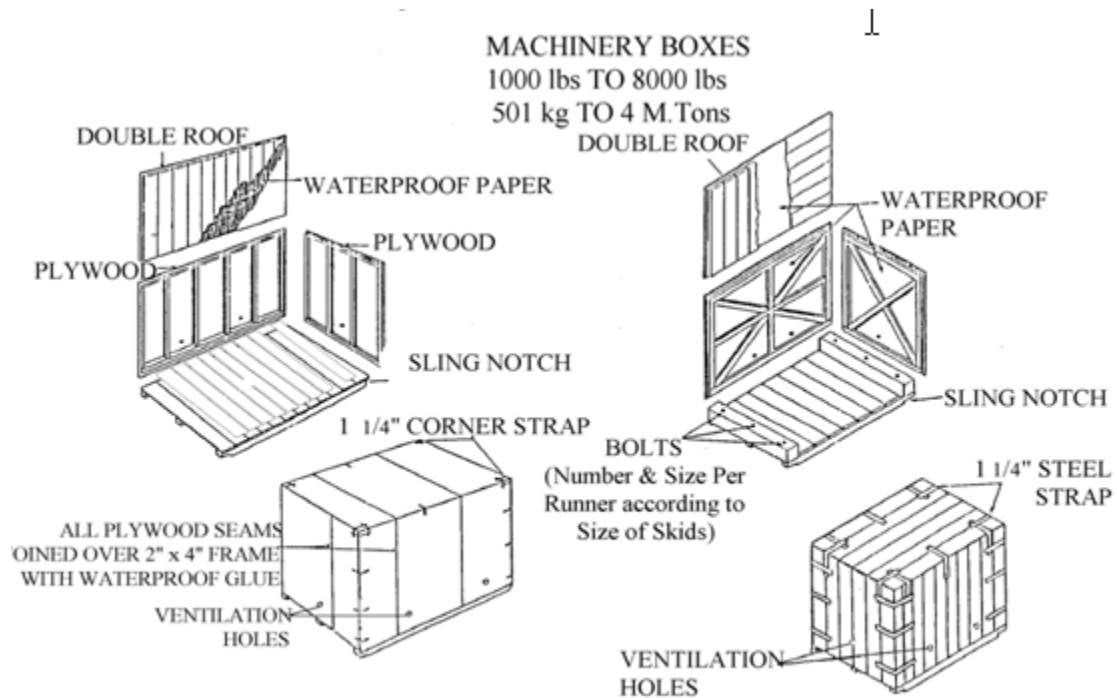
- A. SAMPLE FOR CASES UP TO 500 KG
- B. SAMPLE FOR CASES UP TO 4,000 KG
- C. SAMPLE FOR CASES UP TO 20,000 KG
- D. INTERNATIONAL HANDLING SYMBOLS

Attachment A
Case for Weights up to 500 kg

MACHINERY BOXES

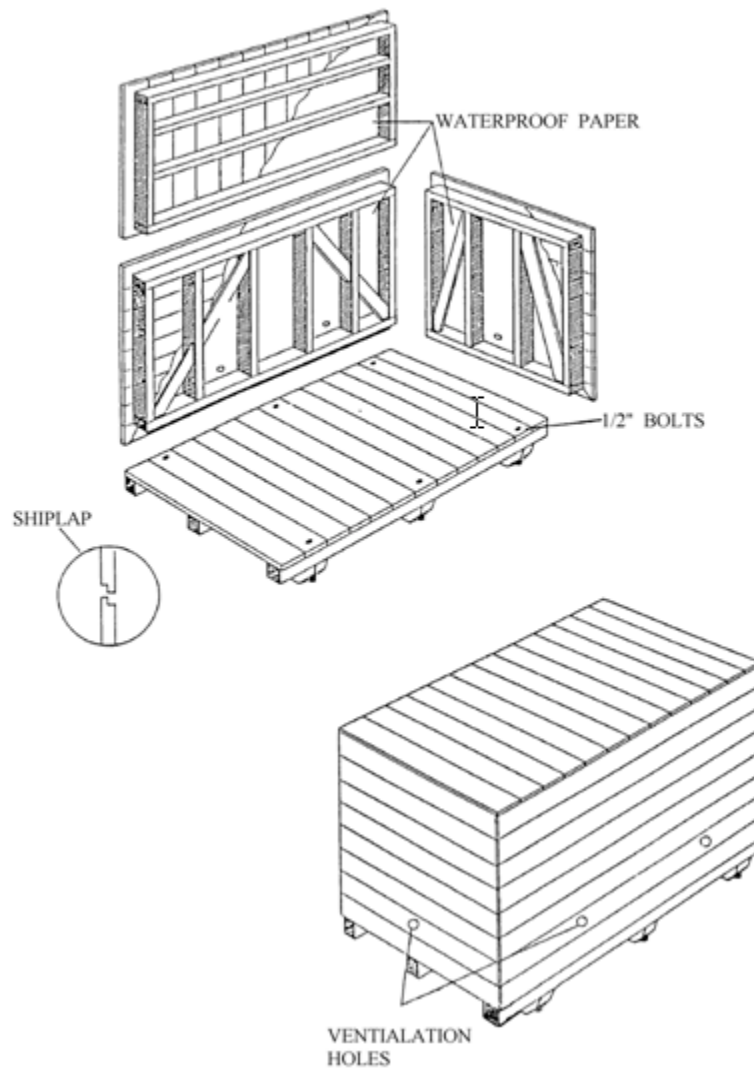


Attachment B
Case for Weights up to 4,000 kg

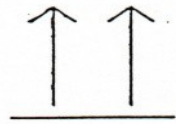


Attachment C
Case for Weights up to 20,000 kg

**HEAVY EQUIPMENT AND HIGH VOLUME
MACHINERY BOXES**



Attachment D
International Handling Symbols



1

THIS SIDE UP



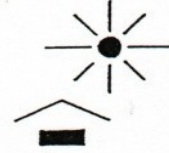
2

FRAGILE GOODS



3

KEEP DRY



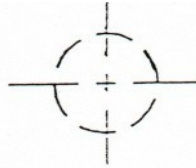
4

PROTECT FROM
SOLAR RADIATION



5

USE NO HOOKS



6

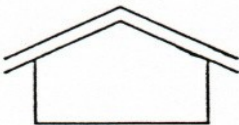
CENTRE OF GRAVITY



7

SLING HERE

STORAGE RECOMMENDATION



A

STORAGE IN
HEATED/VENTILATED
WAREHOUSE AND
WITH SPECIAL
PRECAUTIONS (AIR
CONDITIONING)



B

STORAGE IN
CLOSED WAREHOUSE



C

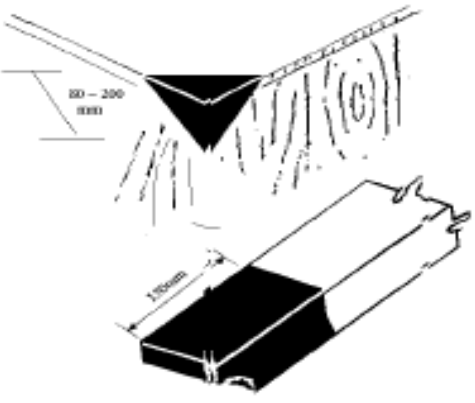
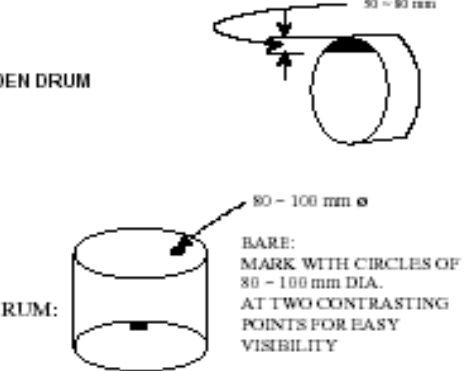
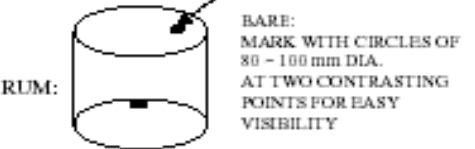
STORAGE IN
OPEN SHED



D

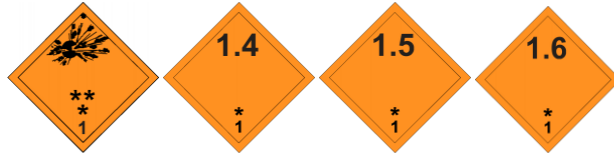
STORAGE IN
OPEN AIR

Attachment E International Symbols

International Symbols		Color Code Marking Procedure	
MARK	DESIGN		
MARKING PLACE			
THIS SIDE UP			
SYMMETRICAL AT UPPER LEFT OF BOTH ENDS			
SLING HERE			
BOTH SIDES			
CENTRE OF BALANCE			
BOTH SIDES			
FRAGILE OR HANDLE WITH CARE			
SYMMETRICAL AT UPPER LEFT OF BOTH ENDS			
KEEP DRY			
UPPER PART OF			
BOTH ENDS			

Attachment F
Hazardous International Symbols
SAFETY MARKS

- Class 1 explosives



- Class 2 gases



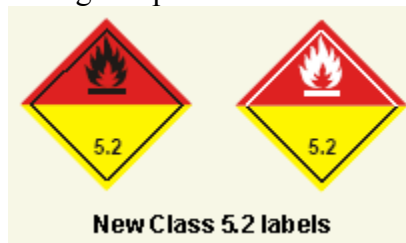
- Class 3 flammable liquids



- Class 4 flammable solids, spontaneously combustibles and substances that, on contact with water, emit flammable gases



- Class 5 oxidizing substances and organic peroxides



New !!!

- Class 6 poisonous (toxic) and infectious substances



- Class 7 radioactive materials



- Class 8 corrosives



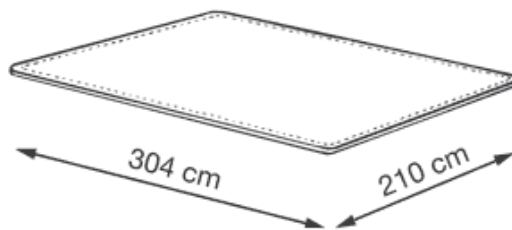
- Class 9 miscellaneous products or substances miscellaneous identified dangerous goods certain specified goods considered dangerous to the environment dangerous wastes



Attachment G
Aircraft Pallet Sizes

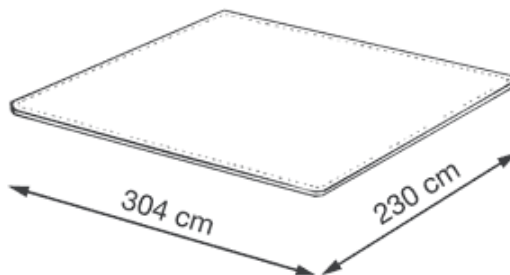
Standard Pallet

Type: Standard Pallet
Code: P1P, PAG, PAJ
Base Dimensions: 318 x 224 cm
Internal Dimensions: 304 x 210 cm
Loadable in: A310-300, A300-600, A340-200,
A340-300, B747-400, B747-200 Combi,
B747-400 Combi, B747-200 SF,
B747-200 F, B737-300 F, DC8-73 F



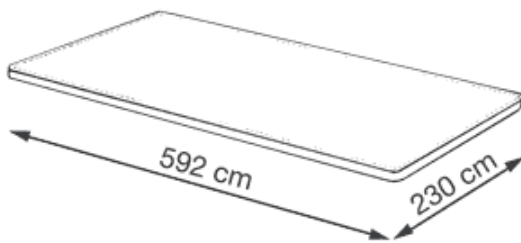
10ft Pallet

Type: 10 ft Pallet
Code: P6P, PQP, PMC
Base Dimensions: 318 x 244 cm
Internal Dimensions: 304 x 230 cm
Loadable in: A310-300, A300-600, A340-200,
A340-300, B747-400, B747-200 Combi,
B747-400 Combi, B747-200 SF, B747-200
F



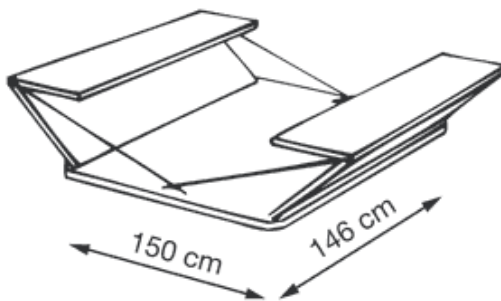
20ft Pallet

Type: 20-ft-Pallet
Code: P7E, PGE
Base Dimensions: 606 x 244 cm
Internal Dimensions: 592 x 230 cm
Loadable in: B747-200 Combi, B747-400 Combi,
B747-200 SF, B747-200 F



A320/A321 Pallet

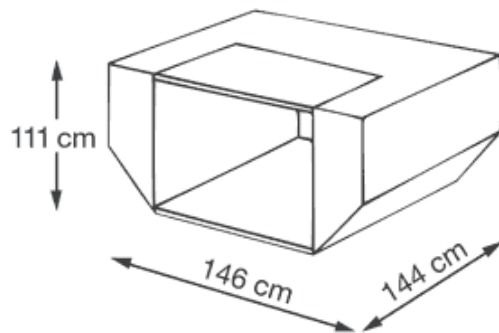
Type: A320/A321 Pallet
Code: PKC
Base Dimensions: 156 x 153 cm
Internal Dimensions: 150 x 146 cm
Loadable in: A320/200, A321/100



Aircraft Container Sizes:

A320/A321 Container

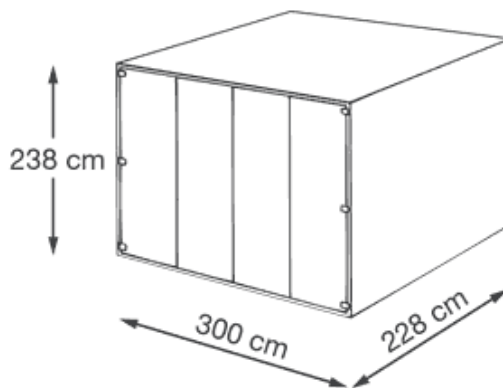
Type: A320/A321 Container
Code: AKH
Base Dimensions: 156 x 153 cm
Height: 114 cm
Volume: 3.5 cbm
Internal Dimensions: 146 x 144 x 111 cm
Loadable in: A320-200, Airbus A321-100



10 ft Container

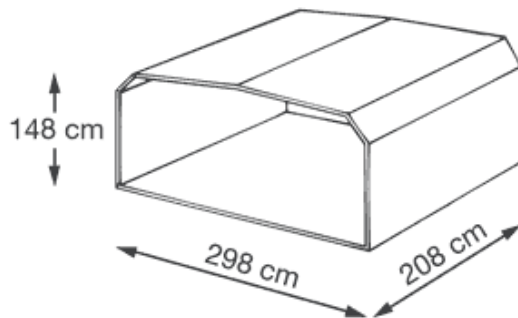
Type: 10 ft Container*
Code: AQ6, AMA, AQA, AMG
Base Dimensions: 318 x 244 cm
Height: 244 cm
Volume: 17 cbm
Internal Dimensions: 300 x 228 x 238 cm
Loadable in: B747-200 Combi, B747-400 Combi,
B747-200 SF, B747-200 F

Flexible doors/solid doors



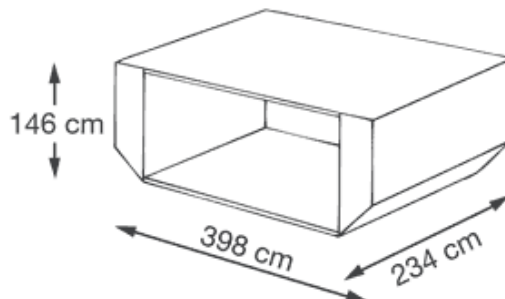
AAN Container

Type: AAN Container
 Code: AAN
 Base Dimensions: 318 x 224 cm
 Height: 156 cm
 Volume: 9 cbm
 Internal Dimensions: 298 x 208 x 148 cm
 Loadable in: A310-300, A300-600, A340-200,
 A340-300, B747-200, B747-400,
 B747-200 Combi, B747-400 Combi,
 B747-200 SF, B747-200 F, B737-300 F,
 DC8-73 F



AMF Container

Type: AMF Container
 Code: AMF
 Base Dimensions: 318 x 244 cm
 Height: 163 cm
 Volume: 13 cbm
 Internal Dimensions: 398 x 234 x 146 cm
 Loadable in: A310-300, A300-600, A340-200,
 A340-300, B747-200, B747-400,
 B747-200 Combi, B747-400 Combi,
 B747-200 SF, B747-200 F



AtkinsRéalis



Candu Energy Inc.
An AtkinsRéalis company
2251 Speakman Drive
Mississauga, Ontario,
L5K 1B2, Canada
 +1 905.823.9040

APPENDIX H
Formal Document Template
ATTACHED



AtkinsRéalis



SNC-Lavalin Nuclear Inc.
Candu Energy Inc.
2251 Speakman Drive
Mississauga, Ontario,
L5K 1B2, Canada
 +1 905.823.9040

- Proprietary -

File No.:					
	Doc. Collection ID	SI	Document Type	Serial No.	Revision No.

